Bachelor Thesis
**(151-0001-10L)**

# CFD Simulation of Hydrogen Propellant Flow in Nuclear Thermal Propulsion Reactor Cores


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# Abstract

This thesis presents a detailed demonstration of OpenFOAM v13 with the $k - \omega$ SST model for CFD of the NERVA NRX-A6 NTR in a single channel in the center region of the reactor. An algorithmic model was devised for the adaptation of axial power profiles to be applicable to different radial positions in the reactor based on validated estimates used in Program ELM in conjunction with model predictions of NRX-A power distributions. Material properties and operating conditions are compiled in the appendix for transparent validation. OpenFOAM case structures designed to accommodate the simulated problem in a pseudo-transient simulation under the given CHT regime documented are in detail for reproducibility and the numerical functionality of the implementation are outlined in a dedicated theory overview. An accurate CAD reconstruction of a late NRX-A type fuel cluster was used for visualization and as a utility for potential future research. Applicability of OpenFOAM V13 to this problem has been found to approximately reproduce results in previous model predictions.

# Acronyms

**CAD** computer-assisted design.

**CFD** computational fluid dynamics.

**CHT** conjugate heat transfer.

**Core Station** reactor axial coordinate measured from core inlet.

**CRT** control room time.

**EoS** equation of state.

**FVM** Finite Volume Method.

**LASL** Los Alamos Scientific Laboratory.

**NERVA** nuclear engine for rocket vehicle application.

**NRX** nuclear rocket experimental.

**NSS** Nuclear sub-System.

**NTR** nuclear thermal rocket.

**RANS** Reynolds-averaged Navier-Stokes.

**SST** shear stress transport.

**WANL** Westinghouse Astronuclear Laboratory.

# Contents

**Abstract** 1

**1 Introduction** 7
- 1.1 Nuclear Space Propulsion Overview and History . . . 7
- 1.2 Problem Description . . . 9

**2 Tool Comparison, Overview and Workflow Description** 10
- 2.1 Historical Methods of Analysis . . . 10
  - 2.1.1 Tools used during the Rover Program and later Methodologies . . . 10
- 2.2 Tools used for the CFD Model in this Thesis and their limitations . . . 12
  - 2.2.1 Computer-assisted Design . . . 12
  - 2.2.2 Meshing of Zone Geometries . . . 12
- 2.3 Computational Fluid Dynamics . . . 12
- 2.4 Overview of the Project Workflow . . . 13

**3 Analytical Theory of Methods used in this Work's OpenFOAM-Model** 14
- 3.1 The PIMPLE Algorithm . . . 14
- 3.2 Density, Momentum, Energy, Pressure, and Flux . . . 16
  - 3.2.1 Density . . . 17
  - 3.2.2 Momentum . . . 17
    - 3.2.2.1 Momentum Diffusion by Viscosity . . . 18
    - 3.2.2.2 Pressure Gradient . . . 19
    - 3.2.2.3 Momentum Predictor and Corrector . . . 19
  - 3.2.3 Conservation of Mass . . . 19
  - 3.2.4 Pressure . . . 21
  - 3.2.5 Energy . . . 21
    - 3.2.5.1 Specific Heat . . . 22
    - 3.2.5.2 Substantial Derivative of Pressure Term . . . 22
    - 3.2.5.3 Dissipation . . . 22
    - 3.2.5.4 Viscous Dissipation . . . 23
    - 3.2.5.5 Source/Sink . . . 23
- 3.3 Turbulence Fields . . . 23
  - 3.3.1 kOmegaSST Turbulence Model . . . 23

**4 Description of the Nuclear sub-System (NSS) 27**
4.1 Overview of the Reactor . . . 27
4.2 Description of the Fuel Cluster . . . 28

**5 Description of the single-channel CFD Model in OpenFOAM 31**
5.1 Equivalent Channel Model . . . 31
5.2 Geometry of equivalent flow channel . . . 32
5.3 Mesh Design . . . 33
5.3.1 Propellant Region . . . 34
5.3.2 Solid Regions . . . 35
5.4 Material Properties . . . 37
5.4.1 Fuel Region . . . 37
5.4.2 Support Region . . . 39
5.4.3 Propellant Region . . . 40
5.5 Fields: Initial and Boundary Conditions . . . 41
5.5.1 Support Region Temperature . . . 42
5.5.2 Fuel Region Temperature and specific Thermal Power . . . 42
5.5.2.1 Temperature . . . 42
5.5.2.2 Thermal Power Distribution . . . 42
5.5.3 Propellant/Coolant Region . . . 47
5.5.3.1 Temperature T . . . 48
5.5.3.2 Velocity U . . . 48
5.5.3.3 Pressure $p$ and $p_{rgh}$ . . . 49
5.5.3.4 Notes on initial and Boundary Conditions of Reynolds-averaged Navier-Stokes (RANS)-related Fields . . . 50
5.5.3.5 Turbulent thermal Diffusivity $\alpha_t$ . . . 50
5.5.3.6 Turbulent kinetic Energy k . . . 51
5.5.3.7 Turbulent kinematic Viscosity $\nu_t$ . . . 51
5.5.3.8 Specific Dissipation Rate $\omega$ . . . 52
5.6 Solver Structure . . . 52
5.6.1 Global Finite Volume Solution Settings . . . 53
5.6.2 Solid Regions . . . 53
5.6.2.1 Finite Volume Solution Settings . . . 53
5.6.2.2 Finite Volume Discretization Schemes . . . 54
5.6.3 Propellant Region . . . 54
5.6.3.1 Finite Volume Solution Settings . . . 54
5.6.3.2 Finite Volume Discretization Schemes . . . 56
5.7 Termination Criterion . . . 56

**6 Results and Discussion 58**
6.1 Temperature . . . 58

6.1.1 Temperature Validation Study . . . 61
6.2 Pressure and Density . . . 62
6.2.1 Pressure Transducer Validation Study . . . 63
6.3 Velocity . . . 64
6.4 Turbulence . . . 67
6.5 Axial Mesh Sensitivity . . . 69

**7 Potential applications for the OpenFOAM Model at Cluster Scale 70**
7.1 Single-element Fuel Cluster Section . . . 70

**8 Conclusion 72**
8.1 Potential Research Applications . . . 73

**References 74**

**Appendices 78**

**A Documentation of Fluid Property Extraction 79**

**B Documentation of System Reference Values 84**
B.1 Overview of NERVA Test Reactors and technical Data . . . 84
B.2 Core Map showing Placement of thermocoupled Fuel Cluster and Cluster Designation Scheme . . . 87
B.3 Original Graphs predicting relative radial Power Distribution in NRX-A reactors to provide data for axial Power Distribution Scaling . . . 89
B.4 Scope, Justification, and Assembly for Use of NRX-A5 Data . . . 92
B.5 NRX-A6 Validation Data . . . 93

**C Parts of the Fuel cluster 100**
C.1 Core Element Drawings . . . 100
C.2 Support Block Reconstruction . . . 102

**D Codes Used 108**
D.1 Python Code used to generate average radial Power Distribution and average Value . . . 108
D.2 Data used in the above code: . . . 115

# Chapter 1
# Introduction

## 1.1 Nuclear Space Propulsion Overview and History

Nuclear space propulsion's prospects for drastic efficiency gains over any chemical propulsion have been well known since its conception. The general conception of nuclear fission propulsion in space is historically attributed to Leo Szilard in 1932, while the nuclear thermal rocket (NTR) is anecdotally claimed to have been first patented by Richard Feynman at the Los Alamos National Laboratory in exchange for 1$, whereas it was first scientifically discussed in a treatise by R. Sperber from the Douglas Aircraft Company in 1947. [1, p.2]
While many alternative variations of nuclear space propulsion are proposed or even being developed, primarily nuclear electric propulsion where a lightweight nuclear reactor powers electric ion engines, which exceed all other existing propulsion methods in propellant efficiency. Furthermore, there are advanced propulsion concepts being researched, such as non-solid-phase thermal propulsion, direct fusion propulsion using various methods of ignition [2], including the comparatively matured though highly dangerous and difficult to scale down nuclear pulsed propulsion developed in the mid-20th century by Freeman Dyson under Project Orion, designed to eject nuclear explosive devices in sequence each creating a plasma blast from a tamper material to push against a springed and dampened pusher plate at the aft of a large spacecraft. [3, Chapter 17] Finally, there is the low thrust and ultra-high specific impulse concept of fission fragment rockets. [4, p.8] The most mature concept is, as of now, solid-core nuclear thermal propulsion as live-fire high-fidelity prototypes have been developed and tested extensively through the Cold War. [5][6][3, p.8–9]
The advantage of an nuclear thermal rocket (NTR) for manned spaceflight over the currently therein ubiquitous chemical propulsion lies in providing adequate thrust for timely orbital ejection and injection while using less than half of the propellant mass (mass to low Earth orbit being the greatest driver of cost in spaceflight to this day) for any given maneuver compared to the most efficient common method of chemical propulsion, the combustion of hydrogen and oxygen. This is achieved as per Tsiolkovsky Rocket equation proportionally by doubling the exhaust velocity, which itself is achieved by lowering propellant molecular mass [7] At extremely high core exit temperatures this advantage can be compounded supra-linearly by dissociation of hydrogen into an atomic gas further lowering average gas particle mass. [8, FIG. 5] One no less important advantage, particularly when relying on in-situ resource utilization specifically of lunar ice deposits, is the conservation of oxygen reserves that would otherwise be copiously and irrecoverably ejected into the vacuum of space through the course of combustive propulsion and are the most vital resource in crewed flight & station-keeping in outer space.

The NERVA tests were conducted under Project Rover, a joint NASA & AEC effort from 1955 to 1970 to develop nuclear propulsion for manned space flight intended for interplanetary applications to follow the Apollo program. [7] Full-scale nuclear prototyping has been crucial in ensuring a maximally safe spaceflight operations and operational modeling of nuclear space propulsion systems. In the era of Project Rover, iterative prototyping of live-reactor tests was the only way of establishing truly reliable benchmarks given the limited computational resources and models available at the time. However, with the sensible environmental concerns of the present day requiring vastly improved environmental safety precautions when testing an NTR prototype terrestrially (i.e. hermetic containment of the reactor and the diversion of exhaust gasses into a dedicated vacuum chamber as opposed to the open atmosphere [3, p. 254–256]), such prototyping has become far more expensive and scrutinized and thus represents the summit of a greater extent of detailed preliminary development now possible using modern computational tools. Thus, expensive and environmentally impactful live-fire prototype engine tests can be reduced to a necessary minimum at the very end of the development process to lend final pre-flight validation to the safety and reliability of a completed system.

While the initial sprint of the space race slowed to a creep due to the overwhelming and unmatched success of the Apollo Program removing the most immediate incentive to develop such a system for manned spaceflight missions for cold war prestige alone, research has been ongoing since and the USSR over the course of subsequent decades until close to its dissolution developed its own NTR prototype designated RD-0410 to similar mission profile of one hour of continuous firing time and 10 restarts. The engine was smaller in scale and thrust compared to NERVA, though of superior thermal efficiency [3, p.8–9] and higher specific impulse due to the use of an expander thermo-cycle, a twisted ribbon fuel element geometry offering higher surface area for heat transfer and higher operating temperatures thanks to the addition of an anti-corrosive agent to the propellant[6] - namely about 1% of hexane. [9] Its design as well as NERVA's is presently being studied by many scientists at various Chinese universities, as Chinese leadership has repeatedly emphasized the economic development of outer space with unmatched scientific and industrial concertion. [10]

The most recent project of an NTR has been DRACO under the commission of DARPA due for spaceflight in 2027, a system similar to NERVA in design though adapted for the use of high-assay low-enriched uranium [11] to limit concerns of nuclear proliferation in space, though the project was promptly cancelled in 2025 because DARPA's space-strategic interests and NASA's scientific interests no longer appeared to justify the cost and effort required to match environmental requirements for live-fire prototyping. Parallel to these developments in the American space sector, the Russian Federation has been undertaking an evolving proposal, that is nevertheless noteworthy for the consistency of ongoing development, to develop a nuclear-electrically driven space tug for cislunar, translunar, and interplanetary cargo applications originally designated TEM for "Transport and Energy Module", intermittently dubbed Project Zeus for its capability of reaching the outer planets such as Jupiter and in its most recent and concrete concept iteration NUKLON envisioning the tug to drop probes onto Venus before heading to Jupiter. It was first commissioned for development around 2010, [12, p.5] with a series of

relevant problems such as radiative droplet cooling for waste heat rejection into space being studied at the same Keldysh Research Center that developed the RD-0410 NTR. [13][6]

## 1.2 Problem Description

This thesis is intended to develop a precise model of the thermohydraulic conditions of the super-critical hydrogen propellant flow inside hot-channel components of a real solid-core NTR running at full power using open-source CFD software OpenFOAM V13.
The precise examination of flow characteristics inside the fuel is crucial for ensuring flow stability to ensure the structural integrity of the lightweight reactor in operation [14, Appendix 2] and limit the corrosion impact of the hot hydrogen near the hot aft end of the reactor. The benchmark chosen for the validation is the NERVA-program's longest running reactor NRX-A6 (running at nominal full power of at least 1120MW [14, p.108] for 62 minutes [5, p.31]) which also featured the most matured fuel element coatings of the entire NRX-A reactor series. [14, p.110][15, p.3-1] The result was a 75-80% reduction in rate of graphite fuel corrosion weight loss and a considerably increased core lifespan compared to previous reactors. [16, p.2]
A study of the radiological impact on the reactor's environment was conducted for this test by the US government's Environmental Protection Agency. [17] Further reduction has since been deemed achievable by use of carbide ceramic fuel elements whose development began towards the end of the Rover program and has been the subject of further study since. It is due to the unique fidelity of the NRX-A6 NTR reactor test with ample data & design information available, that this iteration of the NERVA test series was chosen for this and previous studies for the validation of the computational tools available.

# Chapter 2
# Tool Comparison, Overview and Workflow Description

## 2.1 Historical Methods of Analysis

### 2.1.1 Tools used during the Rover Program and later Methodologies

Already during the original NERVA-program, a number of computational modeling methods were available though each was very limited to a specific domain and those available for fluids operated on much simplified models that made extensive prototyping a necessity. Hereafter follows a short list below for a short overview based on the extensive documentation and citation of documentation in [18, p.2-20 to 2-25]:

**Explicitly Fluid Flow**

- TRACK II: FEM for heat transfer in fluid-cooled solids of arbitrary geometry
- TNT-II: Similar to TRACK II FEM but for heat and fluid flow between NTR sub-systems, but not nuclear computation
- ICAP: For calculations of flow rate, pressure distribution, and surface regression distribution due to corrosion (only for laminar regime)

**Heat Transfer in Fluids**

- TAP-A: FEM for transient or solid-state heat transfer in solids and fluids [19]
- MCAP: For hot-spot temperatures in channel geometries (flexible for various geometries); film here means **thermal** boundary layer

**Meshing**

- AUTOTEM: For meshing irregular 2D geometries into a nodal geometry
- W-MESHER: Improved AUTOTEM for more complex and multi-material 2D geometries

**Neither heat nor Fluid**

- LOWERE (Reference 16): Code to predict corrosion weight loss based on semi-empirical findings
- Core Corrosion & Reliability: Used to calculate failure probabilities

- MCARLO: Code used to automate Monte Carlo neutron transport method for use with TNT-II

It is noteworthy here that aside from empirical correlations there was no accurate numerical model available for the high-velocity flow of the super-critical hydrogen inside the fuel elements. This is because the Finite Volume Method (FVM) was academically theorized during the closing stages of the program and the first turbulence model ($k - \varepsilon$) a few years thereafter. It would take another decade before FVM computational fluid dynamics (CFD) became widely implemented in computational practice and a decade after that before a truly robust turbulence model became available in the form of the $k - \omega$-SST model that allows accurate treatment of both the bulk flow and boundary layer to accommodate the operating conditions in an NTR reactor core. Emrich in 2016 characterized this flow regime within NTR fuel to be *usually fairly turbulent* [3, p.122] and further that this property of the flow inside the fuel is desirable for increased heat transfer to the propellant. A comprehensive overview of common NTR modeling approaches has been summarized by Appel in 2012 as follows:

Table 2.1: Comparison of the abilities of different analysis techniques - source: [20, tab. 1.1].

| **Analysis Objective** | **1. 1D/2D with Empirical Correlations** | **2. Single-Element CFD** | **3. Core-Wide CFD** |
|---|---|---|---|
| Predict cooling channel temperatures and pressures | Has achieved $\pm 5\%$ on wall temperature | Matches SNRE empirical correlations for max fuel temperature to within 100 K | Same capability as Limited CFD. |
| Predict solid stresses | Requires simplifying assumptions on geometry. | Stresses due to coatings/claddings can be calculated. | Same capability as Limited CFD, with additional stresses from adjacent fuel elements and structures. |
| Predict overall engine performance | Assumes an averaging technique for exit flow based on a single channel. | Assumes an averaging technique for exit flow based on a single fuel element. | All flow is modeled explicitly, requiring no assumptions or averaging. |
| Investigate novel cooling configurations | Requires extrapolating correlations beyond their applicability. | Can provide insight into local phenomena. | Same capability as Limited CFD, with additional detail possible. |
| Model the effect of detailed hardware components | Requires a separate, uncoupled analysis. | Requires a separate, uncoupled analysis. | Components are integrated into the simulation. |
| Couple neutronics with temperature reactivity feedback | Not possible with single channel analysis. | Not possible with single fuel element analysis. | Possible with Star-CCM+ and MCNP |

Modeling during the Rover Program was done in the first column of tab. 2.1. The model for a single fuel channel within the NRX-A6 reactor developed in chapter 5 can be classified catego-

rized as between the first two columns of the first row with explicit model assumptions derived and developed to afford valid context and further applicability for the third row, specifically. This characterization of the study at hand is specifically due to the following properties:

- A single propellant channel inside a hot fuel element at the core center is modeled.
- Fluid flow is resolved fully numerically foregoing empirical correlations.
- Geometry, material properties and boundary conditions are sourced and established in detail to achieve maximum model fidelity for the given fluid flow.
- The main focus is the fluid dynamic properties of the problem.

## 2.2 Tools used for the CFD Model in this Thesis and their limitations

All tools used are open-source to minimize cost and provide a workflow easily accessible to any party interested in expanding on this work.

### 2.2.1 Computer-assisted Design

All computer-assisted design (CAD) is done with FreeCAD 1.02. It provides all the functionalities necessary for drafting even complex geometries easily and efficiently, though filleting of complex intersecting edges is unreliable, which is why these feature details are foregone as they are thermodynamically negligible on a single cluster though they may play a minor role for the thermal modeling of a full core where interlocking parts are filleted.

### 2.2.2 Meshing of Zone Geometries

Salome 9.15.0 was used for meshing. It features non-commercial algorithms such as NETGEN and Gmsh which were both used in different meshing iterations. The user must be aware that each meshing algorithm can have instabilities. Parameters are not unit-consistent between algorithms and can sometimes be misleading.

## 2.3 Computational Fluid Dynamics

The OpenFOAM Foundation's latest iteration of OpenFOAM running on Linux, namely v13, is the centerpiece of this undertaking. It contains a vast range of capabilities for many different fluid regimes and state-of-the-art functionalities. Its only downside is that its only comprehensive documentation, the C++ Code Guide is difficult to read without extensive experience with C++ documentation and not always optimally maintained to keep up with the latest features. Every version of OpenFOAM features a large number of tutorials, whose case structures

can be adapted for virtually any use case. Guides are available but cannot possibly cover all functionalities of this application while ensuring everything is up to date. The following is a non-exclusive list of OpenFOAM v13 functionalities utilized in this analysis:

- **foamMultiRun** for conjugate heat transfer (CHT)
- **codeStream** for variable initial fields
- **kOmegaSST** model for momentum transport

## 2.4 Overview of the Project Workflow

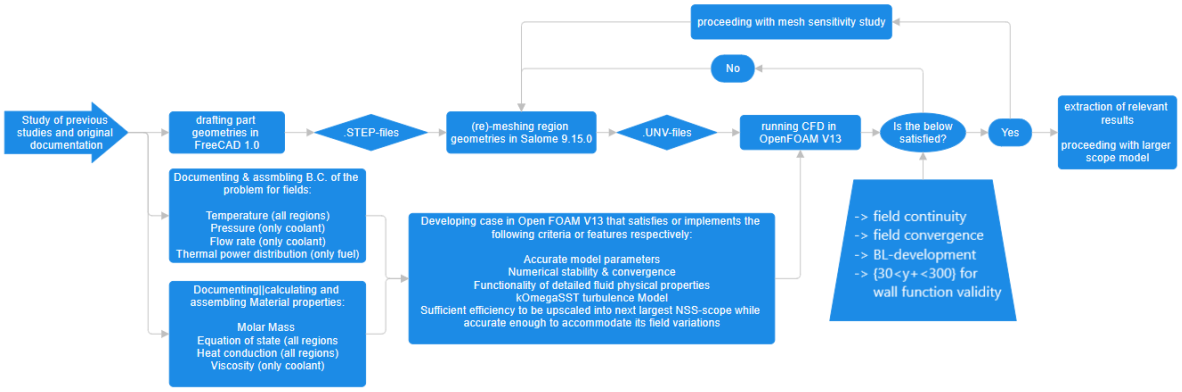



Figure 2.1: Workflow for CFD Model Development

# Chapter 3
# Analytical Theory of Methods used in this Work's OpenFOAM-Model

To keep the scope practical and aligned with the numerical investigation, the following theory review focuses on the governing equations as implemented for multi-region CHT in OpenFOAM v13. These equations are derived directly from formulations in the referenced literature.

## 3.1 The PIMPLE Algorithm

For use in a multi-region case in foamMultiRun, the PIMPLE algorithm combines two algorithms:

- PISO: Pressure Implicit with Split Operator
- SIMPLE: Semi-implicit Method for pressure-linked Equations

By setting time discretization schemes to steady state, it can be used as a SIMPLE steady state algorithm. For reasons specified in chapter 5.6.3.1, the operational mode used is pseudo-transient, meaning the system is simulated with a transient solver structure to arrive at a final steady state that is to model the steady state of the system simulated. PIMPLE pseudo-transient solution functionalities are shown for this use case in the flowchart in fig. 3.1 which is constructed based on [**21**, Chap. 5.21 & 7.1] with corrections analyzing logs from case runs using different corrector settings.

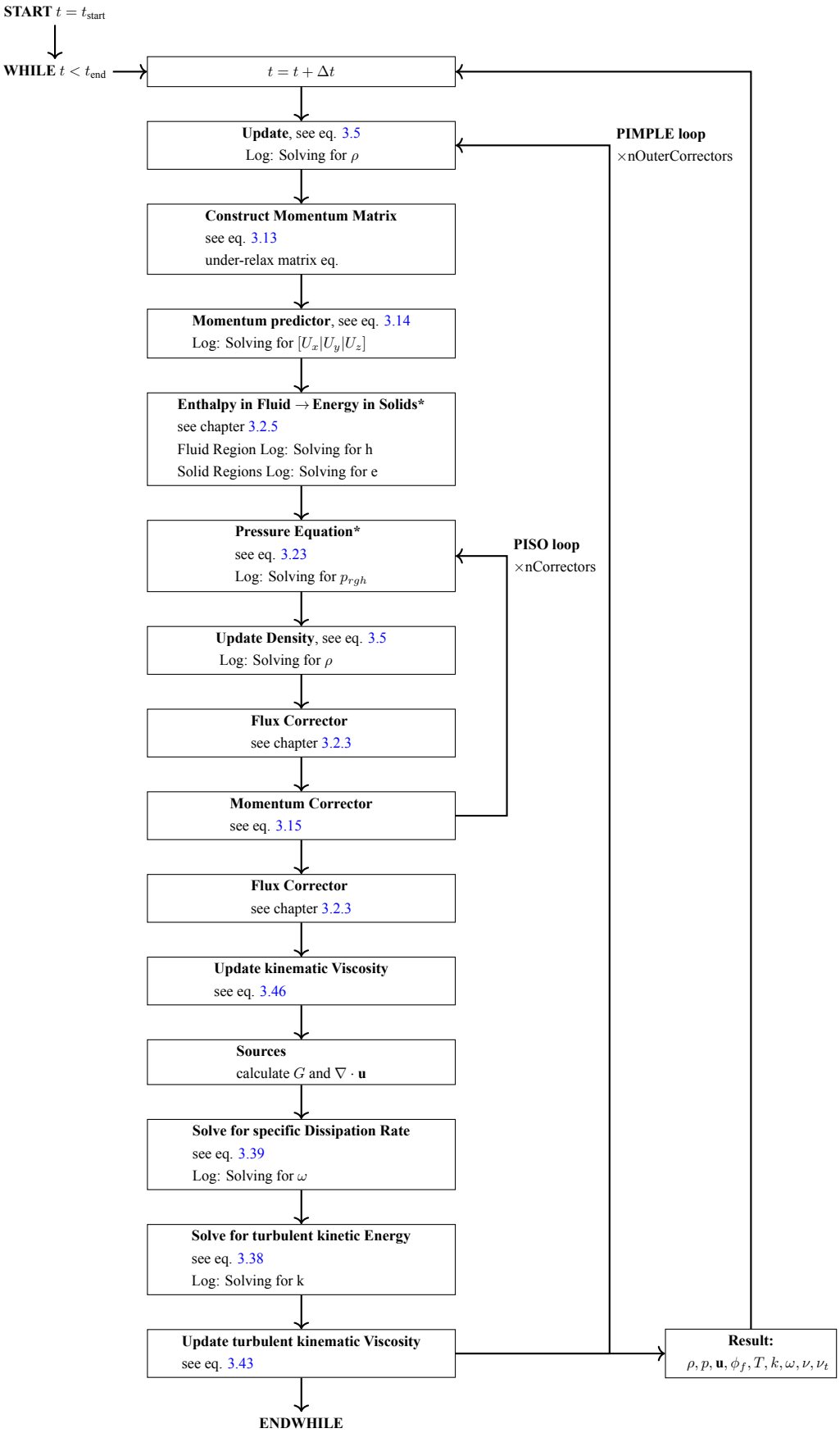



Figure 3.1: Flowchart of the PIMPLE algorithm for this use case.

***Repeated:** ×nNonOrthogonalCorrectors

## 3.2 Density, Momentum, Energy, Pressure, and Flux

The following equations in this section show the functionality of the algorithm and are derived based on ”The Finite Volume Method in Computational Fluid Dynamics” by F. Moukalled, L. Mangani, and M. Darwish. This is done primarily in their original notation summarized and extended in table 3.1 and further illustrated in fig. 3.2.

Table 3.1: Notation overview shown on a generic variable x

| Notation | Meaning |
|---|---|
| $x^o$ | old value from previous time step |
| $x^{(n)}$ | current iterate (known value before correction) |
| $x'$ | correction (increment added to old value) |
| $x^*$ | value estimated in current iteration step |
| $x^{**}$ | corrected output value |
| $x_C$ | cell-specific value |
| $x_F$ | value for a neighbouring cell |
| $\overline{x}$ | interpolated value of $x$ (e.g. face-interpolated from cells) |
| $\mathbf{x}$ | vector $(x, y, z)$ |
| $\underline{x}$ | vector $(x, y, z)$ |
| $\underline{\underline{x}}$ | matrix or tensor |
| $[x]$ | matrix or tensor |

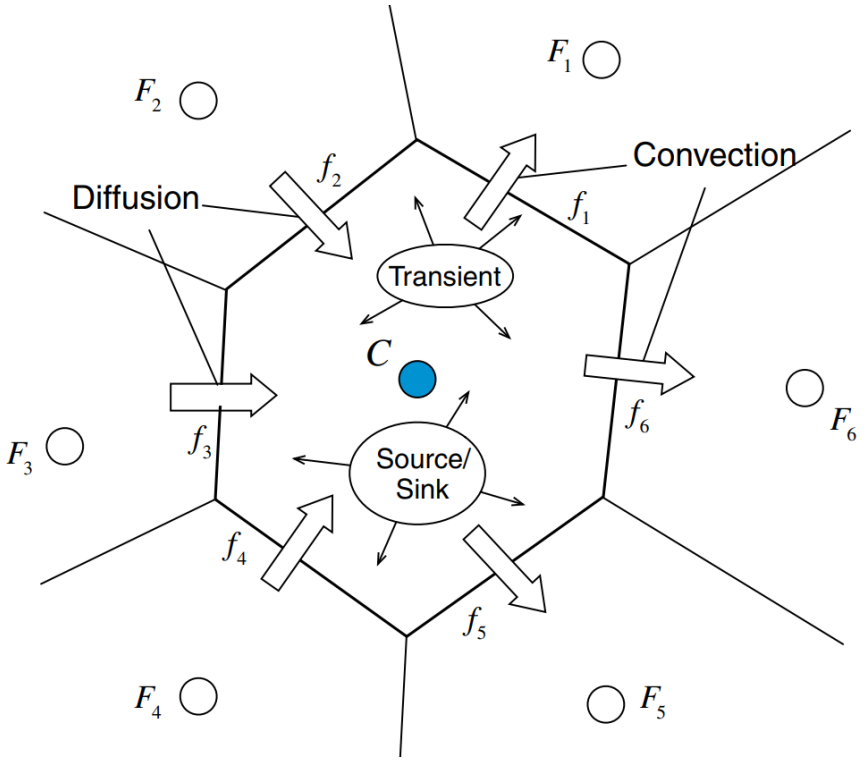



Figure 3.2: Example schematic of a control volume C with its faces ($f_1$ to $f_N$ ) neighbouring finite volume cells (with centers $F_1$ to $F_N$ ) - Source: [22, fig. 16.1]

### 3.2.1 Density

Beginning with the ideal gas law with:

- M: Molar mass[g/mol]
- $\overline{R}$: Ideal gas constant

$$pV = n\overline{R}T = m\frac{\overline{R}}{M}T \tag{3.1}$$

Reforming eq. 3.1 to formulate as per [22, eq. 16.8] ,that:

$$\rho = \frac{m}{V} = \frac{pM}{\overline{R}T} \tag{3.2}$$

Formulating the Taylor-expansion for density from [22, eq. 16.9] as the general method of calculating density in a given iteration:

$$\rho^{**} = \rho\big|_{(p^{(n)}+p')} = \rho^* + \rho' = \rho\big|_{(p^{(n)})} + \frac{\partial\rho}{\partial p}p' \tag{3.3}$$

Inserting eq. 3.2 into eq. 3.3:

$$\rho' = \frac{\partial\rho}{\partial p}p' = \frac{M}{RT}p' \tag{3.4}$$

And finally inserting the resulting $\rho'$ from eq. 3.4 back into eq. 3.3 shows the exact iteration process at each density update for each cell:

$$\rho^{**} = \rho^* + \frac{M}{RT}p' \tag{3.5}$$

Note that for further equations based on eq. 3.4:

$$\frac{M}{RT} = C_\rho \tag{3.6}$$

### 3.2.2 Momentum

As per [22, eq. 15.64] the general momentum balance is written as follows:

$$\underbrace{\frac{\partial}{\partial t}[\rho\mathbf{U}]}_{\text{rate of accumulation}} + \underbrace{\nabla\cdot\{\rho\cdot\mathbf{U}\cdot\mathbf{U}\}}_{convection} = \ldots$$

$$\ldots \underbrace{\nabla\cdot\left\{\mu(T)\cdot(\overbrace{[\nabla\mathbf{U}]+[\nabla\mathbf{U}]^\top}^{\text{symm. velocity tensor}} - \overbrace{\frac{2}{3}\cdot\nabla\cdot\mathbf{U}\cdot\mathbb{I}}^{\text{compressibility}})\right\}}_{\text{momentum diffusion by viscosity}} - \underbrace{\nabla p}_{\text{press. grad.}} + \underbrace{\mathbf{f}_b}_{\text{vol.forces}} \tag{3.7}$$

Note that for the reason given in chapter 5.5.3.3, volumetric forces (last term in eq. 3.7 are neglected in this model.

#### 3.2.2.1 Momentum Diffusion by Viscosity

For a given cell volume, both tensors for the strain rate tensor are established with Gauss' divergence theorem as follows based on [22, eq. 16.65 16.66]:

$$\begin{aligned}\int_{V_C} \nabla \cdot \left\{ \mu \left[ \nabla \mathbf{U} \right]^\top \right\} \mathrm{d}V &= \int_{\partial V_C} \left\{ \mu \left[ \nabla \mathbf{U} \right]^\top \right\} \cdot \mathrm{d}\mathbf{S} \\ &= \sum_{f=1}^{N} \mu(T)_f \cdot \left[ \nabla \mathbf{U} \right]_f^\top \cdot \mathbf{S}_f \\ &= \sum_{f=1}^{N} \mu(T)_f \cdot \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{bmatrix}_f \cdot \begin{pmatrix} S_f^x \\ S_f^y \\ S_f^z \end{pmatrix}\end{aligned} \tag{3.8}$$

where $\mathbf{S}_f$ is the face area vector (magnitude equals face area, direction is the outward face normal)
Further, similarly to eq. 3.8 for the non-transformed velocity gradient:

$$\begin{aligned}\int_{V_C} \nabla \cdot \left\{ \mu \left[ \nabla \mathbf{U} \right] \right\} \mathrm{d}V &= \int_{\partial V_C} \left\{ \mu \left[ \nabla \mathbf{U} \right] \right\} \cdot \mathrm{d}\mathbf{S} \\ &= \sum_{f=1}^{N} \mu(T)_f \cdot \left[ \nabla \mathbf{U} \right]_f \cdot \mathbf{S}_f \\ &= \sum_{f=1}^{N} \mu(T)_f \cdot \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial v}{\partial x} & \frac{\partial w}{\partial x} \\ \frac{\partial u}{\partial y} & \frac{\partial v}{\partial y} & \frac{\partial w}{\partial y} \\ \frac{\partial u}{\partial z} & \frac{\partial v}{\partial z} & \frac{\partial w}{\partial z} \end{bmatrix}_f \cdot \begin{pmatrix} S_f^x \\ S_f^y \\ S_f^z \end{pmatrix}\end{aligned} \tag{3.9}$$

For the compressibility correction in eq. 3.7, a different discretization is used as per [22, eq. 16.5 16.6] following the Gaussian theorem:

$$\int_{V_C} \nabla(\mu(T) \cdot \nabla \cdot \mathbf{U}\, dV = \int_{\partial V_C} \mu(T) \cdot \nabla \cdot \mathbf{U}\, dS = \sum_{f=1}^{n} \mu_f^{(n)} \left\{ \left( \frac{\partial u}{\partial x} \right)_f^{(n)} + \left( \frac{\partial v}{\partial y} \right)_f^{(n)} + \left( \frac{\partial w}{\partial z} \right)_f^{(n)} \right\} \tag{3.10}$$

where each derivative entry is calculated by interpolation between the local cell center with neighbouring cell centers [22, eq. 16.7]:

$$\left( \frac{\partial \phi}{\partial x} \right)_f^{(n)} = g_C \left( \frac{\partial \phi}{\partial x} \right)_C^{(n)} + g_F \left( \frac{\partial \phi}{\partial x} \right)_F^{(n)}, \qquad \phi = u, v, w \tag{3.11}$$

#### 3.2.2.2 Pressure Gradient

As per [22, eqs. 15.67 & 15.68], the pressure gradient can be computationally treated in one of two ways:

$$\begin{aligned}
&\int_{V_C} \nabla p \, dV = (\nabla p)_C \, V_C && \text{(treated as source term similar to volumetric force)} \\
&\int_{V_C} \nabla p \, dV = \int_{\partial V_C} p \, d\mathbf{S} = \sum_{f=1}^{N} p_f \, \mathbf{S}_f && \text{(Surface-flux evaluation based on Gauss-theorem)}
\end{aligned} \tag{3.12}$$

#### 3.2.2.3 Momentum Predictor and Corrector

Based on the source used for the construction of fig. 3.1 the momentum matrix is defined as follows using the terms for rate of accumulation, convection, and viscosity from eq. 3.7:

$$\underline{\underline{M}}(\mathbf{U}) := \underline{\underline{A}} \cdot \underline{u} - \underline{\underline{H}} = \frac{\partial}{\partial t}(\rho \mathbf{U}) + \nabla \cdot (\rho \mathbf{U}\mathbf{U}) - \nabla \cdot \left\{ \mu(T) \cdot \left( [\nabla \mathbf{U}] + [\nabla \mathbf{U}]^{\top} - \frac{2}{3} (\nabla \cdot \mathbf{U}) \, \mathbb{I} \right) \right\} \tag{3.13}$$

where $\underline{\underline{H}}$ is the off-diagonal (explicit) part of the momentum operator.
The momentum predictor is then completed by equating $\underline{\underline{\mathrm{M}}}(\mathbf{U})$ to the pressure gradient term from eq. 3.7:

$$\underline{\underline{A}} \cdot \underline{U} - \underline{\underline{H}} = -\nabla p \tag{3.14}$$

Rearranging eq. 3.14 leads to the momentum corrector:

$$\underline{U} = \underline{\underline{A}}^{-1} \, \underline{H} - \underline{\underline{A}}^{-1} \, \nabla p \tag{3.15}$$

### 3.2.3 Conservation of Mass

The overall iteration step for a given cell as per [22, eq. 16.10] is:

$$\dot{m} = \dot{m}^* + \dot{m}' \tag{3.16}$$

Based on the general conservation of mass equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{U}) = 0 \tag{3.17}$$

Eq. 3.16 is solved implicitly by using a semi-dicretized variant of eq. 3.17 as per [22, eq. 16.11]:

$$0 = \underbrace{\frac{(\rho_C^* + \rho_C' - \rho_C^o)}{\Delta t}}_{\approx \frac{\partial \rho}{\partial t}} V_C \; + \; \underbrace{\sum_{f=1}^{N} \dot{m}_f^*}_{\dot{m}^*} + \underbrace{\sum_{f=1}^{N} \dot{m}_f'}_{\dot{m}'} \tag{3.18}$$

Where:

$$\dot{m}_f^* = \rho_f^* \, \bar{\mathbf{U}}_f^* \cdot \mathbf{S}_f - \rho_f^* \, \bar{D}_f^U \left( \nabla p_f^{(n)} - \overline{\nabla p_f^{(n)}} \right) \cdot \mathbf{S}_f \tag{3.19}$$

with:

- $V_C$: Volume of the cell for which the value is calculated
- $\dot{m}_f$: total mass flux through face $f$ (mass flow rate across the face)
- $\dot{m}_f^*$: predicted (old-iterate) mass flux through face $f$
- $\dot{m}_f'$: mass flux correction
- $\rho_f^*$: density at face $f$ evaluated from the current (old) pressure guess
- $\rho_f'$: density correction at face $f$
- $\mathbf{U}_f^*$: face velocity from the current (old) momentum solution
- $\mathbf{U}_f'$: velocity correction at face $f$ due to pressure correction
- $f$: face index of the finite-volume mesh
- $D_f^U$: face-interpolated inverse diagonal coefficient of the discretized momentum equation

Further based on [22, eq. 16.14] the face mass flux correction for a given cell face ($m_f'$) can be calculated as follows:

$$\begin{aligned}
\dot{m}_f' &= \rho_f^* \, \bar{\mathbf{U}}_f^* \cdot \mathbf{S}_f - \rho_f^* \, \bar{D}_f^u \left( \nabla p_f' - \overline{\nabla p_f'} \right) \cdot \mathbf{S}_f + \left( \frac{\dot{m}_f^*}{\rho_f^*} \cdot \mathbf{S}_f \right) C_{\rho,f} \, p_f' \\
&= -\rho_f^* \, \bar{D}_f^U \, \nabla p_f' \cdot \mathbf{S}_f + \left( \frac{\dot{m}_f^*}{\rho_f^*} \cdot \mathbf{S}_f \right) C_{\rho,f} \, p_f' + \left( \rho_f^* \, \bar{\mathbf{U}}_f^* \cdot \mathbf{S}_f + \rho_f^* \, \bar{D}_f^U \, \overline{\nabla p_f'} \cdot \mathbf{S}_f \right) \\
&= -\rho_f^* \, \bar{D}_f^U \, \nabla p_f' \cdot \mathbf{S}_f + \left( \frac{\dot{m}_f^*}{\rho_f^*} \cdot \mathbf{S}_f \right) C_{\rho,f} \, p_f' - \rho_f^* \, \bar{\mathbf{H}}_f[\mathbf{U}] \cdot \mathbf{S}_f
\end{aligned} \tag{3.20}$$

with additionally:

- $p_f'$: Pressure correction estimated for the given face and scaled with relaxation factor for $p_{rgh}$

But in actual computational practice the last term in eq. 3.20 is typically neglected reducing it to:

$$\dot{m}_f' = -\rho_f^* \, \overline{D_f^U} \, \nabla p_f' \cdot \mathbf{S}_f + \left( \frac{\dot{m}_f^*}{\rho_f^*} \cdot \mathbf{S}_f \right) C_{p,f} \, p_f' \tag{3.21}$$

### 3.2.4 Pressure

Substituting eq. 3.20 into eq. 3.18 as per [**22**, eq. 16.17 & eq. 16.18], the compressible pressure correction equation is:

$$\frac{V_C}{\Delta t}\,C_\rho\,p'_C + \sum_{f=1}^{N}\left\{-\rho_f^* D_f^U \nabla p'_f \cdot \mathbf{S}_f + \left(\frac{\dot{m}_f^*}{\rho_f^*}\right) C_\rho\,p'_f\right\} = \ldots$$
$$\ldots - \left(\frac{\rho_C^* - \rho_C^\circ}{\Delta t} V_C + \sum_{f=1}^{N} \dot{m}_f^*\right) + \sum_{f=1}^{N} H_f[U'] \cdot \mathbf{S}_f \quad (3.22)$$

In PIMPLE as in SIMPLE with one corrector, the last term is neglected to obtain the pressure correction equation:

$$\underbrace{\frac{V_C}{\Delta t}\,C_\rho\,p'_C}_{\text{transient-like term}} + \underbrace{\sum_{f=1}^{N} C_\rho\left(\frac{\dot{m}_f^*}{\rho_f^*}\right) p'_f}_{\text{convection-like term}} - \underbrace{\sum_{f=1}^{N} \rho_f^* D_f^U (\nabla p')_f \cdot \mathbf{S}_f}_{\text{diffusion-like term}} = \underbrace{-\left(\frac{\rho_C^* - \rho_C^\circ}{\Delta t} V_C + \sum_{f=1}^{N} \dot{m}_f^*\right)}_{\text{source-like term}} \quad (3.23)$$

### 3.2.5 Energy

The energy balance in a compressible convecting medium is governed differentially at any given position within a compressible fluid by the following equation as given in [**22**, eq. 16.3]:

$$\underbrace{\frac{\partial}{\partial t}(\rho c_p T)}_{\text{transient term}} + \underbrace{\nabla \cdot (\rho c_p \mathbf{U} T)}_{\text{convection}} = \underbrace{\nabla \cdot (k \nabla T)}_{\text{conduction}} + \underbrace{\rho T \frac{D c_p}{D t}}_{\text{eq. 3.28}} + \underbrace{\frac{D p}{D t}}_{\text{eq. 3.29}} - \frac{2}{3}\underbrace{\mu \Psi}_{\text{eq. 3.30}} + \underbrace{\mu \Phi}_{\text{eq. 3.32}} + \underbrace{\dot{q}_v}_{\text{eq. 3.33}} \quad (3.24)$$

Because in this model, the propellant is considered heat-source free (ignoring neutron interaction with the low-density working fluid) eq. 3.24 reduces to:

$$\underbrace{\frac{\partial}{\partial t}(\rho c_p T)}_{\text{rate of accumulation}} + \underbrace{\nabla \cdot (\rho c_p \mathbf{U} T)}_{\text{convection}} = \underbrace{\nabla \cdot (k \nabla T)}_{\text{conduction}} + \underbrace{\rho T \frac{D c_p}{D t}}_{\text{eq. 3.28}} + \underbrace{\frac{D p}{D t}}_{\text{eq. 3.29}} - \frac{2}{3}\underbrace{\mu \Psi}_{\text{eq. 3.30}} + \underbrace{\mu \Phi}_{\text{eq. 3.32}} \quad (3.25)$$

For the fuel region using isochoric specific heat $c_v$ and removing all convective and dissipative terms retaining the heat source eq. 3.24 reduces to;

$$\underbrace{\frac{\partial}{\partial t}(\rho c_v T)}_{\text{rate of accumulation}} = \underbrace{\nabla \cdot (k \nabla T)}_{\text{conduction}} + \underbrace{\rho T \frac{D c_v}{D t}}_{\text{eq. 3.28}} + \underbrace{\dot{q}_v}_{\text{eq. 3.33}} \quad (3.26)$$

and finally for the support region additionally without the heat source term, eq. 3.26 reduces to:

$$\underbrace{\frac{\partial}{\partial t}(\rho c_v T)}_{\text{rate of accumulation}} = \underbrace{\nabla \cdot (k \nabla T)}_{\text{conduction}} + \underbrace{\rho T \frac{Dc_v}{Dt}}_{\text{eq. 3.28}} \tag{3.27}$$

The following sections show how each is applied volumetrically to a given cell.

#### 3.2.5.1 Specific Heat

$$\begin{aligned} \int_{V_C} \rho T \frac{Dc_p}{Dt}\, dV &= \rho_C^{**} T_C^{(n)} \left(\frac{Dc_p}{Dt}\right)_C V_C \\ &= \rho_C^{**} T_C^{(n)} \left[\frac{c_p^{(n)} - c_p^{\circ}}{\Delta t} + u_C^{**} \left(\frac{\partial c_p}{\partial x}\right)_C^{(n)} + v_C^{**} \left(\frac{\partial c_p}{\partial y}\right)_C^{(n)} + w_C^{**} \left(\frac{\partial c_p}{\partial z}\right)_C^{(n)}\right] V_C \end{aligned} \tag{3.28}$$

#### 3.2.5.2 Substantial Derivative of Pressure Term

This term appears only in compressible fluid regions:

$$\begin{aligned} \int_{V_C} \frac{Dp}{Dt}\, dV &= \left(\frac{Dp}{Dt}\right)_C^{*} V_C \\ &= \left[\frac{p_C^{*} - p_C^{\circ}}{\Delta t} + u_C^{**} \left(\frac{\partial p}{\partial x}\right)_C^{*} + v_C^{**} \left(\frac{\partial p}{\partial y}\right)_C^{*} + w_C^{**} \left(\frac{\partial p}{\partial z}\right)_C^{*}\right] V_C \end{aligned} \tag{3.29}$$

#### 3.2.5.3 Dissipation

$$\begin{aligned} \int_{V_C} \mu \Psi\, dV &= \mu_C^{(n)} \Psi_C^{**} V_C \\ &= \mu_C^{(n)} \left[\left(\frac{\partial u}{\partial x}\right)_C^{**} + \left(\frac{\partial v}{\partial y}\right)_C^{**} + \left(\frac{\partial w}{\partial z}\right)_C^{**}\right]^2 V_C \end{aligned} \tag{3.30}$$

where the dilation invariant is generally defined as:

$$\Psi = (\nabla \cdot \mathbf{U})^2 = \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}\right)^2 \tag{3.31}$$

#### 3.2.5.4 Viscous Dissipation

$$\begin{aligned}\int_{V_C} \mu\Phi \, dV &= \mu_C^{(n)} \Phi_C^{**} V_C \\ &= \mu_C^{(n)} \Bigg\{ 2 \left[ \left( \frac{\partial u}{\partial x} \right)_C^2 + \left( \frac{\partial v}{\partial y} \right)_C^2 + \left( \frac{\partial w}{\partial z} \right)_C^2 \right]^{**} \\ &\quad \ldots + \left[ \left( \frac{\partial u}{\partial y} \right)_C^{**} + \left( \frac{\partial v}{\partial x} \right)_C^{**} \right]^2 \\ &\quad \ldots + \left[ \left( \frac{\partial u}{\partial z} \right)_C^{**} + \left( \frac{\partial w}{\partial x} \right)_C^{**} \right]^2 \\ &\quad \ldots + \left[ \left( \frac{\partial v}{\partial z} \right)_C^{**} + \left( \frac{\partial w}{\partial y} \right)_C^{**} \right]^2 \Bigg\} V_C \end{aligned} \tag{3.32}$$

#### 3.2.5.5 Source/Sink

This is where volumetric heat is added in this case in the fuel region:

$$\int_{V_C} \dot{q}_v \, dV = (\dot{q}_v)_C \, V_C \tag{3.33}$$

## 3.3 Turbulence Fields

In the following section, the governing equations of the turbulence model are presented at the continuous (differential) level. Details of the numerical discretization and solver implementation are not treated here.

### 3.3.1 kOmegaSST Turbulence Model

The $k-\omega$ -shear stress transport (SST) is a RANS model for turbulence that combines both the $k-\omega$ and the $k-\varepsilon$ turbulence models for near-wall and bulk fluid behaviour respectively to overcome the following limitations of each model:

- The wall treatment of the $k$–$\omega$ model is affected by free-stream velocity.
- $k$–$\varepsilon$ is less robust in its wall treatment and its validity depends strongly on the $y^+$ value of the wall-adjascent cell.

The functionality of the combined SST model is shown in is shown in fig. 3.3 below:

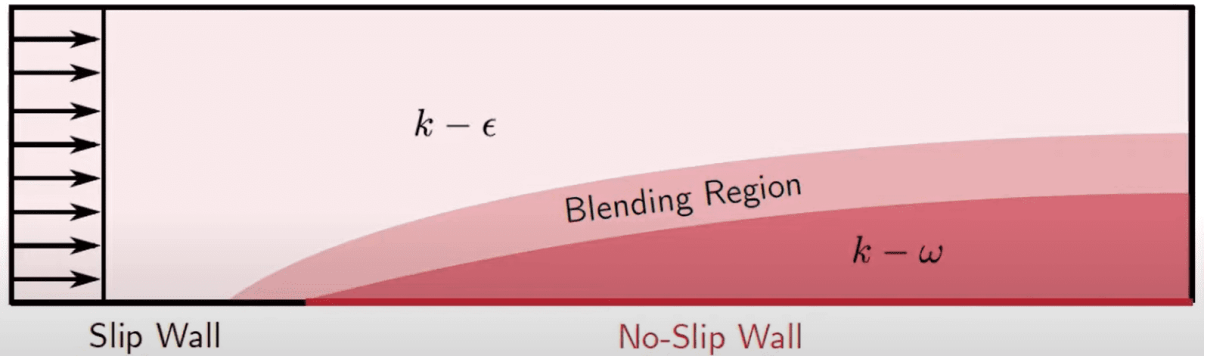



Figure 3.3: kOmegaSST model blending $k-\omega$ and $k-\varepsilon$ - Source: SimScale

The model is constructed in the paper by F.R. Menter[23] that first conceived it as follows:
**Original $k$–$\omega$ model:**

$$\frac{D\rho k}{Dt} = \tau_{ij}\frac{\partial u_i}{\partial x_j} - \beta^*\rho\omega k + \frac{\partial}{\partial x_j}\left[(\mu + \sigma_{k1}\mu_t)\frac{\partial k}{\partial x_j}\right] \tag{3.34}$$

$$\frac{D\rho\omega}{Dt} = \frac{\gamma_1}{\nu_t}\tau_{ij}\frac{\partial u_i}{\partial x_j} - \beta_1\rho\omega^2 + \frac{\partial}{\partial x_j}\left[(\mu + \sigma_{\omega1}\mu_t)\frac{\partial \omega}{\partial x_j}\right] \tag{3.35}$$

**Transformed $k$–$\varepsilon$ model:**

$$\frac{D\rho k}{Dt} = \tau_{ij}\frac{\partial u_i}{\partial x_j} - \beta^*\rho\omega k + \frac{\partial}{\partial x_j}\left[(\mu + \sigma_{k2}\mu_t)\frac{\partial k}{\partial x_j}\right] \tag{3.36}$$

$$\frac{D\rho\omega}{Dt} = \frac{\gamma_2}{\nu_t}\tau_{ij}\frac{\partial u_i}{\partial x_j} - \beta_2\rho\omega^2 + \frac{\partial}{\partial x_j}\left[(\mu + \sigma_{\omega2}\mu_t)\frac{\partial \omega}{\partial x_j}\right] + 2\rho\sigma_{\omega2}\frac{1}{\omega}\frac{\partial k}{\partial x_j}\frac{\partial \omega}{\partial x_j} \tag{3.37}$$

**Blended kOmegaSST model:**

$$\frac{D\rho k}{Dt} = \tau_{ij}\frac{\partial u_i}{\partial x_j} - \beta^*\rho\omega k + \frac{\partial}{\partial x_j}\left[(\mu + \sigma_{k}\mu_t)\frac{\partial k}{\partial x_j}\right] \tag{3.38}$$

$$\frac{D\rho\omega}{Dt} = \frac{\gamma}{\nu_t}\tau_{ij}\frac{\partial u_i}{\partial x_j} - \beta\rho\omega^2 + \frac{\partial}{\partial x_j}\left[(\mu + \sigma_{\omega}\mu_t)\frac{\partial \omega}{\partial x_j}\right] + 2\rho(1-F_1)\sigma_{\omega2}\frac{1}{\omega}\frac{\partial k}{\partial x_j}\frac{\partial \omega}{\partial x_j} \tag{3.39}$$

where for eqs. 3.38 and 3.39 based on the 1998 revision by D.C. Wilcox [24]:

- $k$ : turbulent kinetic energy per unit mass
- $\omega$ : specific dissipation rate in the original formulation
- $\tau_{ij}$ : Reynolds stress tensor, representing turbulent momentum transport
- $\mu_t$ : turbulent (eddy) viscosity, given in the SST model, see eq. 3.43
- $\beta^* = 0.09$ : model constant controlling destruction of $k$
- $\beta = 0.075$ : model constant for the dissipation of $\omega$

- $\gamma$ : production coefficients in the $\omega$-equation for the $k$–$\omega$, transformed $k$–$\varepsilon$, and blended SST models respectively, see eq. 3.45.
- $\sigma_k$ : turbulent Prandtl-number for diffusion of $k$
- $\sigma_\omega$ : turbulent Prandtl-number for diffusion of $\omega$
- $\sigma_{\omega 2}$: turbulent Prandtl-number for diffusion of $\omega$ for the $k-\varepsilon$ term.
- $F_1$ : blending function activating the $k$–$\omega$ formulation in the near-wall region and the transformed $k$–$\varepsilon$ formulation in the free stream, see fig. 3.3
- $\nu_t$ : turbulent kinematic eddy viscosity
- $\mu$ : molecular (laminar) dynamic viscosity.
- $u_i$ : velocity components in Cartesian directions $x_i$
- $x_j$ : spatial coordinate in direction $j$

The equations to define the variables are sourced from the tutorial script "Turbulence modeling in OpenFOAM: Theory and Applications" by J. Guerrero and adapted to the original notation of Menter's paper.
Two relations apply between $\omega$ and $\varepsilon$ for the coupling of both models. The first is shown in the following equation:

$$\omega = \frac{\varepsilon}{k \cdot \beta^*} \tag{3.40}$$

This relation is critical for the ability to construct both models for conjoined use with either $\varepsilon$ or $\omega$. Further, there is an approximation as per [**24**, eq. 4.8]:

$$\varepsilon \approx \frac{k^{3/2}}{L} \tag{3.41}$$

with $L$ being the characteristic length of the problem.
The Reynolds stress tensor $\tau_{ij}$ is defined similarly to the velocity tensor in eq. 3.7 with the Boussinesq-hypothesis, assuming isotropy of turbulent stresses;

$$\underline{\underline{\tau_{ij}}} = \mu_t \cdot \left([\nabla \mathbf{U}] + [\nabla \mathbf{U}]^T\right) - \frac{2}{3}\rho\, k\, \mathbb{I} \tag{3.42}$$

with turbulent viscosity being treated like molecular viscosity in the fluid and defined based on $k$ and $\omega$ typically using a limiting function for near-wall behaviour:

$$\nu_t = \frac{\mu_t}{\rho} = \frac{k}{\omega} f_{\text{lim}} \tag{3.43}$$

It can then be found by equating the scalar coefficients from the first right-hand term in eq. 3.39 with that from Wilcox:

$$\frac{\gamma}{\nu_t} = \alpha \frac{\omega}{k} \Rightarrow \gamma = \alpha \frac{\omega}{k} \cdot \nu_t \text{ , with: } \alpha = \frac{5}{9} \tag{3.44}$$

Then inserting eq. 3.43 into eq. 3.44, it follows that:

$$\gamma = \frac{5}{9} \cdot f_{lim} \tag{3.45}$$

The viscosity update in PIMPLE is conducted by adding turbulent viscosity from eq. 3.43 to molecular viscosity:

$$\nu = \nu_{mol} + \nu_t = \frac{\mu}{\rho} + \frac{k}{\omega} f_{\text{lim}} \tag{3.46}$$

Turbulent thermal diffusivity $\alpha_t$ is defined by the following equation:

$$\alpha_t = \frac{\nu}{\text{Pr}} + \frac{\nu_t}{\text{Pr}_t} \tag{3.47}$$

where:

$$Pr = \frac{\nu \cdot c_p}{\kappa} \tag{3.48}$$

Further, dimensionless wall distance is defined as per [**24**, eq. 3.101]:

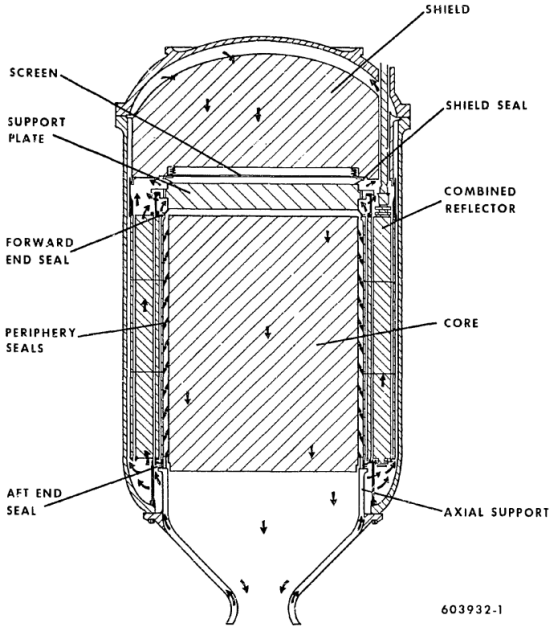
$$y^+ = y \cdot \frac{u_\tau}{\nu} \text{ , with: } u_\tau = \sqrt{\frac{\tau_w}{\rho}} \tag{3.49}$$

Lastly, dimensionless velocity as per [**24**, eq. 3.102] is approximated using the classical wall of the law in:

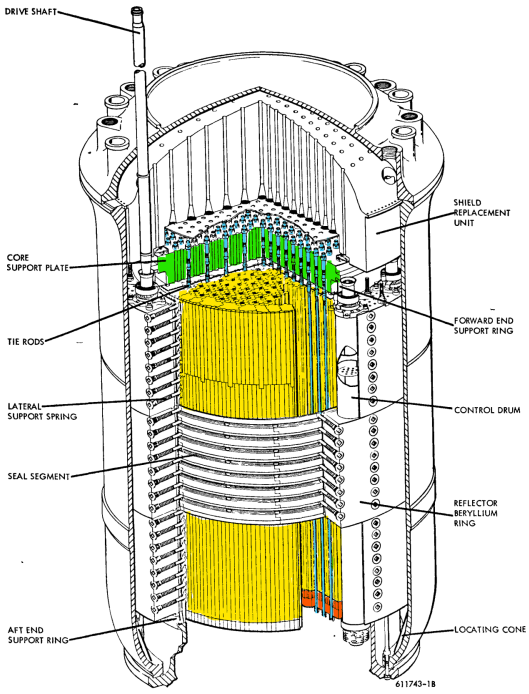
$$U^+ \approx \frac{1}{\kappa_{Karman}} ln(y^+) + B \text{ , with: } B \approx 5 \tag{3.50}$$

# Chapter 4
# Description of the Nuclear sub-System (NSS)

## 4.1 Overview of the Reactor




(a) Total NRX-A NTR-system flow diagram (source: [25, fig. 2-1])




(b) Cutaway of NRX-A6 test reactor (source: [26, fig. A1-1])
Green: core support plate
Yellow: reactor core
Blue: tie rods
Orange: support blocks

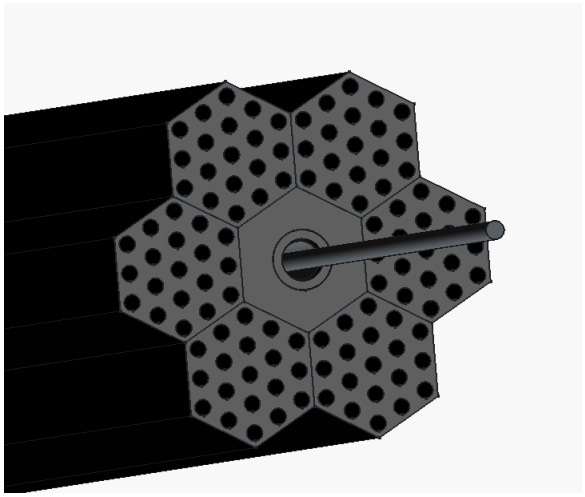
Figure 4.1: Flow schematic (a) and reactor cutaway

A flow schematic of the NTR system is shown in fig. 4.1. This reactor did not feature a flight-ready turbopump but was supported by external pump facilities on the test stand. A constructional cutaway drawing is shown in fig. 4.1b. Marked yellow therein encompassed by the neutron reflector ring with its rotating control drums, is the reactor core consisting of a matrix of honeycomb elements made of graphite. To prevent axial movement due to wall friction of the propellant, the clustered elements sit against interlocking support blocks (orange) which themselves are supported by the tie rods made of Inconel-718 [14, p. 105] reaching all through the support block and reactor (detail shown in fig.4.2c for the hot end and fig. 4.3 for the cold end) suspended from the support plate. The core was fueled for low non-uniformity and fuel elements featured improved coating to minimize the formation of cracks and/or pinholes from

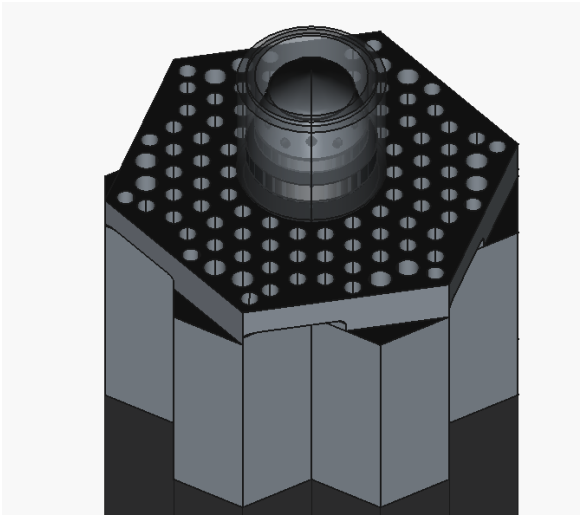
uneven thermal expansion of core elements leading to excessive local corrosion. [15] Fuel channel passages and tie rod annuli draw their coolant flows from the same plenum between the support plate and the core. In the NRX-A series as in all previous Project Rover tests, the cooler tie rod annulus exhausts into the nozzle chamber where it decreases propulsive efficiency due to decreased exhaust temperature. In later test reactors such as the large Phoebus 2a, 2-pass tie tubes were used instead drawing their coolant from a different cycle point before feeding it back into the core inlet plenum. [27, t=2:39] This renders NRX-A6's tie rod design thermo-hydraulically obsolete though the relative completeness of its documentation still makes it the primary choice for NTR research since the Rover program. The most thermally critical components are the fuel elements at the center region of the reactor and the fluid-solid interface between fuel element passages and the coolant passing through them to the nozzle chamber where the highest temperatures of the entire system are found particularly towards the hot end as an inherent operational condition of the engine concept. This is where the greatest mass loss from corrosion occurs. [15, fig. 3-3]

## 4.2 Description of the Fuel Cluster

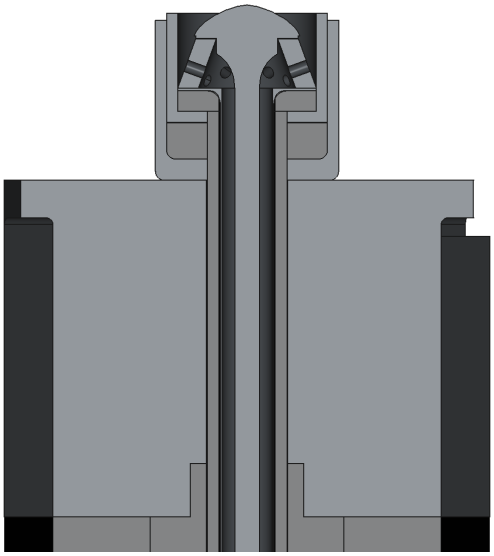
For each cluster over one support block and tie rod, 6 fuel elements with 19 propellant channels each are arranged around one central unfueled element containing the tie rod as seen in fig. 4.2a. The tie rod annulus is surrounded concentrically by a pyrolytic graphite sleeve, which has a significant impact on the heat transferred from the fuel elements to the tie rod annulus. The cold end hardware is included schematically for completeness in fig. 4.3. Note that it was found after all simulations were concluded that the cold end hardware contained a short extension to the channels, though it would likely not cause any major difference to the result due to its short length and negligible heating. The radii for the tie rod and hardware surrounding it through the length of the core were taken from WANL-TNR-209, table B-1 on page B-6. [28]



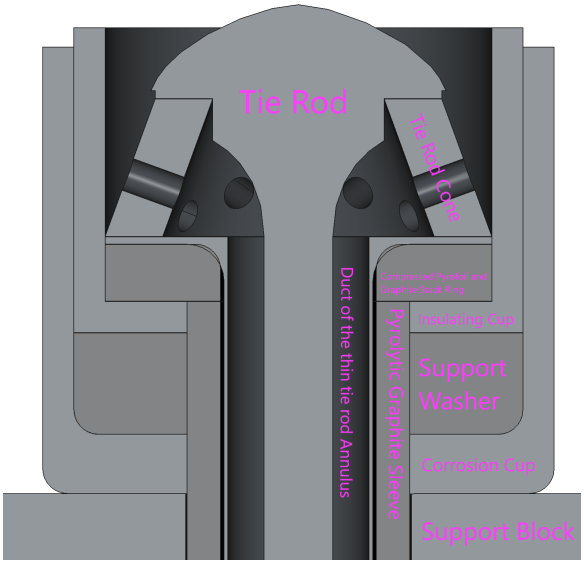
(a) Fuel cluster cold end without the hardware shown in fig. 4.3



(b) NRX-A fuel cluster hot end with "skirtless" support block and hardware (partially transparent) thermohydraulically equivalent to NRX-A6



(c) Cutaway view of the cluster hot end




(d) Support block hardware detail

Figure 4.2: Reconstructed NRX-A series fuel cluster geometry rendered in FreeCAD, showing cold-end and hot-end configurations.

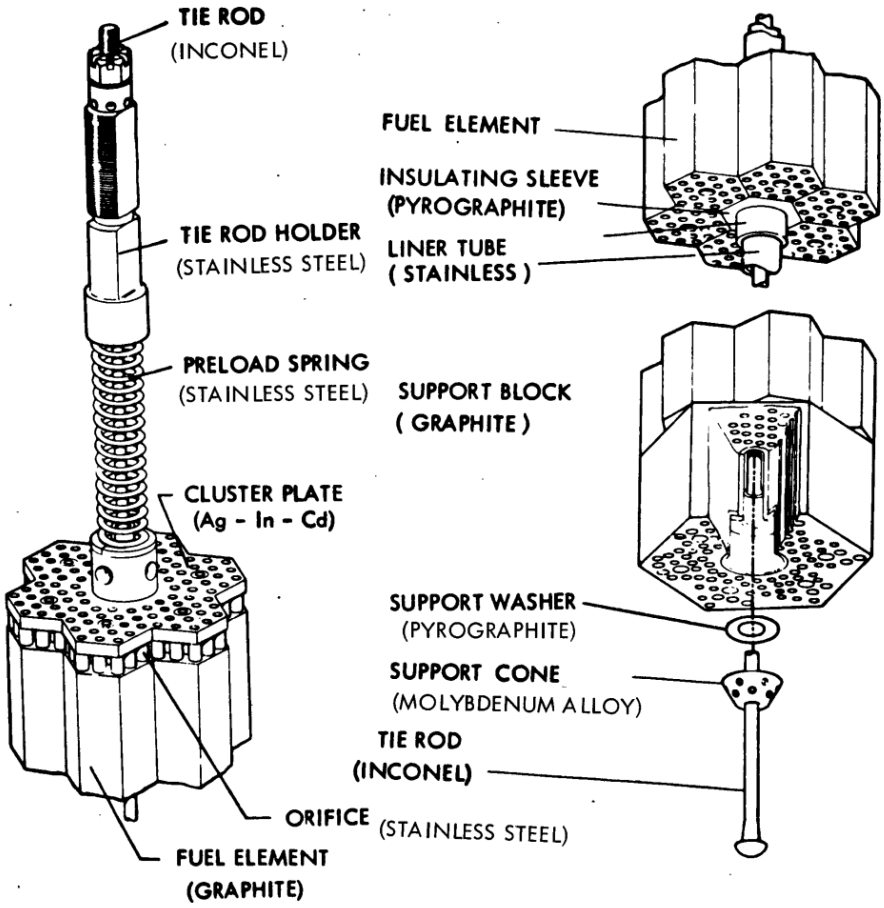



FIGURE 3-3

REGULAR FUEL CLUSTER

Figure 4.3: Schematic of the fuel cluster cold end hardware - Source: [28, fig. 3-3]

# Chapter 5
# Description of the single-channel CFD Model in OpenFOAM

To develop a robust and efficient CFD model from the bottom up, all while gaining command of the functionalities of OpenFOAM, the simplest physical model conceivable for heat transfer from fuel to coolant was chosen. It was originally intended to integrate this model into a more extensive model featuring an entire fuel element and later a full fuel cluster, though the time did not permit this. These considerations are thus relegated to the applications section following the completion of this model study.
The model is a pseudo-transient CFD simulation, where transient PIMPLE solvers are used for their improved stability progressing from an initial state centered around estimated hot-end and cold-end states with a number of fields linearly interpolated between them at the initial time as described in section 5.5 to allow for a well-posedness of the field solutions before ultimately converging on an accurate steady state describing the fluid and solid region fields within the model.

## 5.1 Equivalent Channel Model

Emrich Jr. (2016) devised a concept that is commonly used when modeling a single channel of the hexagonal NERVA fuel element independently from the wider system to create a simplified model wherein to examine coolant flow properties, cross-sectional asymmetries in the wall are eliminated using the equivalent channel concept. It functions thus, that the geometric cross-section of the entire hexagonal fuel element is partitioned into hexagonal sections each of which belongs to one coolant channel at its center. Then the hexagonal section is replaced with a circular section with a matching surface area and the problem is assumed to be azimuthally symmetric in terms of heat flux. This process is outlined schematically in fig. 5.1.

Figure 5.1: Equivalent flow channel concept - Source: [3, p.119]

This simplification step incurs a degree of inaccuracy for the solid temperature. Specifically, results shown in fig. 5.2 from the previous OpenFOAM validation study by Wang & Kotlyar indicate that around the outer hexagonal perimeter of a given single-channel section, temperature can vary by up to 30 K, while thermal equipotentials become increasingly circular closer to the channel, though with some asymmetry at the hot end.

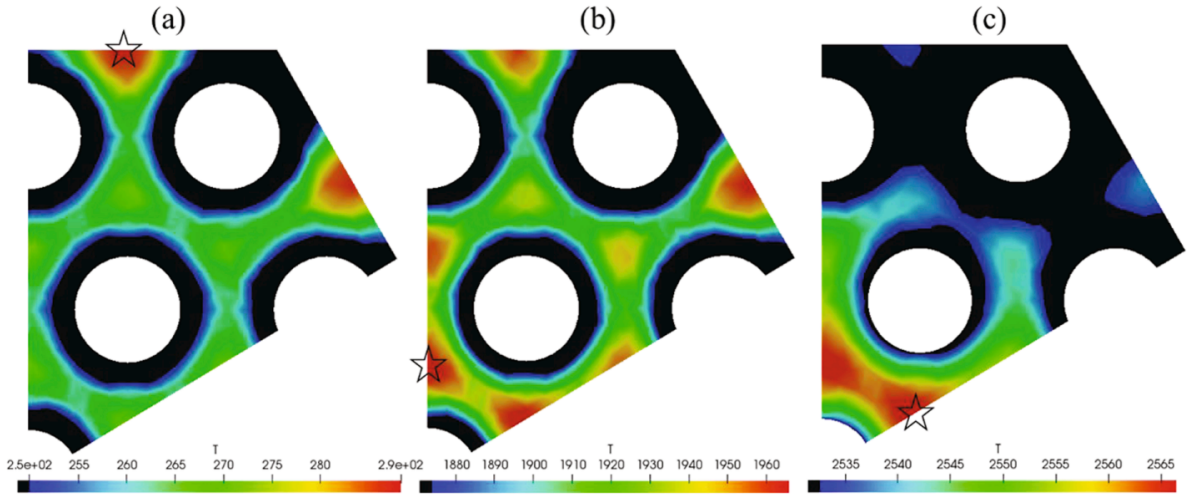



Figure 5.2: Cross-sectional temperature variability inside Fuel Element
(a): cold end
(b) axial center
(c) hot end
source: [29, fig. 10]

Due to the aforementioned observations, an equivalent single-channel model is deemed sufficient for a primarily thermohydraulic analysis such as the one intended here. In contrast, studies focused on peak fuel temperature or safety margins require modeling at least an entire fuel element.

The present model represents a channel in the central region of the reactor in order to minimize bias from core-radial temperature gradients. Owing to the low angular variability around a given channel and the suppression of core-radial gradients, the temperature at the outer boundary of the equivalent channel is assumed to be suitable for axial comparison with experimentally measured temperatures from thermocouples located in the fuel as well as unfueled elements.

## 5.2 Geometry of equivalent flow channel

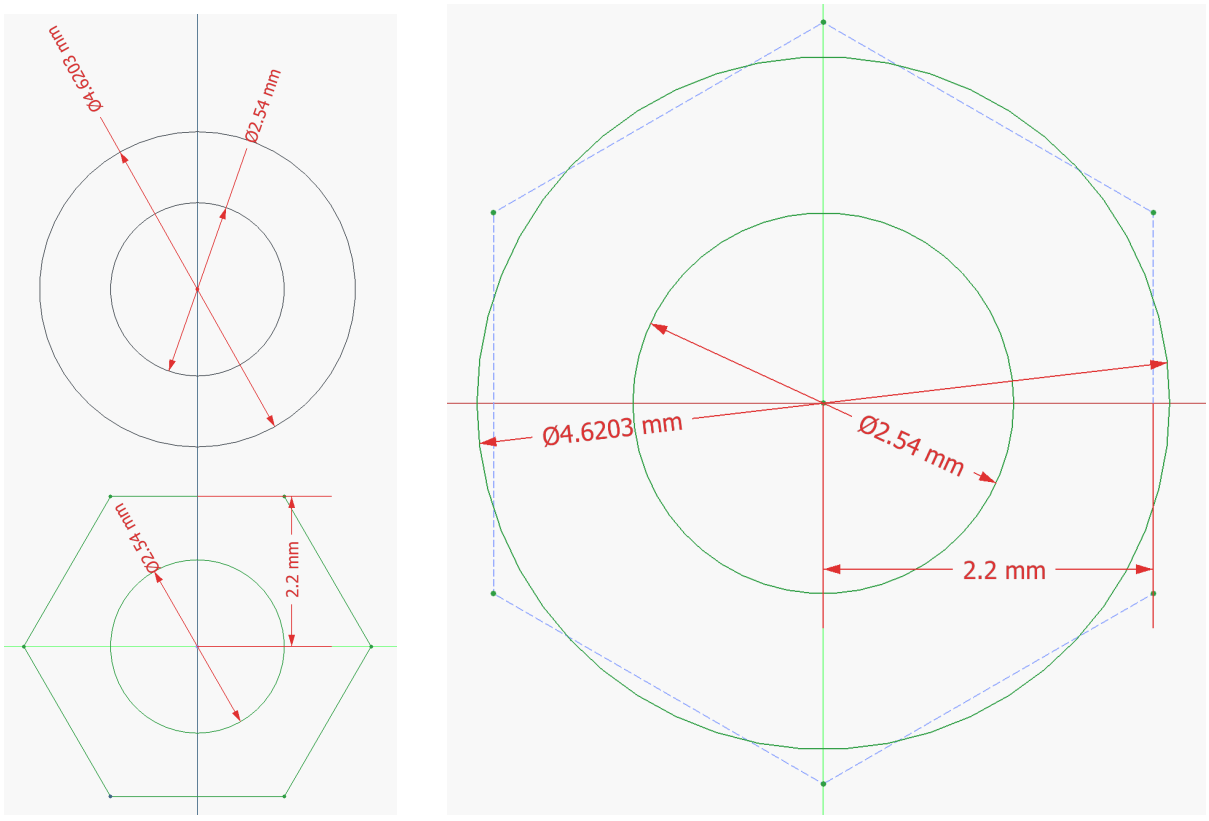
Using measurements compiled from Appendix C.1, a single circular equivalent cell is constructed. To calculate the geometric cross-section of an equivalent channel, the area of a regular hexagon is taken and the cross-sectional area of the propellant channel is subtracted:

$$A_{equivalent} = \frac{\sqrt{3}}{2} \cdot d^2_{channel-pitch} - \frac{\pi}{4} d^2_{channel} = \frac{\sqrt{3}}{2} \cdot (4.4mm)^2 - \frac{\pi}{4}(2.54mm)^2 = 11.6992mm^2 \tag{5.1}$$

For the outer diameter of the equivalent circular fuel channel with an identical inner diameter:

$$A_{equivalent} = \frac{\pi}{4}(d_o^2 - d_i^2) \rightarrow \sqrt{\frac{4}{\pi}A_{equivalent} + d_i^2} = d_o = \sqrt{\frac{4}{\pi}11.6992mm^2 + (2.54mm)^2} = 4.6203mm \tag{5.2}$$

In figure 5.3, the original geometry of the fuel section and the equivalent section are shown in direct comparison:




(a) Juxtaposition

(b) Superposition

Figure 5.3: Comparison of the hexagonal coolant-channel cross section and its equivalent circular representation in FreeCAD.

## 5.3 Mesh Design

Throughout the development of this CFD model, a number of different mesh profiles have been used to iteratively increase model robustness and accuracy. The meshing for each region is constrained by its boundary conditions and co-determined by a single global objective:

- The overall number of cells must be minimized to allow for maximum computational efficiency.

Mesh designs were meshed in 2D onto the cross-sectional geometry shown in the last section and were then extruded to the full length of each region with a specified axial mesh density. Each region had the following lengths established in Appendices C.1 & C.2:

Table 5.1: Axial extents of the modeled reactor regions

| Region | z-Length | Information |
|---|---|---|
| Fuel Region | 1320.8 mm (52 in) | spans from cold inlet to hot end — |
| Support Region | 38.1 mm (1.5 in) | spans from fuel hot end to exhaust chamber — |
| Propellant / Coolant Region | 1358.9 mm (53.5 in) | spans coaxially through Fuel & Support |

### 5.3.1 Propellant Region

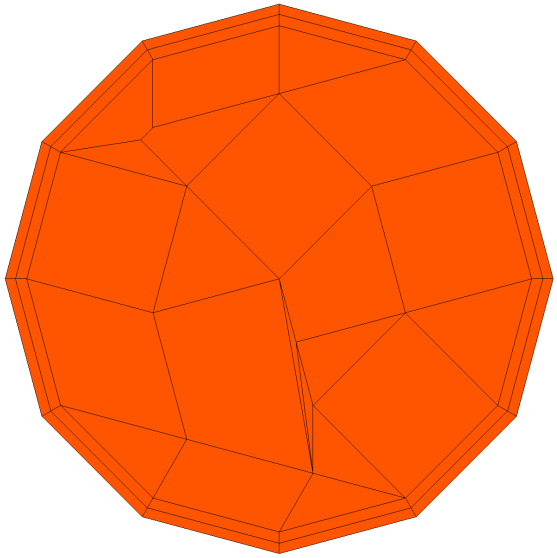
This region is the determining one for the entire problem, its constraints are as follows:

- Axial refinement has to be small enough to allow pressure field stability and convergence.
- Cross-sectional bulk refinement has to be sufficiently fine to avoid introducing unphysical angular asymmetries, continuous chords as well as axial asymmetries.
- The outermost cell layer of the fluid region must satisfy $30 \leq y^{+} \leq 300$ for the wall functions of the turbulence model fields to be valid

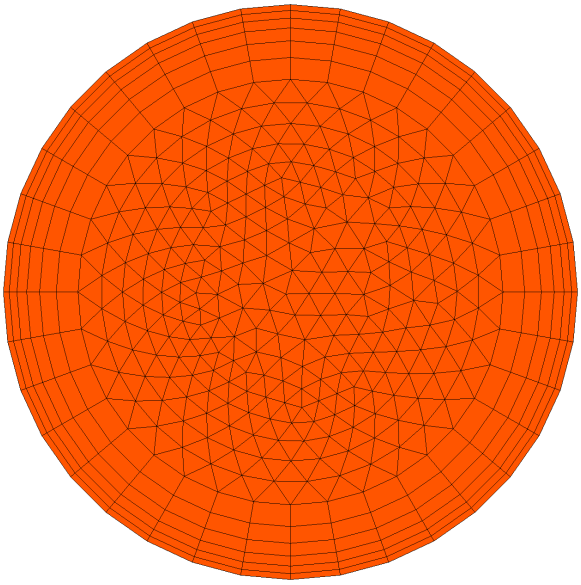
and it entails the following objectives:

- Axial refinement must be as coarse as possible to allow for a practical time step size with a Courant number smaller than 1 for a pseudo-transient simulation.
- Thickness of the outermost bulk layer and the innermost viscous layer must be minimized.
- Angular periodicity of the bulk mesh is to be avoided to prevent biases in axial resolution.
- Mesh non-orthogonality must be minimized.

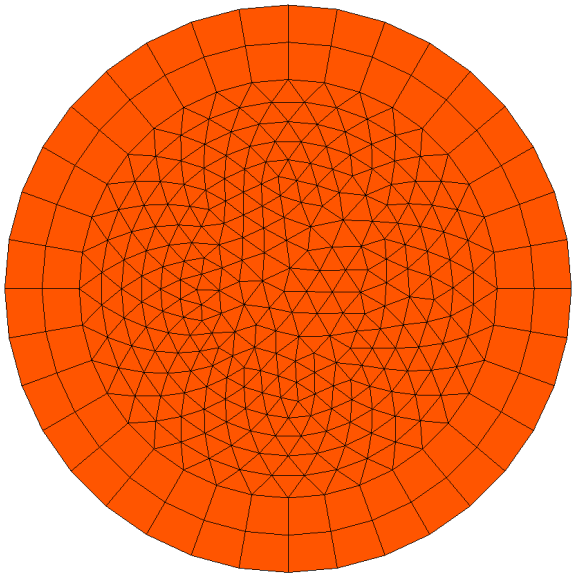
The wall function approach is chosen - as opposed to targeting full boundary layer resolution by targeting $\{y^{+} = 1\}$ - to afford increased robustness for the large field ranges expected and to minimize the number of cells necessary. In ensuring pressure convergence, axial refinement has been centered around 1.27mm (1/20th inch). This allows for an even partition of all regions across the entire length of the channel. As for cross-sectional profiles, these underwent the following evolution:



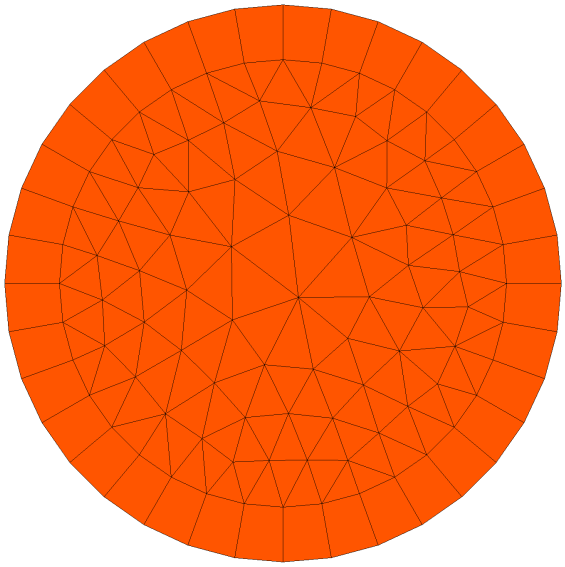
(a) Iteration 1.0: Initial low-resolution for efficient debugging



(b) Iteration 2.0: Attempt to fully resolve the boundary layer with $y^+ \approx 1$



(c) Iteration 3.0: Insufficient $y^+$ margin to ensure wall-function validity across entire wall length; resolution contrast between wall and bulk is excessive.



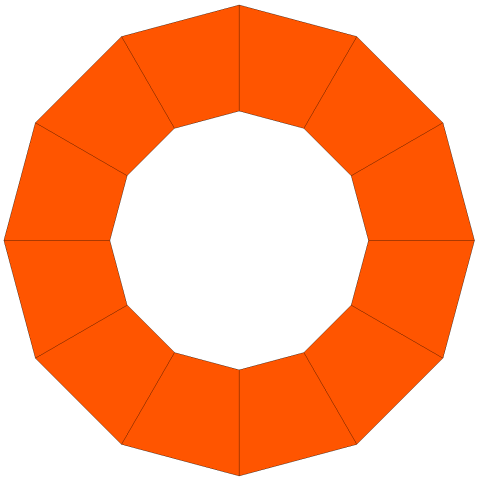
(d) Iteration 4.0: Adequate $y^+$ margin along the full pipe length with smooth transition to the bulk and angular aperiodicity;
final mesh used

Figure 5.4: Progression of fluid-region mesh refinement from initial debugging configuration to the final production mesh; each generated using NETGEN-algorithm + viscous layers hypothesis

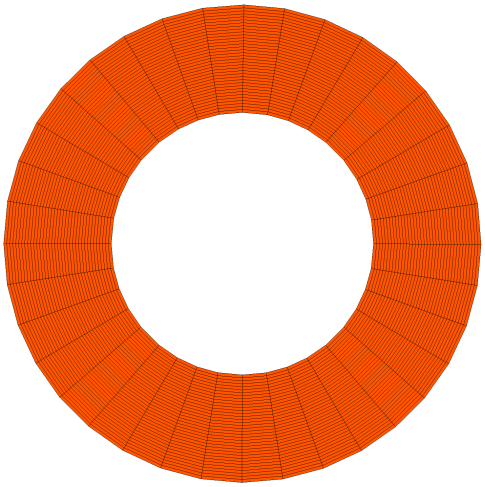
### 5.3.2 Solid Regions

Both solid regions use the same meshing profile because their outer boundary conditions are identical, except for the thermal power source profile that pertains only the fuel region. Their meshing constraints were as follows:

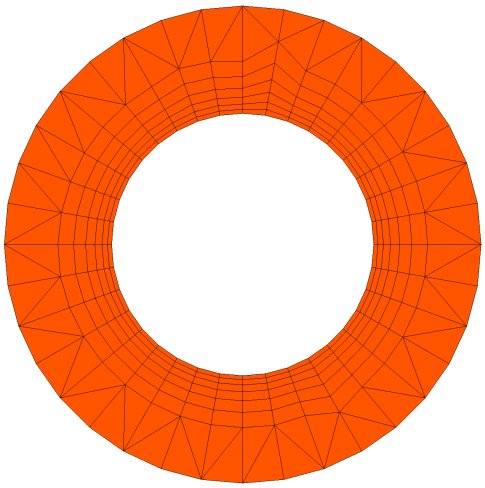
- zeroGradient boundary condition of the solid region perimeter must be satisfied to ensure an accurate energy balance.
- Axial refinement must match that of the fluid region to ensure a continuous axial coupled temperature distribution of the solid-fluid interface.
- Inner layers must allow for accurate modeling of radial temperature profile towards the fluid-solid interface.



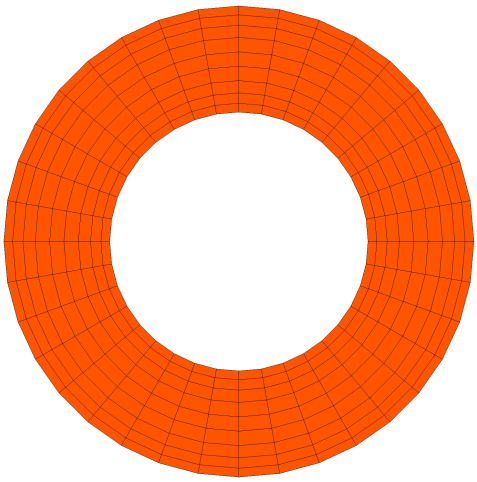
(a) Iteration 1.0: Initial low-resolution for efficient debugging



(b) Iteration 2.0: High resolution affords accurate solution but is computationally costly



(c) Iteration 3.0 using NETGEN algorithm: breakdown of zeroGradient boundary condition at the outer edge



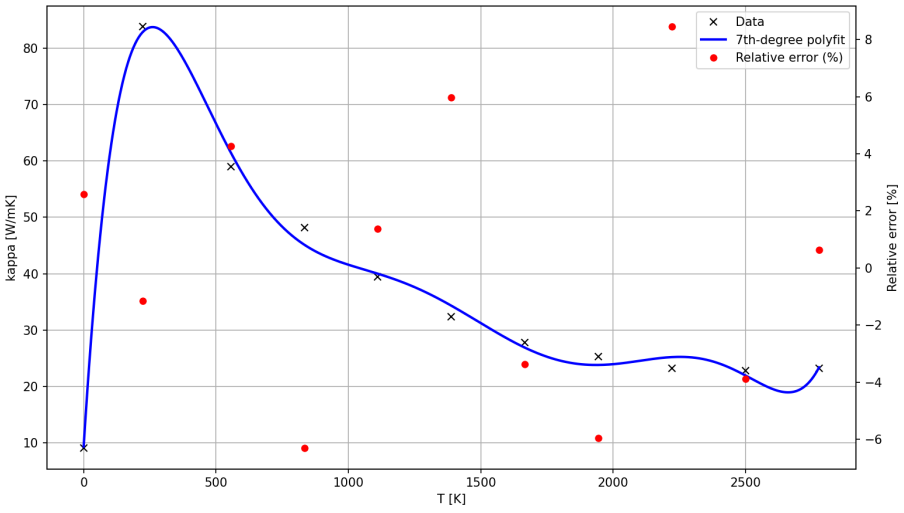
(d) Iteration 4.0: Custom cross-section with stretch factor 1.18 from both edges, yielding optimal accuracy–efficiency trade-off;
(final configuration used)

Figure 5.5: Progression of solid-region mesh iterations from initial low-resolution debugging setup to the final production configuration

## 5.4 Material Properties

### 5.4.1 Fuel Region

Fuel Elements were made of graphite with fuel particles embedded into the graphite matrix during the extrusion process. Fuel content for each element was adapted to the power profile required for the reactor. Exact empirical values for temperature-dependent thermal conductivity of the fuel elements used are difficult to specify with precision as they were manufactured at multiple plants (WANL's WAFF and Oak Ridge's Y-12 facility) with slightly varying thermal properties as part of the fuel development process, while post-mortem analysis established comparisons between elements in terms of corrosion and cracking behaviour. [30] It is noteworthy that during the manufacture of NRX-A6 the Y-12 plant encountered difficulties in sourcing powder of consistent quality for the extrusion of fuel elements. [31, p.10] The overall inconsistency of fuel element thermal conductivity throughout the NERVA program due to the highly experimental nature of the fuel used and its resulting variance in production quality makes this one critical parameter a point of unquantifiable uncertainty in this analysis.
Two estimates were found for the thermal conductivity of the fuel elements. The first is for a standard graphite fuel element with loaded 630mg of nuclear fuel beads per $cm^3$ with data provided above 555K $(1000°R)$ and the second is from a PROTK material model used for a thermal computational simulation of reactor components, which is here used to estimate $\kappa(T)$ below the lower cutoff temperature of the aforementioned empirical model. [32, p. A-1-101 to A-1-104 ] The compilation of both datasets shown explicitly in the following polynomial fit (of 7th-order compatible with a physicalProperties-dictionary in OpenFOAM v13 as are all following polynomial fits) in fig. 5.6:




$$\begin{aligned}\kappa(T) = \; & 3.19935732 \cdot 10^{-20}\, T^7 \\ & - 3.42194774 \cdot 10^{-16}\, T^6 \\ & + 1.47930845 \cdot 10^{-12}\, T^5 \\ & - 3.30794303 \cdot 10^{-9}\, T^4 \\ & + 4.05518685 \cdot 10^{-6}\, T^3 \\ & - 2.63206744 \cdot 10^{-3}\, T^2 \\ & + 7.47937033 \cdot 10^{-1}\, T \\ & + 9.37349442\end{aligned}$$

Figure 5.6: Polynomial fit for fuel element thermal conductivity

As for specific heat capacity of graphite, the IAEA's Nuclear Graphite Knowledge Base is referred to wherein an industrial source specifies that all graphites exhibit essentially the same specific heat at non-cryogenic temperatures, therefore also applies to the support block material

described in the subsequent chapter. [33, p. 4-14] The plot points taken from the graph from this source combined with those taken from a graph cited by C.C.Sorell et al. results in the polynomial fit in fig. 5.7.

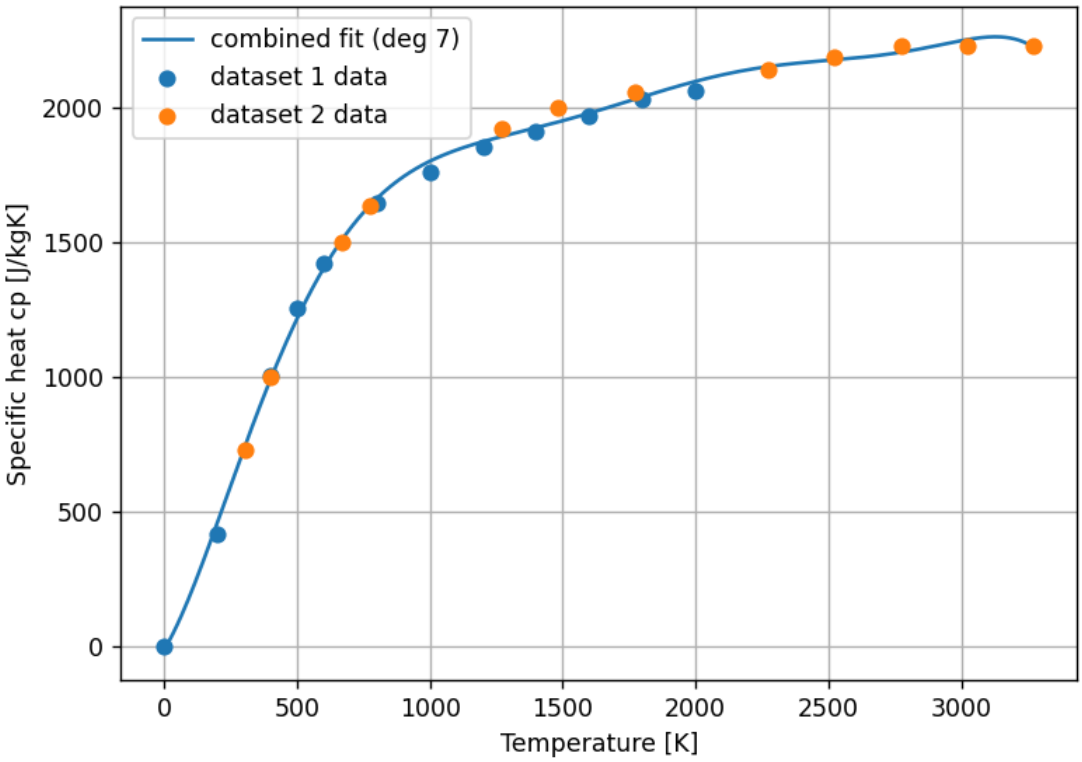



Figure 5.7: Specific Heat of Graphite
Dataset 1: [33, p. 4-14]
Dataset 2: [34, fig.6]

However, because the actual transient behaviour of the system during startup and power shifts is not the subject of this simulation, an arbitrarily lower $c_v$ shown in table 5.2 is chosen to cut computation time per time step in half as well as shorten simulated time until steady state substantially. This choice is justified as follows with the generalized heat conduction equation in solids:

$$\rho c_v(T)\,\frac{\partial T}{\partial t} = \nabla \cdot (\kappa(T)\,\nabla T) + \dot{q}''' \tag{5.3}$$

with $\frac{\partial T}{\partial t} = 0$ in making the equation independent of $c_v$ when in steady state:

$$0 = \nabla \cdot (\kappa\,\nabla T) + \dot{q}''' \tag{5.4}$$

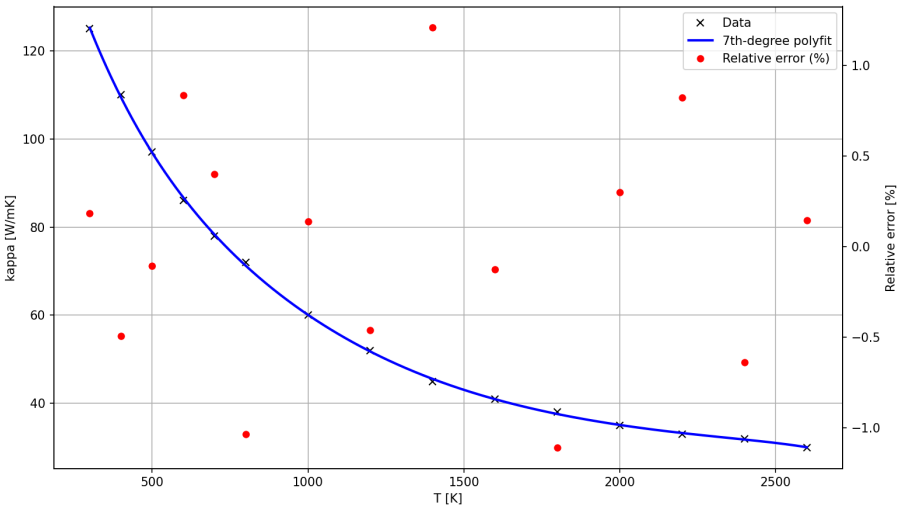
In summary, the material properties of the fuel material are implemented in a materialProperties-dictionary for OpenFOAM as follows in tab. 5.2:

Table 5.2: Fuel region thermophysical model keywords

| Keyword | Value | Information |
|---|---|---|
| type | heSolidThermo | Solid region |
| mixture | pureMixture | Homogenous graphite extrusion |
| transport | polynomialSolid | with single constant $500\frac{J}{kgK}$, alternatively fig. 5.6 |
| thermo | ePolynomial | Internal Energy using specific heat in fig. 5.7 |
| equationOfState | rhoConst | Constant density of $2236\frac{kg}{m^3}$ from [35, table A1-1] |
| specie | specie | molWeight = 12g/mol for carbon |
| energy | sensibleInternalEnergy | Isochoric behaviour, internal energy e is used to find T |

### 5.4.2 Support Region

Support blocks were made of ATJ-grade Graphite. [36, p.1] As per last chapter on fuel material, all properties of the graphites used here are identical except for thermal conductivity $\kappa(T)$. Using more extensive data available for ATJS-graphite, a lower-porosity variant of ATJ, tabulated in reference [37, Table XIX] the following polynomial fit was created:




$$\begin{aligned}\kappa(T) = &-1.93727752 \cdot 10^{-21}\, T^7 \\ &+2.22090849 \cdot 10^{-17}\, T^6 \\ &-1.08196756 \cdot 10^{-13}\, T^5 \\ &+2.92151965 \cdot 10^{-10}\, T^4 \\ &-4.82591161 \cdot 10^{-7}\, T^3 \\ &+5.19312622 \cdot 10^{-4}\, T^2 \\ &-3.86096723 \cdot 10^{-1}\, T \\ &+2.05231624 \cdot 10^{2}\end{aligned}$$

Figure 5.8: Polynomial fit for ATJ-S graphite thermal conductivity, with relative error at the data points

For the thermal conductivity, V. Kachur tabulated two values for $\kappa(T)$ of ATJ [38] seen below in table 5.3 which includes an error analysis showing that heat conductivities of ATJ-S and ATJ converge towards the high operating temperatures of the support block at full reactor power justifying the use of this property as a modeling choice for the given regime:

Table 5.3: Thermal Conductivity of ATJ and error analysis compared with ATJ-S

| Temperature [K] | $\kappa$ [W/mK] | $\Delta\kappa$ [W/mK] | relative error [%] |
|---|---|---|---|
| 1373 | 51.8816 | 5.3816 | 11.57 |
| 1673 | 43.0952 | 3.4952 | 8.826 |

Accurate low-temperature thermal conductivity of the support region material is not of interest in this analysis as that region is placed under the high-temperature thermal regime of the hot end.

### 5.4.3 Propellant Region

The terms propellant and coolant are subsequently used synonymously as the latter was the keyword used in the CFD model.
Below in table 5.4 the overall model choices for the hydrogen propellant are shown:

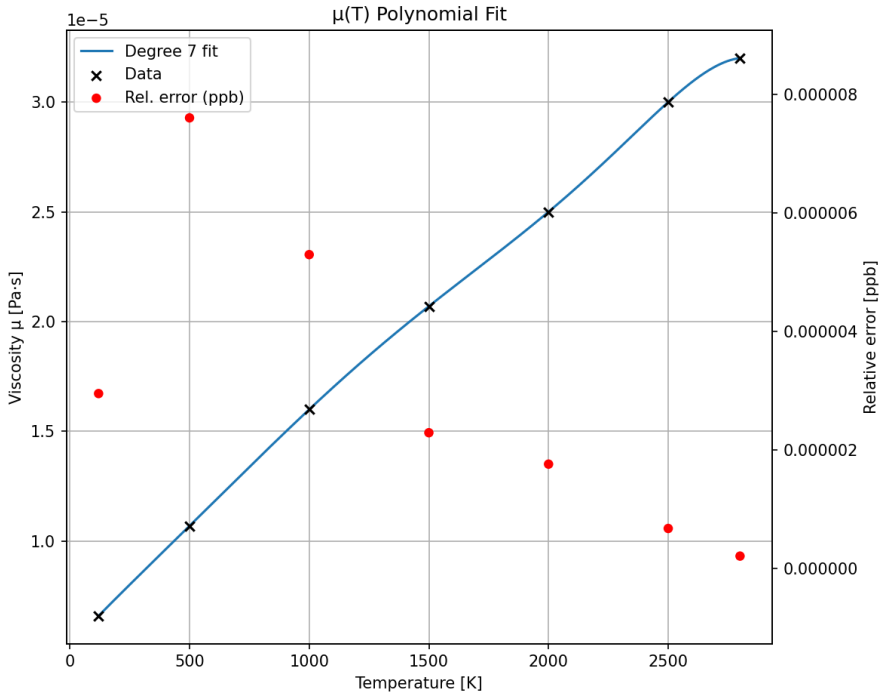
Table 5.4: thermoType-block for the physicalProperties dictionary of the coolant region

| **Keyword** | **Value** | **Information** |
|---|---|---|
| type | heRhoThermo | Solid region |
| mixture | pureMixture | Single super-critical gas phase flow |
| transport | polynomial | $\mu(T)$ in fig. 5.9 & $\kappa(T)$ fig. 5.10 |
| thermo | hPolynomial | Enthalpy using $c_p$ in eq. 5.4.3 |
| equationOfState | perfectGas | Equation of State, not calorically, see Appendix fig. A.2 |
| specie | specie | using molWeight: $2.016 \frac{g}{mol}$ |
| energy | sensibleEnthalpy | compressible gas meaning $h \neq e$ |

The origin of the data used is specified in Appendix A. Data points for figures are hand-drawn from the diagrams therein, with the exception of $c_p$ which is directly copied from fig. A.1 by Brad Appel, which contains the following 5th degree polynomial fit:

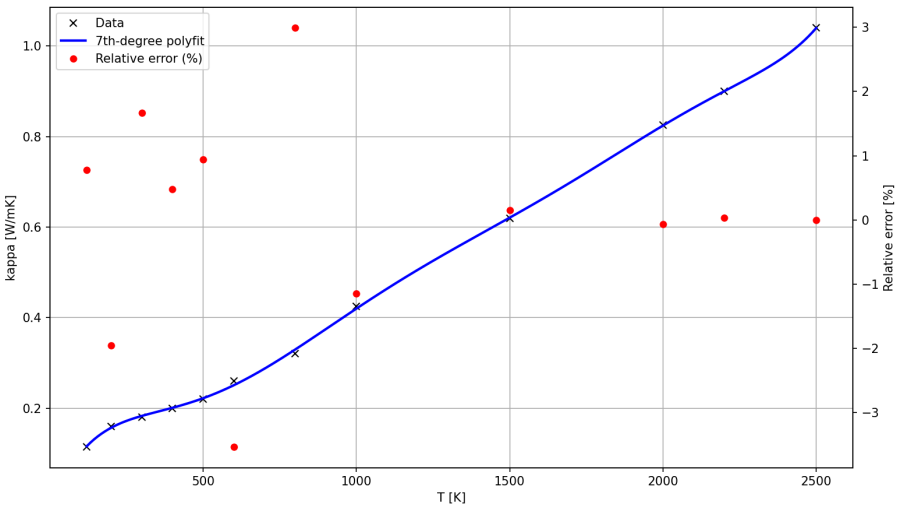
$$
\begin{aligned}
c_p(T) = \; & -5.4377 \cdot 10^{-13}\, T^5 \\
& +5.8506 \cdot 10^{-9}\, T^4 \\
& -2.1053 \cdot 10^{-5}\, T^3 \\
& +3.3433 \cdot 10^{-2}\, T^2 \\
& -2.1912 \cdot 10^{1}\, T \\
& +1.9304 \cdot 10^{4}
\end{aligned}
$$

The polynomial fits used for the propellant properties specified in table 5.4 are shown.




$$\begin{aligned}\mu(T) = &-1.12725482 \cdot 10^{-28}\, T^7 \\ &+5.46808833 \cdot 10^{-25}\, T^6 \\ &-8.51850148 \cdot 10^{-23}\, T^5 \\ &-3.10420209 \cdot 10^{-18}\, T^4 \\ &+5.23317492 \cdot 10^{-15}\, T^3 \\ &-3.42387097 \cdot 10^{-12}\, T^2 \\ &+1.17103350 \cdot 10^{-8}\, T \\ &+5.23566483 \cdot 10^{-6}\end{aligned}$$

Figure 5.9: Seventh-degree polynomial fit for hydrogen dynamic viscosity as a function of temperature at pressures between 4 & 5 MPa, including relative error compared to data points




$$\begin{aligned}\kappa(T) = &\ 1.50659402 \cdot 10^{-22}\, T^7 \\ &-1.42587311 \cdot 10^{-18}\, T^6 \\ &+5.43636993 \cdot 10^{-15}\, T^5 \\ &-1.06349833 \cdot 10^{-11}\, T^4 \\ &+1.11988120 \cdot 10^{-8}\, T^3 \\ &-6.00527931 \cdot 10^{-6}\, T^2 \\ &+1.72197823 \cdot 10^{-3}\, T \\ &-2.15399530 \cdot 10^{-2}\end{aligned}$$

Figure 5.10: Seventh-degree polynomial fit for hydrogen thermal conductivity as a function of temperature at pressures between 4 & 5 MPa, including relative error compared to data points

## 5.5 Fields: Initial and Boundary Conditions

Historically, and also throughout this thesis, the working fluid of a ntr} is referred to as the *propellant*. In the OpenFOAM implementation, due to the reactor-centric scope of the model and for brevity in extensive code development, the same medium is labeled *Coolant*. Consequently, code objects quoted verbatim that refer to the propellant may carry the *Coolant* designation.

Furthermore, in both the propellant and fuel regions, initial fields are linearly varied along the length of the channel using the *codeStream* functionality within OpenFOAM. This is done for different physical reasons in each region.

### 5.5.1 Support Region Temperature

Due to its comparatively short length, a simple uniform hot-end temperature is chosen as an initial condition for this region taken from an NRX-A5 measurement as shown and justified in appendix B.4. This results in an OpenFOAM-dictionary containing the entries below in table 5.5

Table 5.5: Boundary Conditions & Initial Conditions for Support Region

| Patch name [patch type] | B.C. Type | Initial Value [K] | Information |
|---|---|---|---|
| internalField | I.C.: uniform | 2379.444 | From appendix B.9 |
| Support_to_Coolant [wall] | B.C.: coupledTemperature | 2379.444 | Solid-fluid interface |
| Support_to_Fuel [patch] | B.C.: coupledTemperature | 2379.444 | Solid-solid interface |
| Support_to_Neighbour[patch] | B.C.: zeroGradient | 2379.444 | channel non-communication assumed |
| Support_to_Chamber [patch] | B.C.: zeroGradient | 2379.444 | Heat transfer assumed negligible due to low $\Delta T$ and modelling complexity |

### 5.5.2 Fuel Region Temperature and specific Thermal Power

#### 5.5.2.1 Temperature

In this region, initial axial fuel temperature is interpolated between the control room time (CRT)=14600s measurement of Station 1 temperature 25.4mm (1in) downstream from the inlet in the core center region (lowest value in the range shown in appendix tab. B.1) and the initial temperature of the support block. This is done to accelerate the generally stiff solid temperature transient and allow for the rapid development of physically realistic thermal conditions within the Propellant. Below in tab. 5.6 is a summary of initial field values inside the fuel region:

Table 5.6: Boundary Conditions & Initial Conditions for Fuel Region

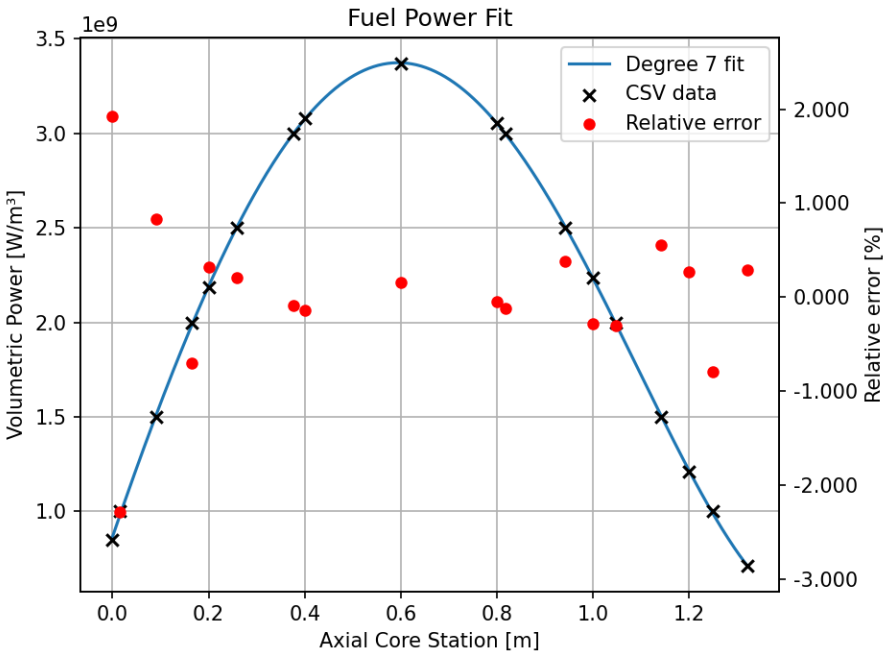
| Patch name [patch type] | B.C. Type | Initial Value [K] | Information |
|---|---|---|---|
| internalField | I.C.: codeStream | [277,...,2379.444] | linear in z∈[0,...,1.3208]m |
| Fuel_to_Coolant [wall] | B.C.: coupledTemperature | 1254 | global average overwritten in first time step |
| Fuel_to_Support [patch] | B.C.: coupledTemperature | 2379.444 | solid-solid interface, same as support block |
| Fuel_to_Neighbour[patch] | B.C.: zeroGradient | 1254 | global average overwritten in first time step |
| Fuel_to_Plenum [patch] | B.C.: zeroGradient | 277 | from tab. B.1 |

#### 5.5.2.2 Thermal Power Distribution

Wang and Kotlyar used an axial power distribution shown fitted and with a polynomial in fig. 5.12, which was deducted calculated based on data for the computer program ELM by J.T. Walton. [**29**, fig.6][**39**]. ELM uses the axial power distribution for a channel of average power within the core as is noted in the beginning of its computer code shown in fig. 5.11 below:

```
      PROGRAM ELM
CC
CC    THIS PROGRAM PERFORMS AEROTHERMODYNAMIC ANALYSIS ON H2 GAS COOLED
CC    NUCLEAR REACTOR CORE FUEL ELEMENTS.  THE ANALYSIS IS BASED ON
CC    CONSERVATION OF ENERGY EQUATIONS WITH EXTENSIVE USE OF CORRELATIONS
CC    FOR THE CONVECTIVE HEAT TRANSFER COEFFICIENT, FRICTION FACTOR, AND
CC    ELEMENT AXIAL POWER DISTRIBUTION.  THE ANALYSIS IS FOR AN AVERAGE
CC    COOLANT CHANNEL, SUCH AS AT A POINT IN THE CORE WHERE THE RADIAL
CC    POWER OVER AVERAGE POWER RATIO IS UNITY.
```

Figure 5.11: Description of axial power profile noted on the first lines of Walton's code - Source: [39, p. 14]




$$
\begin{aligned}
q'''(z) = \; & 4.26533771 \cdot 10^{8}\, z^{7} \\
& - 2.28114575 \cdot 10^{9}\, z^{6} \\
& + 5.71012082 \cdot 10^{9}\, z^{5} \\
& - 3.85483810 \cdot 10^{9}\, z^{4} \\
& - 2.65005044 \cdot 10^{9}\, z^{3} \\
& - 3.42210349 \cdot 10^{9}\, z^{2} \\
& + 7.43574730 \cdot 10^{9}\, z \\
& + 8.66332569 \cdot 10^{8}
\end{aligned}
$$

Figure 5.12: Seventh-degree polynomial representation of axial distribution of volumetric power based on points taken from Wang's & Kotlyar's graph

First, it is to be verified whether $q'''(z)$ is for an average fuel element inside the core.
Using $A_{equivalent}$ from eq. 5.1, the axial power distribution is integrated for the volume of a single fuel element with 19 channels:

$$
\begin{aligned}
P_{\text{AvgFuelElement}}[W] &= 19 A_{\text{equivalent}}[m^2] \int_{0m}^{1.3208m} q'''(z) \left[\tfrac{W}{m^3}\right] \, \mathrm{d}z[m] \\
&= 19 A_{\text{equivalent}} \cdot \int_{0}^{1.3208} \begin{pmatrix} 4.26533771 \cdot 10^{8} \\ -2.28114575 \cdot 10^{9} \\ 5.71012082 \cdot 10^{9} \\ -3.85483810 \cdot 10^{9} \\ -2.65005044 \cdot 10^{9} \\ -3.42210349 \cdot 10^{9} \\ 7.43574730 \cdot 10^{9} \\ 8.66332569 \cdot 10^{8} \end{pmatrix} \cdot \begin{pmatrix} z^7 \\ z^6 \\ z^5 \\ z^4 \\ z^3 \\ z^2 \\ z \\ 1 \end{pmatrix} \mathrm{d}z \\
&= 19 \cdot 11.6992 \cdot 10^{-6} m^2 \cdot 3.1478 \cdot 10^{9} \left[\tfrac{W}{m^2}\right] \\
&= 699'708 W
\end{aligned}
\tag{5.5}
$$

Then knowing from chapter 4.2 that each regular fuel cluster has 6 such fueled elements and using the map for the entire core in appendix fig. B.4 and the periphery in fig. B.5 :

- 1 sector-less regular cluster in the very core center, designated 000
- 6 sectors, each with:
  - 33 regular clusters in the intermediate zone
  - 65 fuel elements in the peripheral subsector J

This brings the number of total fuel elements to:

$$
n_{FuelElementsTotal} = 6 + 6 \cdot (33 \cdot 6 + 65) = 1584 \tag{5.6}
$$

Multiplying the result of eq. 5.6 with the result from eq. 5.5 estimates the total power of the reactor:

$$
n_{FuelElementsTotal} \cdot P_{\text{AvgFuelElement}} = 1584 \cdot 699'708 W = 1108 \cdot 10^{9} W = 1108 MW \tag{5.7}
$$

Compared to the average nominal power estimate from appendix B.1, this reverse-prediction of nominal reactor thermal power has a relative error of:

$$
100\% \cdot \frac{1140 MW - 1108 MW}{1140 MW} = 2.81\% \tag{5.8}
$$

The result, showing good agreement, indicates that the axial power distribution is consistent with an average fuel element rather than a central hot channel. Consequently, representing a hot channel in the core center region requires application of an additional scaling factor, the magnitude of which must be established by further analysis.

To scale the that average axial power distribution adequately for a channel in the thermocoupled cluster 1A1 placed adjacent to the center cluster (000) in the aforementioned core map, the data from the predictive estimates for radial power distribution are used with one crucial modification based on the following quote from the report in appendix B.3:

*"Figures 4-5 and 4-6 show a comparison of these NRX-A6 channel power factors with those calculated by the same method for the NRX-A3 design. This comparison indicates that the major portion of the loading scheme used in these NRX-A6 calculations will probably be very close to that selected for the final design. The channel power factors in the perimeter of the central core region which are above the central values will be lowered by a reduction in loading. Figures 4-7 and 4-8 show a comparison of the operating fissions per gram distributions derived in the same calculations."* - WANL-TME-1290, Oct. 1965

Given that low non-uniformity was crucial in preventing excessive local differences in thermal expansion as they could cause cracking of the fuel elements[**15**, p. 2-1], it is inferred from the above quote that the perimeter peak in nominal radial volumetric power distributions seen in fig. B.8a and B.8b was in the actual reactor brought down to approximately match the specific power in the reactor center.

Appendix D shows the code and data used to generate the subsequent plots and results. Its functionality is outlined in the subsequent analysis of this section.
It is for this reason that in the following estimation, the data point for the relative value of the peripheral peak in relative specific power for NRX-A6 is taken from the NRX-A3 estimate in the same original figures. That set of assumptions results in the following estimated relative radial power distributions for estimates made across core radius for two different core azimuths taken from and specified in fig. 5.13.

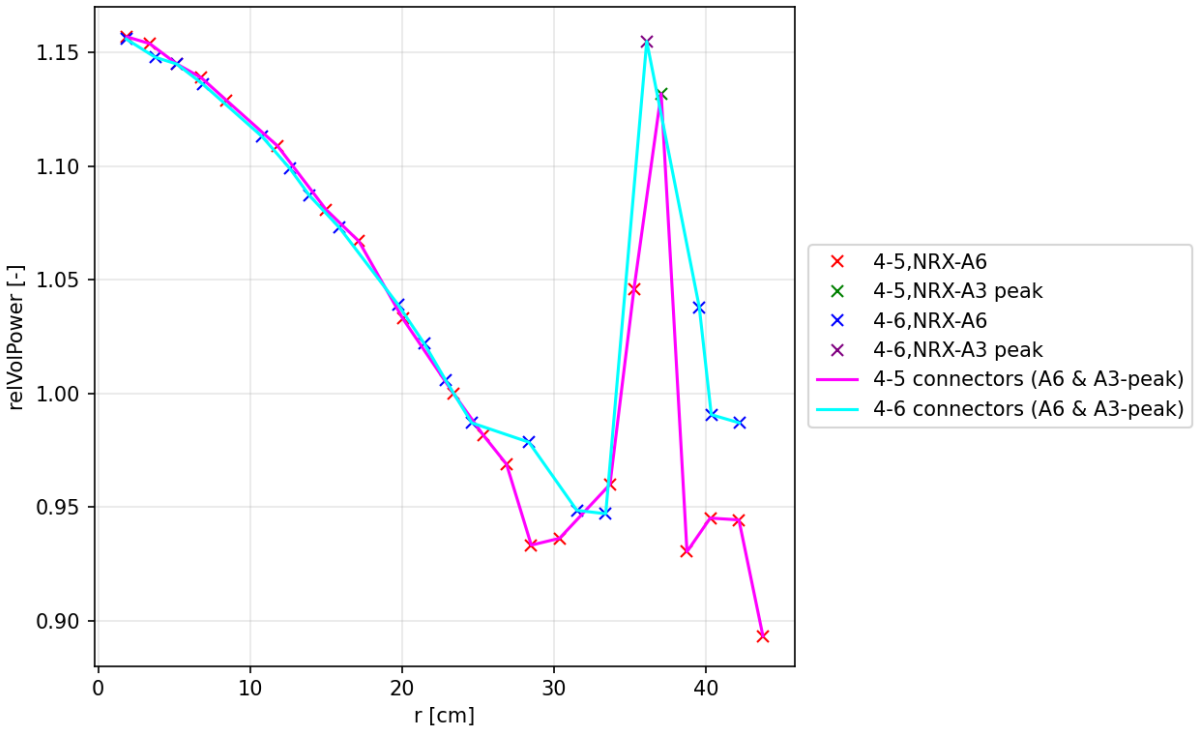



Figure 5.13: Relative radial power distributions predicted for two different azimuths within NRX-A6 supplemented with respective NRX-A3 values for peripheral peak estimation and linear connectors added

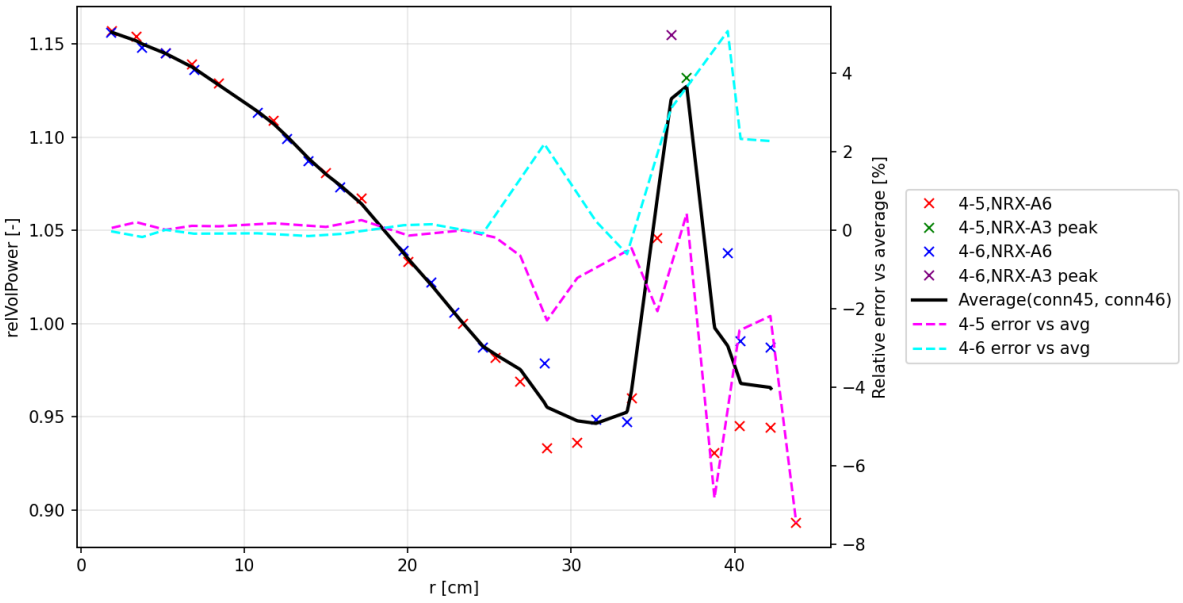



Figure 5.14: Black line shows arithmetic mean of both linear connector graphs; red, green, blue and purple show original data points; cyan and magenta dotted lines show relative error for the averaging process compared to original connector graphs in fig. 5.13

The black graph itself is then finally itself averaged from the core center to edge using the mean value integral:

$$P_{glob,avg} = \frac{1}{r_1 - r_0} \int_{r_0}^{r_1} P_{rel}(r)\,\mathrm{d}r = 1.04662717 \tag{5.9}$$

Then the value of the black graph at the desired radius is divided by the value determined by eq. 5.9, to find the approximate factor to scale the average power distribution from fig. 5.12 appropriately by a factor $F(r)$. In case of the hot channel at the core center:

$$F(r) = \frac{P_{rel}(r)}{P_{glob,avg}} \Rightarrow F(r=0) = \frac{1.1572}{1.04662717} = 1.105647 \tag{5.10}$$

This method makes the power distribution theoretically applicable to any channel in the core, though the inherent uncertainty and growing error towards the periphery must be accounted for, also with regards to the angular power scalloping shown in fig. B.7, when using this approach.

### 5.5.3 Propellant/Coolant Region

The thermo-hydraulic problem is subsequently posed in principle fluid boundary conditions as follows.
At the inlet, there is constant:

- density
- mass flow rate
- temperature
- turbulence parameters

At the outlet, there is constant:

- pressure

Quantities of interest include:

- Pressure drop and distribution between inlet and outlet
- Temperature distribution along and across the pipe in the propellant
- Axial and radial Velocity distribution of the propellant
- Reynolds numbers at inlet and outlet
- Axial and radial distribution of turbulence fields $\alpha_t$, $\nu_t$, $k$, and $\omega$

The purpose of the initial linear variation of the Propellant fields T, U, and $p/p_{rgh}$ in the axial direction is to minimize the initial buildup of pressure waves traversing between the inlet and outlet boundaries. These waves are caused in the initial time steps by the correction of the inevitable deviation of the initial fields from the initial equilibrium of the flow field. Mitigating this has been crucial for initial transient stability because a buildup of waves from ill-posed initial field conditions can cause pressure field values to go locally negative with subsequent failure of enthalpy calculation.

#### 5.5.3.1 Temperature T

Same method from is used to set initial propellant temperature to rise axially linearly between the fixed inlet plenum temperature for the inlet towards the same temperature as that of the support block:

Table 5.7: Temperature field boundary and initial conditions for the propellant

| Patch name [patch type] | B.C. Type | Initial Value [K] | Information |
|---|---|---|---|
| internalField | I.C.: codeStream | [128.89,...,2379.444] | linear in z∈[0,...,1.3589]m |
| Coolant_to_Fuel [wall] | B.C.: turbulentTemperatureCoupledBaffleMixed[1] | 1253.945 | overwritten in 1st time step |
| Coolant_to_Support [wall] | B.C.: turbulentTemperatureCoupledBaffleMixed[1] | 1253.945 | overwritten in 1st time step |
| Outlet_Boundary [patch] | B.C.: zeroGradient | 2379.444 | from appendix B.9 |
| Inlet_Boundary [patch] | B.C.: fixedValue | 128.89 | from appendix B.1 |

[1]compressible::turbulentTemperatureCoupledBaffleMixed in OpenFOAM v13

#### 5.5.3.2 Velocity U

The following initial and boundary conditions are specified in tab. 5.5 and subsequently explained:

Table 5.8: Velocity initial and boundary conditions for propellant velocity

| Patch name [patch type] | B.C. Type | Initial Value [m/s] | Information |
|---|---|---|---|
| internalField | codeStream | [21.6,...,500] | linear in z∈[0,...,1.3589]m |
| Coolant_to_Fuel [wall] | B.C.: noSlip | (0 0 21.6) | overwritten in 1st time step |
| Coolant_to_Support [wall] | B.C.: noSlip | (0 0 21.6) | overwritten in 1st time step |
| Outlet_Boundary [patch] | B.C.: flowRateInletVelocity | (0 0 500) | estimate overwritten in 1st time step |
| Inlet_Boundary [patch] | B.C.: pressureInletOutletVelocity | (0 0 21.6) | $\dot{m}$: eq. 5.11 |
| | | | $\rho$: eq. 5.12 |

Using detailed flowrate data from appendix B.4, based on the fact that each fuel cluster has 6 fuel elements with 19 channels each:

$$\dot{m}_{Channel} = \frac{\dot{m}_{Cluster} - \dot{m}_{TieRodAnnulus}}{6 \cdot 19} = \frac{[0.121426677 - 6.622 \cdot 10^{-3}]\frac{kg}{s}}{6 \cdot 19} = 1.007 \cdot 10^{-3}\frac{kg}{s} \tag{5.11}$$

Using the molecular weight of hydrogen to calculate the density at the inlet using the ideal gas in eq. 5.12.

$$pv = RT = \frac{\overline{R}}{M_{H_2}}T \Rightarrow \rho_{inlet} = \frac{1}{v} = \frac{M_{H_2} \cdot p}{\overline{R}T} = \frac{2.016 \cdot 10^{-3}\frac{kg}{mol} 4.889 \cdot 10^{6} Pa}{8.314\frac{J}{molK} 128.89 K} = 9.1977\frac{kg}{m^3} \tag{5.12}$$

The same is then estimated for the initial density es hot end outlet using chamber pressure and outflow temperature from from appendix B.10 and B.9 respectively:

$$\rho_{outlet} = \frac{M_{H_2} \cdot p}{\overline{R}T} = \frac{2.016 \cdot 10^{-3}\frac{kg}{mol} 4.033 \cdot 10^{6} Pa}{8.314\frac{J}{molK} 2379.4449 K} = 0.411\frac{kg}{m^3} \tag{5.13}$$

It is then noted that due to the conservation of mass for any given channel cross-section along z:

$$\dot{m} = \rho U A \Rightarrow U = \frac{\dot{m}}{\rho A} \text{ with: } A = \frac{\pi}{4} d_{channel}^2 = \frac{\pi}{4}(0.00254m)^2 = 5.0671 \cdot 10^{-6} m^2 \quad (5.14)$$

Which using the results from eqs. 5.12, 5.13, and 5.11 gives the initial estimates for the inlet and outlet velocity for the axial interpolation of flow velocity:

$$U_{inlet} = \frac{\dot{m}}{\rho_{inlet} A} = \frac{1.007 \cdot 10^{-3} \frac{kg}{s}}{9.1977 \frac{kg}{m^3} 5.0671 \cdot 10^{-6} m^2} = 21.61 \frac{m}{s} \quad (5.15)$$

$$U_{outlet} = \frac{\dot{m}}{\rho_{outlet} A} = \frac{1.007 \cdot 10^{-3} \frac{kg}{s}}{0.411 \frac{kg}{m^3} 5.0671 \cdot 10^{-6} m^2} = 483.5 \frac{m}{s} \quad (5.16)$$

Note that $U_{outlet}$ has been rounded up to $500 \frac{m}{s}$ in the actual implementation.

#### 5.5.3.3 Pressure $p$ and $p_{rgh}$

The relationship between both fields is shown in the following equation:

$$p_{rgh} = p - \rho \vec{g} \cdot \vec{h} \quad (5.17)$$

Where by default:

- $\vec{g} = (0, 0, -9.81) \left[\frac{m}{s^2}\right]$ standard gravity on Earth surface, where test took place
- $\vec{h}$ is the height measured from the top surface of the fluid

The primary field PIMPLE uses to solve all equations is $p_{rgh}$. p is then calculated in each time step with the following equation:

$$p = p_{rgh} + \rho \vec{g} \cdot \vec{h} \quad (5.18)$$

For this problem specifically, assuming maximum density and minimum applied pressure throughout, it can be found that maximally:

$$\frac{\rho_{max} |\vec{g} \cdot \vec{h}|}{p_{min}} = \frac{\rho_{max} |g_z \cdot h_{z,max}|}{p_{min}} = \frac{9.177 \frac{kg}{m^3} \cdot 9.81 \frac{m}{s^2} \cdot 1.3589m}{4.033 \cdot 10^6 Pa} = 3.033 \cdot 10^{-5} \quad (5.19)$$

Where:

- $\rho_{max}$: inlet density (global maximum) calculated in eq. 5.12
- $p_{min}$: chamber pressure at outlet (global minimum)
- $h_{z,max}$: total height of the channel

Because the ratio resulting from eq. 5.19 is exceedingly small below any engineering consideration, it can be safely assumed that $\vec{g} = 0$, making $p = p_{rgh}$, eliminating any potential cause of failure in the initial transient resulting from the relationship between the two fields and their initial conditions.
Inlet and outlet pressures are taken from measurement No. 1 in appendix fig. B.10 because as visible in fig. B.11, it was axially closer to the reactor outlet within the nozzle chamber.
Here follow initial and boundary conditions for both fields where initial values are identical while boundary conditions differ due to the aforementioned relationship between the fields:

Table 5.9: Initial and boundary conditions for pressure p

| **Patch name [patch type]** | **B.C. Type** | **Initial Value [Pa]** | **Information** |
|---|---|---|---|
| internalField | I.C.: codeStream | [4.889·10$^6$,...,4.033·10$^6$] | linear in z∈[0,...,1.3589]m |
| Coolant_to_Fuel [wall] | B.C.: calculated | 4.033·10$^6$ | overwritten in 1st time step |
| Coolant_to_Support [wall] | B.C.: calculated | 4.033·10$^6$ | overwritten in 1st time step |
| Outlet_Boundary [patch] | B.C.: calculated | 4.033·10$^6$ | from appendixB.10 |
| Inlet_Boundary [patch] | B.C.: calculated | 4.889·10$^6$ | from appendix B.10 |

Table 5.10: Initial and boundary conditions for pressure $p_{rgh}$

| **Patch name [patch type]** | **B.C. Type** | **Initial Value [Pa]** | **Information** |
|---|---|---|---|
| internalField | I.C.: codeStream | [4.889·10$^6$,...,4.033·10$^6$] | linear in z∈[0,...,1.3589]m |
| Coolant_to_Fuel [wall] | B.C.: zeroGradient | 4.033·10$^6$ | overwritten in 1st time step |
| Coolant_to_Support [wall] | B.C.: zeroGradient | 4.033·10$^6$ | overwritten in 1st time step |
| Outlet_Boundary [patch] | B.C.: fixedMean | 4.033·10$^6$ | from appendixB.10 |
| Inlet_Boundary [patch] | B.C.: fixedFluxPressure | 4.889·10$^6$ | from appendix B.10 |

#### 5.5.3.4 Notes on initial and Boundary Conditions of RANS-related Fields

Inlet values for all of the subsequent turbulence fields with the exception of $\alpha_t$, which derives from other fields as shown in eq. 3.3.1, are assumed to be fixed to account approximately for the flow around the rim of the fuel inlet imparting some degree of turbulence at the beginning of the channel.
How the initial internal field estimate for the field $\alpha_t$ in the following section was arrived at, was unfortunately not recoverably documented. However, because the value is overwritten in the first PIMPLE iteration, this is of no consequence to the final result as long as the initial estimate is not excessive to cause thermal instability.

#### 5.5.3.5 Turbulent thermal Diffusivity $\alpha_t$

The turbulent Prandtl-number is in industry practice empirically by default:

$$Pr_t = 0.85 \pm 0.05 \tag{5.20}$$

Because $\alpha$ is a quantity calculated quantity, all initial values are overwritten, minimizing the relevance of the initial condition for the final result.

Table 5.11: Initial and boundary conditions for turbulent thermal diffusivity $\alpha_t$

| Patch name [patch type] | B.C. Type | Initial Value [$m^2/s$] | Information |
|---|---|---|---|
| internalField | I.C.: uniform | $4.167 \cdot 10^{-5}$ | overwritten in 1st time step |
| Inlet_Boundary [patch] | I.C.: calculated | $4.167 \cdot 10^{-5}$ | overwritten in 1st time step |
| Outlet_Boundary [patch] | I.C.: calculated | $4.167 \cdot 10^{-5}$ | overwritten in 1st time step |
| Coolant_to_Fuel [wall] | B.C.: compressible::alphatWallFunction | $4.167 \cdot 10^{-5}$ | turbulent Prandtl number $\mathrm{Pr}_t = 0.85$ |
| Coolant_to_Support [wall] | B.C.: compressible::alphatWallFunction | $4.167 \cdot 10^{-5}$ | turbulent Prandtl number $\mathrm{Pr}_t = 0.85$ |

#### 5.5.3.6 Turbulent kinetic Energy k

The the value for the fixed value boundary condition at the inlet as well as the equivalent estimate for the initial values of the internal field were calculated using the following estimation method[40]:

$$k_{est} = \frac{3}{2} U^2 = \frac{3}{2} \left(0.05\, U_{\text{ref}}\right)^2 = \frac{3}{2} \left(0.05 \cdot 21.6\,\text{m/s}\right)^2 = 1.75\ \text{m}^2/\text{s}^2 \tag{5.21}$$

It is assumed that

Table 5.12: Initial and boundary conditions for turbulent kinetic energy $k$

| Patch name [patch type] | B.C. Type | Initial Value [$m^2/s^2$] | Information |
|---|---|---|---|
| internalField | I.C.: uniform | 1.75 | overwritten in 1st time step |
| Inlet_Boundary [patch] | B.C.: fixedValue | 1.75 | see eq. 5.21 |
| Outlet_Boundary [patch] | B.C.: zeroGradient | 1.75 | overwritten in 1st time step |
| Coolant_to_Fuel [wall] | B.C.: kqRWallFunction | 1.75 | overwritten in 1st time step |
| Coolant_to_Support [wall] | B.C.: kqRWallFunction | 1.75 | overwritten in 1st time step |

#### 5.5.3.7 Turbulent kinematic Viscosity $\nu_t$

It is common practice to estimate turbulent viscosity at approximately $1 - 10\times$ molecular viscosity. This is done here using eq. 3.2 with the averages of the initial conditions for temperature and pressure:

$$\rho(p_{avg},\, T_{avg}) = \frac{p_{avg} M}{\overline{R} T} = \frac{4.461 \cdot 10^6 Pa \cdot 2.016 \cdot 10^{-3} \frac{kg}{mol}}{8.314 \frac{J}{mol \cdot K} \cdot 1314 K} = 0.8232 \frac{kg}{m^3} \tag{5.22}$$

Then using average the average over the range interpolated in fig. 5.9, an estimate for average kinematic viscosity is arrived at as follows:

$$\nu_{avg} = \frac{\mu_{avg}}{\rho(p_{avg},\, T_{avg})} = \frac{1.75 \cdot 10^{-5} Pa \cdot s}{0.8232 \frac{kg}{m^3}} = 2.126 \cdot 10^{-5} \frac{m^2}{s} \tag{5.23}$$

Based on which it can easily seen that for the estimate used in tab. 5.14:

$$\{\nu_t = 2.5 \cdot 10^{-5}\} \gtrsim \{\mu_{avg} = 2.126 \cdot 10^{-5} \frac{m^2}{s}\} \tag{5.24}$$

Table 5.13: Initial and boundary conditions for turbulent kinematic viscosity $\nu_t$

| **Patch name [patch type]** | **B.C. Type** | **Initial Value** [$m^2 \cdot s$] | **Information** |
|---|---|---|---|
| internalField | I.C.: uniform | $2.5 \cdot 10^{-5}$ | overwritten in 1st time step |
| Inlet_Boundary [patch] | B.C.: fixedValue | $2.5 \cdot 10^{-5}$ | equal to internalField |
| Outlet_Boundary [patch] | B.C.: zeroGradient | $2.5 \cdot 10^{-5}$ | overwritten in 1st time step |
| Coolant_to_Fuel [wall] | B.C.: nutUSpaldingWallFunction | $2.5 \cdot 10^{-5}$ | overwritten in 1st time step |
| Coolant_to_Support [wall] | B.C.: nutUSpaldingWallFunction | $2.5 \cdot 10^{-5}$ | overwritten in 1st time step |

#### 5.5.3.8 Specific Dissipation Rate $\omega$

Specific dissipation rate is estimated using the following method. Inserting eq. 3.41 into eq. 3.40, it can be found that with characteristic length being hydraulic diameter $D_h$:

$$\omega \approx \frac{k^{3/2}}{k \cdot \beta^* \cdot D_h} = \frac{\sqrt{k}}{\beta^* \cdot D_h} \tag{5.25}$$

Then using $k_{est}$ from eq. 5.21 and an inlet factor $I = 0.07$:

$$\omega_{est} \approx \frac{\sqrt{k_{est}}}{\beta\, I\, D} = \frac{\sqrt{1.75 \frac{m^2}{s^2}}}{0.09 \cdot 0.07 \cdot 2.54 \cdot 10^{-3} m} = 8.27 \cdot 10^4 \text{ s}^{-1} \tag{5.26}$$

Table 5.14: Initial and boundary conditions for turbulent kinematic viscosity $\nu_t$

| **Patch name [patch type]** | **B.C. Type** | **Initial Value** [$1/s$] | **Information** |
|---|---|---|---|
| internalField | I.C.: uniform | 82669 | overwritten in 1st time step |
| Inlet_Boundary [patch] | B.C.: fixedValue | 82669 | see eq. 5.26 |
| Outlet_Boundary [patch] | B.C.: zeroGradient | 82669 | overwritten in 1st time step |
| Coolant_to_Fuel [wall] | B.C.: omegaWallFunction | 82669 | overwritten in 1st time step |
| Coolant_to_Support [wall] | B.C.: omegaWallFunction | 82669 | overwritten in 1st time step |

## 5.6 Solver Structure

See fig. 3.1 and description reference for overview of how each of the finite volume solution settings are used.

### 5.6.1 Global Finite Volume Solution Settings

Table 5.15: Active fvSolution PIMPLE Configuration (OpenFOAM v13)

| Section | Parameter | Value | Description |
|---|---|---|---|
| PIMPLE | momentumPredictor | yes | Enables momentum predictor |
| | correctPhi | yes | Enforces flux consistency for mass conservation |
| | nOuterCorrectors | 1 | Repeats Pimple Loop for a given time step |
| | rhoMin | 0.2 | minimum fluid density clamp |
| | rhoMax | 666 | maximum fluid density clamp |
| | transient | no | Steady-state formulation (pseudo-transient) |
| | consistent | yes | Uses SIMPLE-style pressure correction |
| ResidualControl | $p_{rgh}$ | $1 \cdot 10^{-6}$ | Pressure residual convergence threshold |
| | U | $1 \cdot 10^{-6}$ | Velocity residual convergence threshold |
| | h | $1 \cdot 10^{-6}$ | Enthalpy residual convergence threshold |
| | omega | $1 \cdot 10^{-6}$ | Turbulence variable residual threshold |

### 5.6.2 Solid Regions

For both the fuel and support regions, the discretization schemes and matrix solver arrays are chosen identically.

#### 5.6.2.1 Finite Volume Solution Settings

Table 5.16: fvSolution configuration for fuel and support Regions

| Section | Parameter | Value |
|---|---|---|
| solvers: e | solver | smoothSolver |
| | smoother | symGaussSeidel |
| | tolerance | $1 \cdot 10^{-6}$ |
| | relTol | 0 |
| | minIter | 1 |
| | maxIter | 100 |
| solvers: eFinal | solver | smoothSolver |
| (is skipped, see section PIMPLE) | smoother | symGaussSeidel |
| | tolerance | $1 \cdot 10^{-6}$ |
| | relTol | 0 |
| | minIter | 1 |
| | maxIter | 100 |
| PIMPLE | nNonOrthogonalCorrectors | 0 |
| relaxationFactors: equations | e | 0.5 |

#### 5.6.2.2 Finite Volume Discretization Schemes

Table 5.17: fvSchemes configuration for fuel and support regions

| Section | Parameter | Value |
|---|---|---|
| ddtSchemes | default | backward |
| gradSchemes | default | Gauss linear uncorrected |
| divSchemes | default | none |
| laplacianSchemes | default | Gauss linear uncorrected |
| interpolationSchemes | default | linear |
| snGradSchemes | default | uncorrected |

### 5.6.3 Propellant Region

#### 5.6.3.1 Finite Volume Solution Settings

The serial conjunction of the PBiCGStab solver for the initial solution and the GAMG solver in the non-orthogonal corrector loop was chosen to parry the initial instability of the pressure equation. The PBiCGStab solver is comparatively stable but offers sub-optimal convergence for this problem. The GAMG solver allows for superior convergence of the $p_{rgh}$ residuals but can cause time-step continuity errors to diverge if the initial residuals are too large. Hence, an optimal solution is achieved in which PBiCGStab resolves the large initial residuals and final convergence of the pressure equation is attained using GAMG. For this reason, a pseudo-transient approach was used instead of a steady-state simulation, since in OpenFOAM v13 non-orthogonal corrector loops are unavailable when using PIMPLE with a steady-state time-discretization scheme.

Table 5.18: fvSolution configuration for propellant region

| **Section** | **Parameter** | **Value** |
|---|---|---|
| solvers: $p_{rgh}$ | solver | PBiCGStab |
| | preconditioner | DIC |
| | tolerance | $1 \cdot 10^{-3}$ |
| | relTol | 0.0 |
| | minIter | 1 |
| | maxIter | 1000 |
| solvers: $p_{rgh}Final$ | solver | GAMG |
| | tolerance | $5 \cdot 10^{-5}$ |
| | relTol | $1 \cdot 10^{-2}$ |
| | smoother | GaussSeidel |
| | nPreSweeps | 1 |
| | nPostSweeps | 1 |
| | nFinestSweeps | 1 |
| | cacheAgglomeration | true |
| | nCellsInCoarsestLevel | 20 |
| | agglomerator | faceAreaPair |
| | mergeLevels | 1 |
| | minIter | 1 |
| | maxIter | 100 |
| solvers: $\rho$ & $\rho$Final | solver | diagonal |
| solvers{ | solver | PBiCGStab |
| h & hFinal, | preconditioner | DILU |
| U & UFinal | tolerance | $1 \cdot 10^{-6}$ |
| } | relTol | 0.0 |
| | minIter | 1 |
| | maxIter | 100 |
| solvers{ | solver | PBiCG |
| k & kFinal, | preconditioner | diagonal |
| $\omega$ & $\omega$Final | tolerance | $1 \cdot 10^{-6}$ |
| } | relTol | 0.0 |
| | minIter | 1 |
| | maxIter | 100 |
| PIMPLE | nCorrectors | 1 |
| | nNonOrthogonalCorrectors | 1 |
| relaxationFactors: equations | $\vec{U}, h, k,$ & $\omega$ | 0.5 |

#### 5.6.3.2 Finite Volume Discretization Schemes

Table 5.19: fvSchemes configuration for propellant region

| Section | Parameter | Value |
|---|---|---|
| wallDist | method | meshWave |
| ddtSchemes | default | backward |
| gradSchemes | default | Gauss linear |
| divSchemes | default | none |
| | $\text{div}(\phi, \vec{U})$ | Gauss upwind |
| | $\text{div}(\phi, h)$ | Gauss upwind |
| | $\text{div}(\phi, k)$ | Gauss upwind |
| | $\text{div}(\phi, K)$ | Gauss upwind |
| | $\text{div}(\phi, \omega)$ | Gauss upwind |
| | div(((rho*nuEff)*dev2(T(grad(U))))) | Gauss linear |
| laplacianSchemes | default | Gauss linear corrected |
| interpolationSchemes | default | linear |
| snGradSchemes | default | corrected |

## 5.7 Termination Criterion

Similarly to eq. 5.5 but with the axial volumetric power integral scaled using the fuel cross-sectional area for the equivalent channel from eq. 5.1 and the result for $F$ from eq. 5.10 the total thermal power of the simulated system is calculated as follows:

$$\begin{aligned}
\dot{Q}_{\text{equivalentChannel}}[W] &= A_{\text{equivalent}}[m^2] \cdot F \cdot \int_{0m}^{1.3208m} q'''(z) \left[\tfrac{W}{m^3}\right] \, \mathrm{d}z[m] \\
&= A_{\text{equivalent}} \cdot F \cdot \int_{0}^{1.3208} \begin{pmatrix} 4.26533771 \cdot 10^{8} \\ -2.28114575 \cdot 10^{9} \\ 5.71012082 \cdot 10^{9} \\ -3.85483810 \cdot 10^{9} \\ -2.65005044 \cdot 10^{9} \\ -3.42210349 \cdot 10^{9} \\ 7.43574730 \cdot 10^{9} \\ 8.66332569 \cdot 10^{8} \end{pmatrix} \cdot \begin{pmatrix} z^7 \\ z^6 \\ z^5 \\ z^4 \\ z^3 \\ z^2 \\ z \\ 1 \end{pmatrix} \mathrm{d}z \\
&= \cdot 11.6992 \cdot 10^{-6} m^2 \cdot 1.105647 \cdot 3.1478 \cdot 10^{9} \left[\tfrac{W}{m^2}\right] \\
&= 40'717W
\end{aligned} \tag{5.27}$$

The criterion is then formulated using $\dot{Q}_{\text{equivalentChannel}}$ from the above equation with all its other variables logged in regular intervals of simulated time throughout the PIMPLE run and $\dot{m}_{inlet} =$

$\dot{m}$ at all times due to inlet boundary condition for U in chapter 5.5.3.2:

$$100\% \cdot \frac{|\dot{m}_{outlet} - \dot{m}_{inlet}|}{\dot{m}_{inlet}} \leq 0.1\% \tag{5.28}$$

$$100\% \cdot \frac{\left|\dot{m}_{inlet} h_{inlet} + \dot{Q}_{equivalentChannel} - \dot{m}_{outlet} h_{outlet}\right|}{\dot{Q}_{equivalentChannel}} \leq 3.45\% \tag{5.29}$$

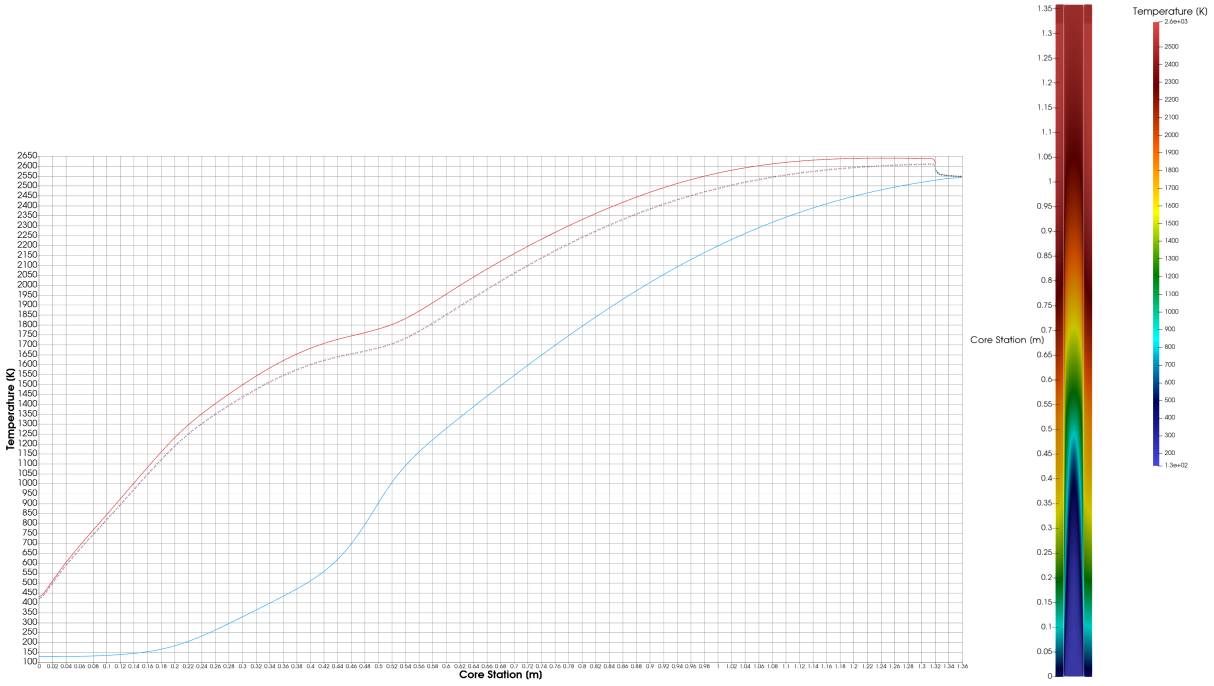
The threshold of $3.45\%$ for eq. 5.29 was arrived at empirically as marginal simulated time for a $0.001\%$ decrease in error increased to several simulated seconds. Shown in tab. 5.20 are quantities logged for the evaluation of simulation progress and additional fields written for the extraction of validation results:

Table 5.20: Actively logged or written quantities

| **field** | **written \|\| logged** | **writeControl** | **writeInterval** |
|---|---|---|---|
| $\phi_{\text{in}}$ | logged | timeStep | 20 |
| $\phi_{\text{out}}$ | logged | timeStep | 20 |
| $h_{\text{in,bulk}}$ | logged | timeStep | 20 |
| $h_{\text{out,bulk}}$ | logged | timeStep | 20 |
| $p_{\text{rgh,min}}$ | logged | timeStep | 20 |
| $p_{\text{rgh,max}}$ | logged | timeStep | 20 |
| $T_{\text{min,Propellant}}$ | logged | timeStep | 20 |
| $T_{\text{max,Propellant}}$ | logged | timeStep | 20 |
| $T_{\text{min,Fuel}}$ | logged | timeStep | 20 |
| $T_{\text{max,Fuel}}$ | logged | timeStep | 20 |
| $T_{\text{min,Support}}$ | logged | timeStep | 20 |
| $T_{\text{max,Support}}$ | logged | timeStep | 20 |
| $\rho_{\text{min}}$ | logged | timeStep | 20 |
| $\rho_{\text{max}}$ | logged | timeStep | 20 |
| $y^+$ | written | runTime | $5 \cdot 10^{-2}[s]$ |
| $\tau_w$ | written | runTime | $5 \cdot 10^{-2}[s]$ |

# Chapter 6
# Results and Discussion

## 6.1 Temperature




(a) Axial temperature distribution at region interfaces - Legend:
blue solid: propellant bulk center ($r = 0\,\text{mm}$);
blue dashed: propellant–solid interface ($r = 1.27\,\text{mm}$);
red solid: fuel outer boundary ($r = 0.23\,\text{mm}$);
red dotted: fuel–propellant interface ($r = 1.27\,\text{mm}$);
black solid (semi-transparent): support outer boundary ($r = 0.23\,\text{mm}$);
black dotted: support–propellant interface ($r = 1.27\,\text{mm}$).

(b) Temperature field (aspect ratio scaled down $\times 16$ & fluid-solid boundary marked white for visibility)

Figure 6.1: Axial temperature profiles and corresponding temperature field color map through centerline cut.
The diametral

The region interface graphs in fig. 6.1a indicate that temperature continuity at the fluid-solid boundary is satisfied throughout the axial length of the channel, which is consistent with physically correct CHT. It can further be seen that the radial temperature difference in the solid regions is greatest in the axial center of the fuel around the channel, which is reflected in detail in the radial profile in fig. 6.3c. This figure also shows that the mesh does satisfy the zero-gradient boundary condition for the maximum radial temperature gradient present.

The radial temperature range in the support block section is very small as indicated by fig. 6.2 also visible from the coincidence of its inner and outer boundary temperatures in fig. 6.1a.
Propellant bulk temperature lags far below fuel temperature in the first 1/3 of the axial length continuously increasing in axial direction. A distinct increase in axial gradient appears for the bulk propellant temperature at core station $z = 0.48 \pm 0.08$ m at the center of which an inflection point is reached. This increase corresponds with a drop in the axial temperature gradient of the fuel, indicating an increase in heat transferred to the bulk of the propellant. The color map on fig. 6.1b shows that isotherms are of conical shape near the inlet and become increasingly blunt downstream. This shows the development of a thermal boundary layer that appears to reach full development at approximately the axial center of the channel.
There is a sharp inflection point visible on fig. 6.1a at the interface between the fuel and support region that appears as a small discontinuity of 25 K hinting at a large transfer of heat from the fuel to the support region which makes the boundary between the fuel and support faintly visible at the top of fig. 6.1b. The sharp temperature difference between these regions reflects both the fact that heat is generated only in the fuel region and some of it is transferred through the support block into the propellant and the fact that in this temperature range the thermal conductivity $\kappa$ is greater in the support material than in the fuel material. At the hot end, all temperatures converge on the propellant exit temperature.

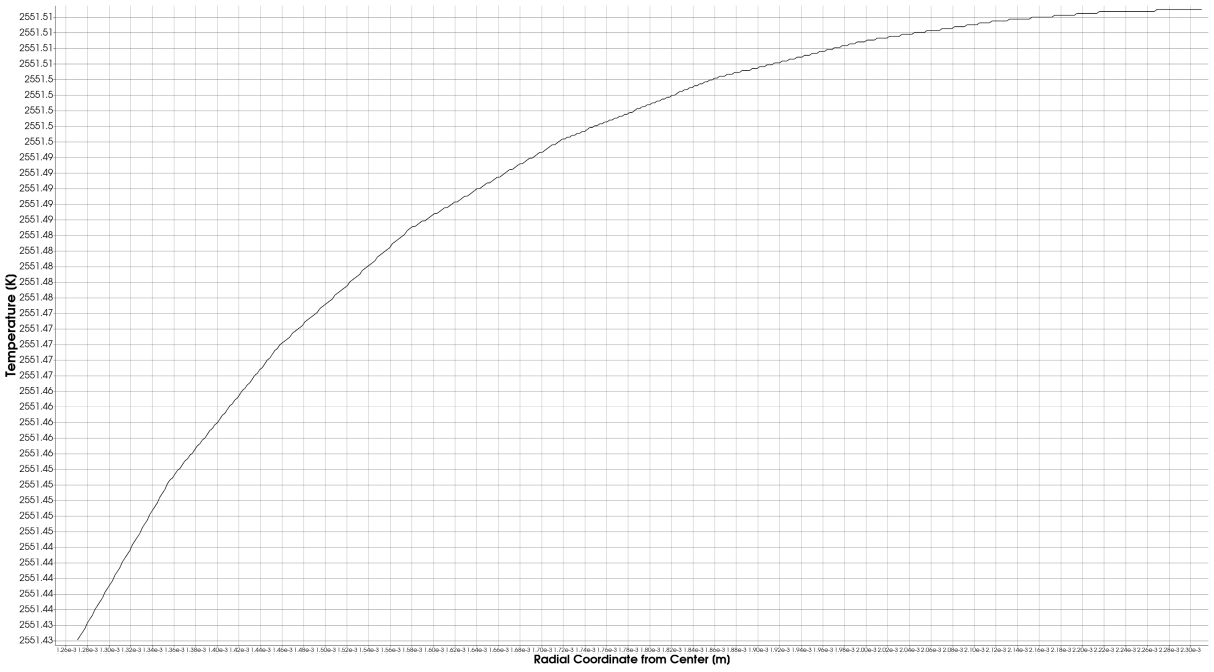
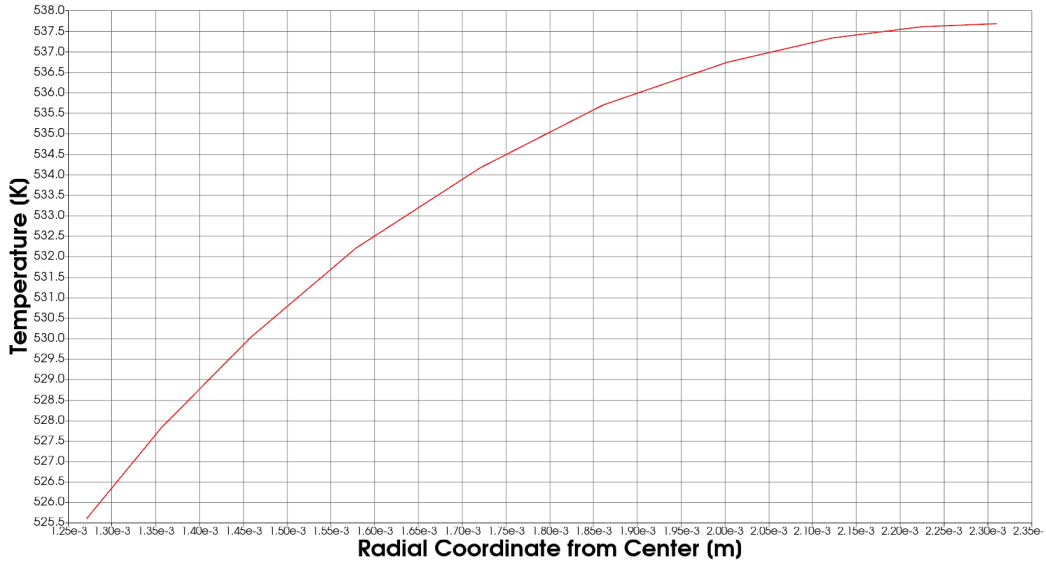
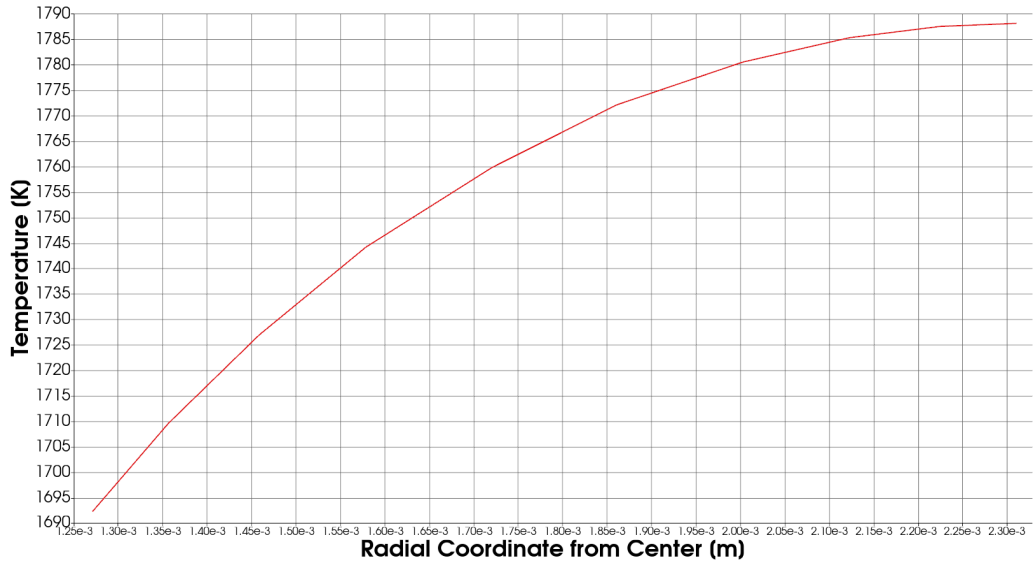


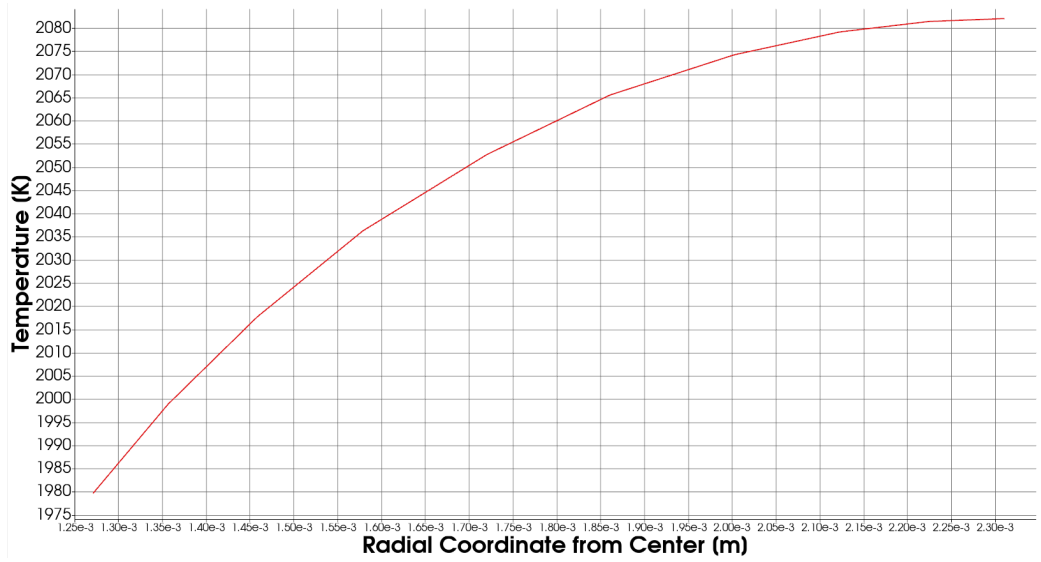

Figure 6.2: Radial temperature distribution in the axial center of the support region (z=134cm)




(a) at core station $z = 2.54$ cm




(b) at core station $z = 50.8$ cm




(c) at core station $z = 66$ cm

Figure 6.3: Fuel temperature distributions shown radially from channel centerline across various axial positions along the channel

| Model Probe Region | Axial Position | OpenFOAMv13 [K] | Experiment [K] | Thermocouple Location | Appendix Source |
|---|---|---|---|---|---|
| Fuel_to_Neighbour | 2.54cm (1in ) | 538.7 | $324 \pm 46$ | Fuel Element | tab. B.1 |
| Fuel_to_Neighbour | 50.8cm (20in) | 1788 | $1986.1 \pm 5.6$ | Unfueled El. | tab. B.1 |
| Fuel_to_Neighbour | 66.05cm (26in) | 2083 | $2172 \pm 11$ | Unfueled El. | tab. B.2 |
| Outlet_Boundary | 135.89cm | $2544 \pm 2.6$ | $2422 \pm 97$ | Cluster Exit Gas | tab. B.1 |

Table 6.1: Comparison of regions and positions between OpenFOAM simulation and experimental data.

| Source of Value | Region | Axial Position | Temperature [K] |
|---|---|---|---|
| Calculated Curve | Fuel Element | 50.8cm (20in) | 1720 |
| OpenFoam v13 | Fuel (Station 20) | 50.8cm (20in) | 1788 |
| Calculated Curve | Fuel Element | 66.05cm (26in) | 2027.8 |
| OpenFoam v13 | Fuel (Station 26) | 66.05cm (26in) | 2083 |
| Calculated Curve | Fuel axial Peak | 116.84cm (46in) | 2544.4 |
| Calculated Curve | Unfueled El. ax. Peak | 111.76cm (44in) | 2583.3 |
| OpenFoam v13 | Fuel Region ax. Peak | 124.14cm | 2640.4 |

Table 6.2: Comparison with Temperature calculated curves in fig. B.12 for measured core stations and peak values

### 6.1.1 Temperature Validation Study

Tab. 6.1 and 6.2 indicate that the modeled temperature values at axial measurement locations fall between the values measured in the adjacent unfueled element and the corresponding values estimated for the fuel elements during and following the reactor test respectively.
Compared with values from the coeval calculation for the fueled and unfueled elements, the present model predicts a peak temperature that is higher by $76.55 \pm 19.45$ K (rel. error: $3.0 \pm 0.8$ %) and shifted axially downstream by $9.7 \pm 2.5$ cm. Tab. 6.1 indicates that the model overestimates the temperature at axial position $z = 2.54$ cm by $214.7 \pm 46$ K (rel. error: $66 \pm 8.2$ %), which provides strong evidence that the data used for the interpolation of $\kappa(T)$ are of insufficient accuracy for the lower half of the operational temperature range.
Further, tab. 6.1 shows a small overprediction of cluster exit gas temperature, measuring maximally 27.6 K above the range of measurements, which supports sufficient thermal validity of the propellant model for further analysis.

## 6.2 Pressure and Density

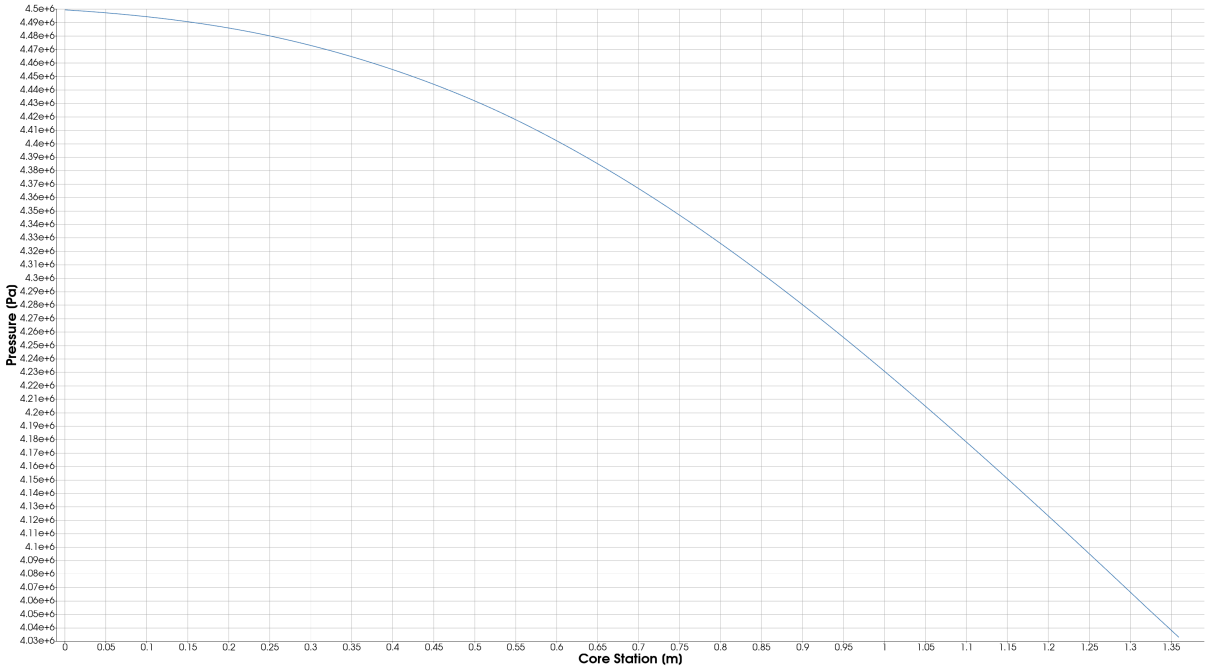
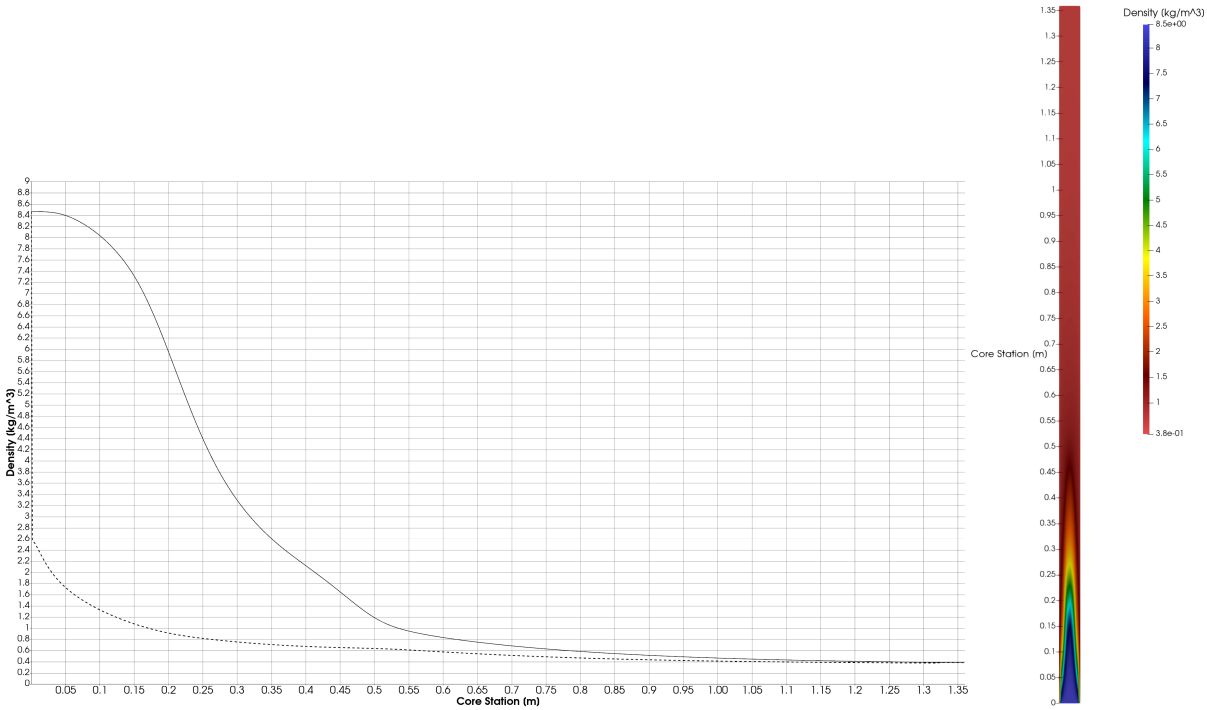



Figure 6.4: Axial pressure distribution in the channel - No color map is presented for pressure because the field is radially isobaric.




(a) Axial propellant density distribution at region interfaces - Legend:
black solid: propellant bulk center ($r = 0\,\text{mm}$);
black dashed: propellant–solid interface ($r = 1.27\,\text{mm}$);

(b) Density color map (aspect ratio scaled down $\times 16$ for visibility)

Figure 6.5: Axial graph and color map of diametral cut

Fig. 6.5a viewed in conjunction with fig. 6.4 indicates a sudden isobaric expansion at the fluid-solid interface in the immediate vicinity of the inlet, which propagates towards the bulk of the fluid as the flow propagates through the channel. The density both in the bulk and the fluid-solid interface converge rapidly on the outlet value, such that the flow becomes quasi-isochoric in the downstream half of the channel. The overall range of densities shows good agreement with the predictions made for the inlet and outlet in chapter 5.5.3.2.

### 6.2.1 Pressure Transducer Validation Study

Table 6.3: Model pressure values at inlet and outlet compared with reactor measurements from fig. B.10

| Position | OpenFOAM v13 | Experiment |
|---|---|---|
| Inlet Pressure | 4.499MPa | 4.889MPa |
| Outlet Pressure (fixed) | 4.033MPa | 4.033MPa |
| Pressure Drop | 0.466MPa | 0.856MPa |

Based on the values in 6.3, the model underestimates inlet pressure and pressure drop by 0.39MPa, which for the absolute inlet pressure results in a relative error of 7.9 % and for the pressure drop a rel. error of 45.6 %. This underestimation can have the following sources:

- Pressure losses occurring at the outlet due to its geometry (discontinuous increase of hydraulic diameter) affecting the flow upstream of the pressure measurement requiring adaptation of the outlet pressure boundary condition
- inherent error of the turbulence model under use The density range matches predictions made for the initial values with of the chosen meshing and field parameters
- error in the equation of state not being corrected for the compressibility factor of super-critical hydrogen (Z=1.044) that can be interpolated as a convergence value in fig. A.2 for the temperature range in this problem.

## 6.3 Velocity

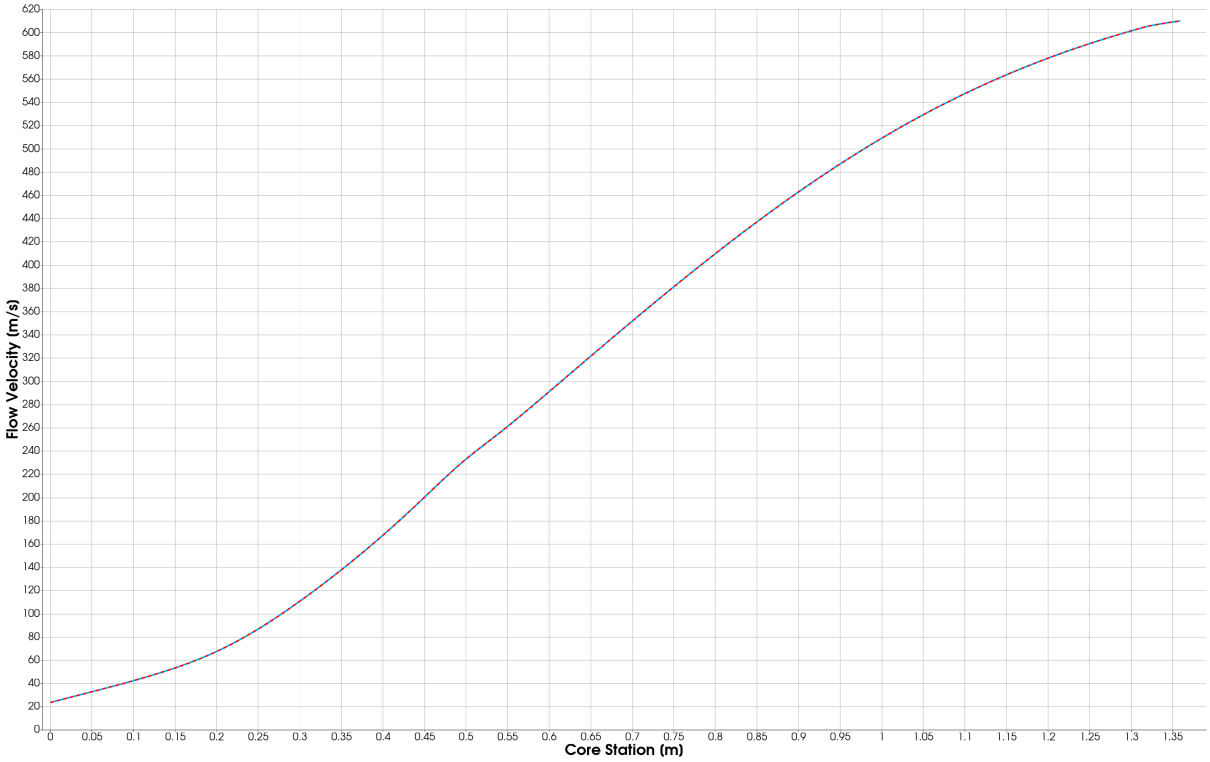



Figure 6.6: Legend:
blue solid: $|\vec{U}|$ in propellant bulk center ($r = 0\,\text{mm}$);
red dashed: $U_z$ propellant–solid interface ($r = 0\,\text{mm}$);

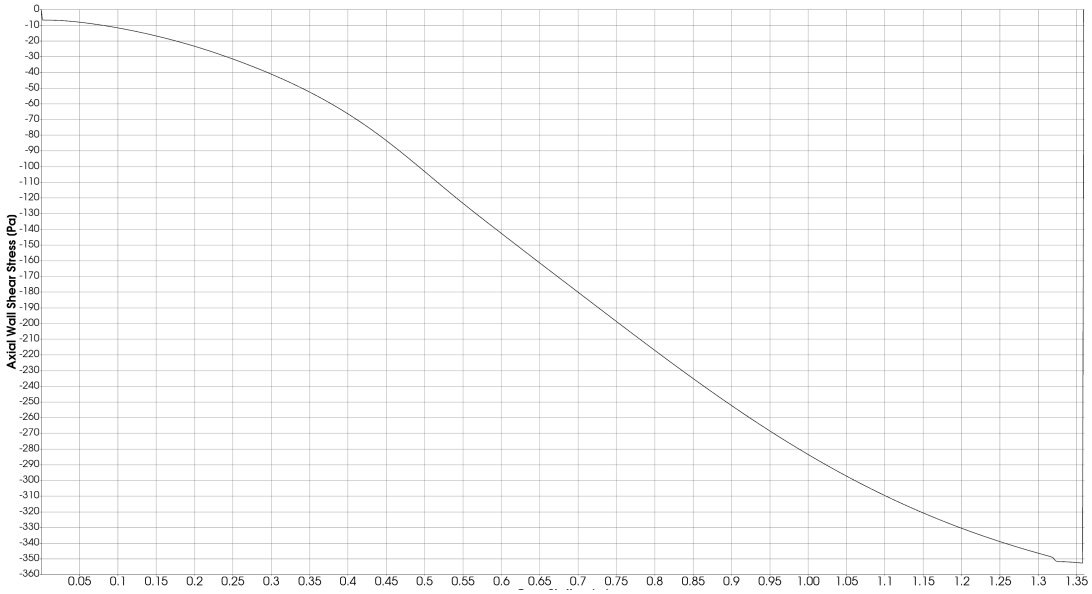



Figure 6.7: Axial distribution of wall shear stress on the fluid

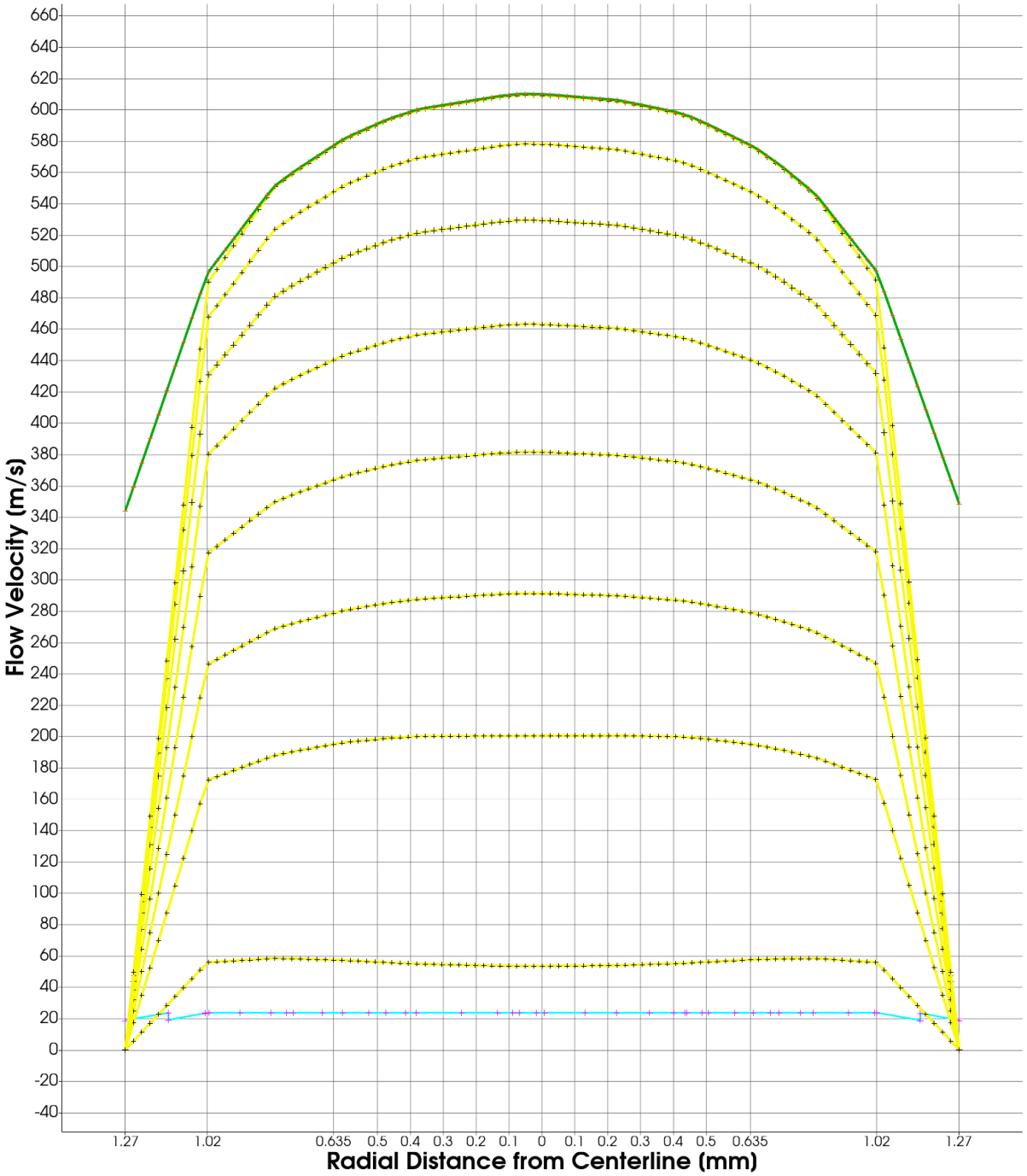
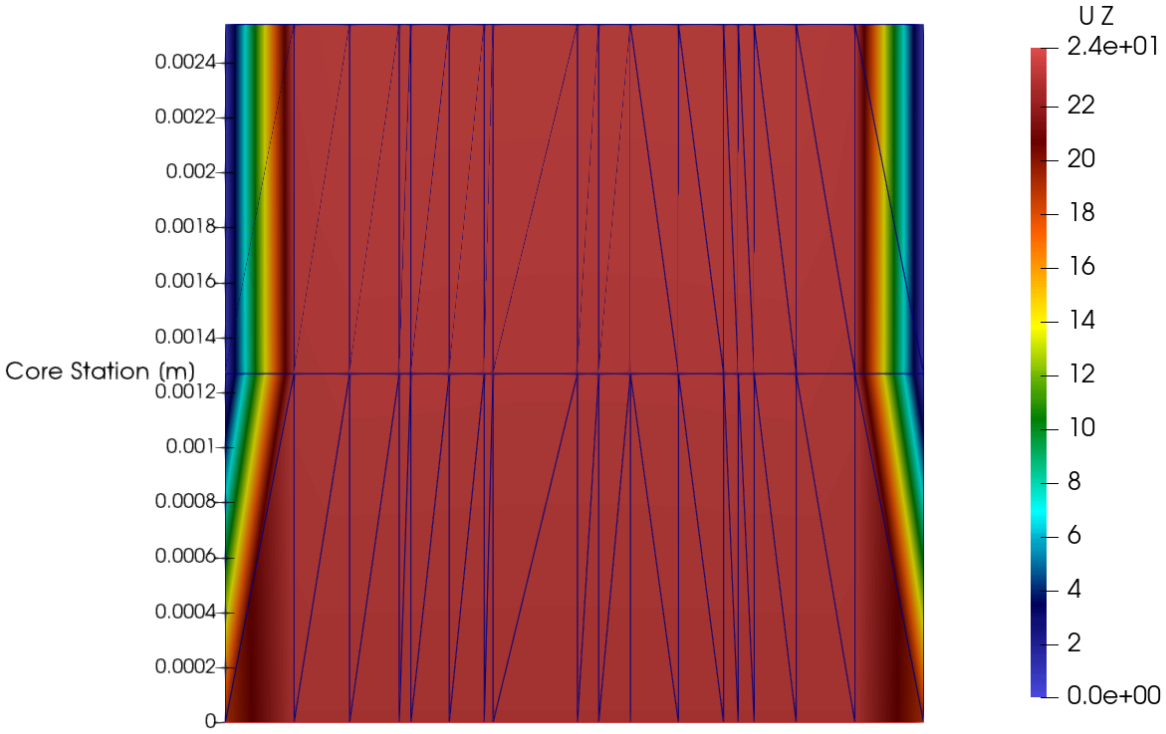
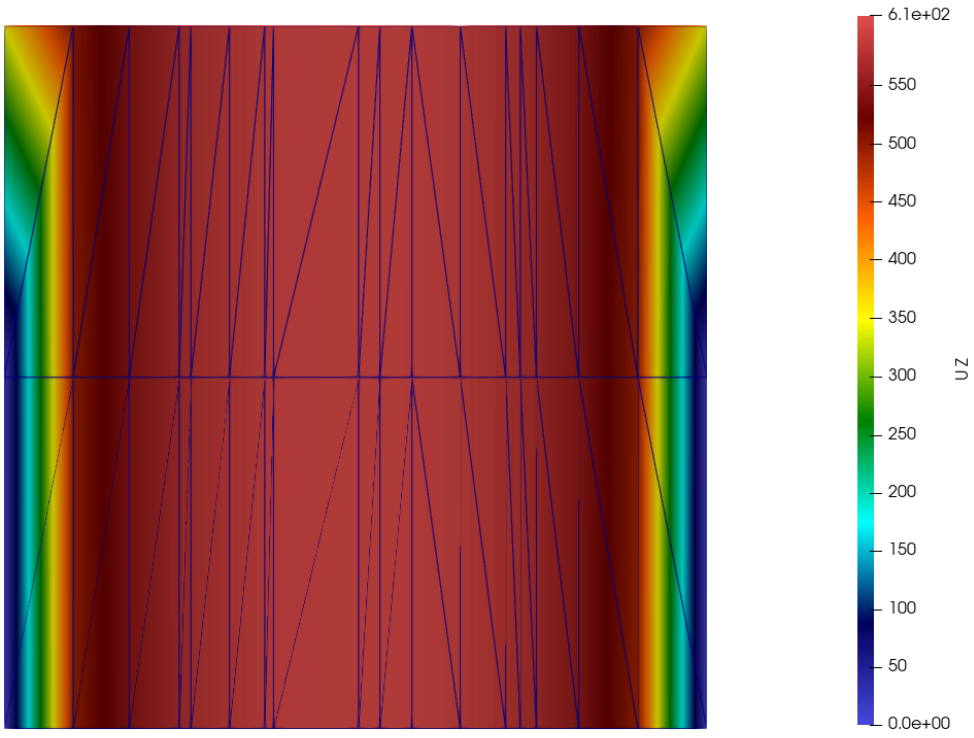



Figure 6.8: Radial Velocity Profiles - Legend:
cyan line: $|\vec{U}|$ at inlet at ($z = 0$ m)
magenta plus: $U_z$ at inlet at ($z = 0$ m)
yellow line: $|\vec{U}|$ at axial stations $z = n \times 0.15$ m, $\quad n = 1, 2, \ldots, 9$
black plus: $U_z$ at axial stations $z = n \times 0.15$ m, $\quad n = 1, 2, \ldots, 9$
green line: $|\vec{U}|$ at outlet at ($z = 1.3589$ m)
orange plus: $U_z$ at outlet at ($z = 1.3589$ m)




(a) Development of the velocity boundary layer over the first 2 cells at the inlet




(b) Dissolution of the velocity boundary layer over the last 2 cells at the outlet

Figure 6.9: Color map of $U_z$ through centerline cut at the inlet and outlet (true aspect ratio)

Congruency of $|\vec{U}|$ and $U_z$ in fig. 6.6 and fig. 6.8 confirms that the velocity field is uniaxial throughout the model. By comparing the result in fig. 6.6 with appendix fig. A.5, it can be seen that the flow operates in a low Mach number regime. Fig. 6.6 and 6.7 suggest a quasi-

linear relationship between bulk velocity and wall shear stress. Fig. 6.8 indicates that the no slip condition is satisfied throughout the length of the channel between the inlet and the outlet as well as that near-wall treatment consistent with a kOmegaSST wall function in the wall-adjacent cells ranging across $r \in [1.02, 1.27]$ mm. Fig. Fig. 6.9 shows that for the development and dissolution of the velocity boundary layer at the inlet and outlet respectively, continuity is satisfied.

## 6.4 Turbulence

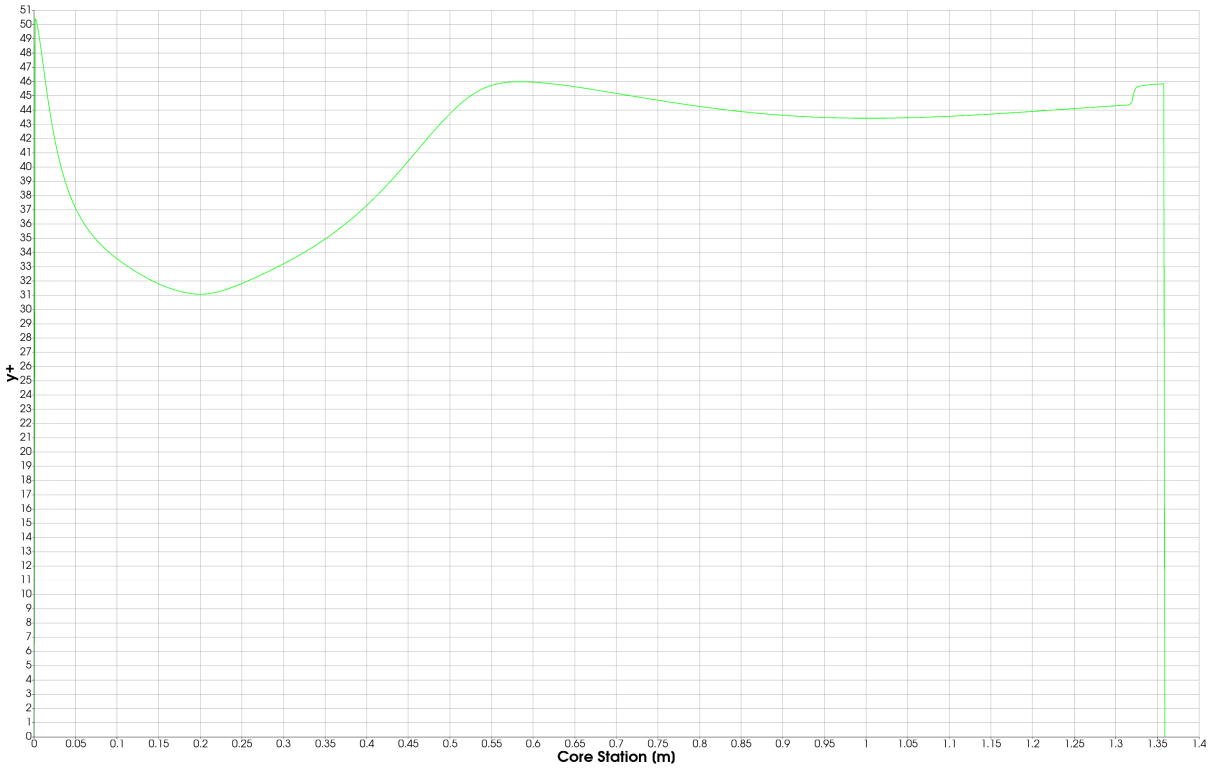
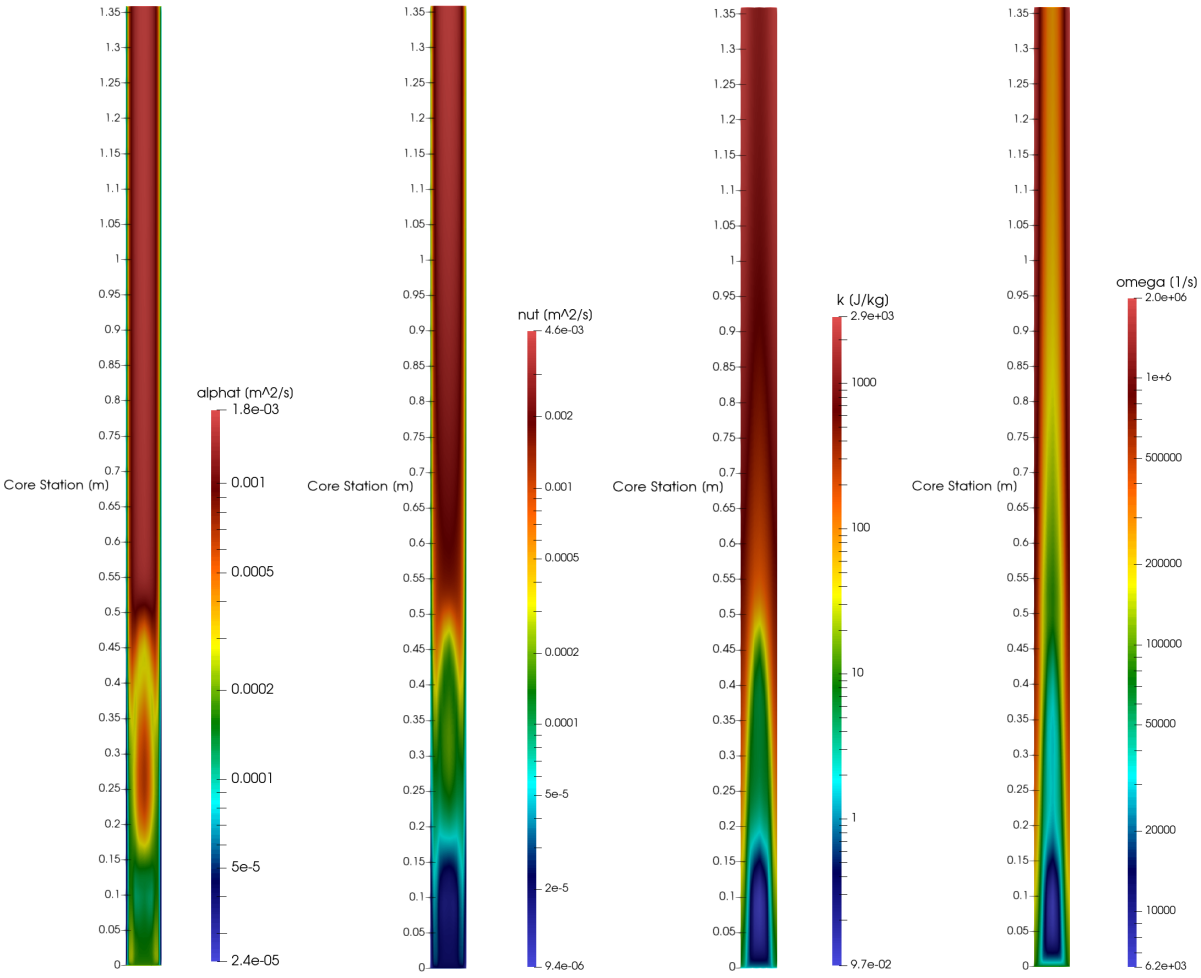



Figure 6.10: Axial distribution of $y^+$ along the fluid-solid interface




(a) $\alpha_t$ (b) $\nu_t$ (c) $k$ (d) $\omega$

Figure 6.11: Color maps of all turbulence fields for the kOmegaSST-model (aspect ratio scaled down $\times 20$ for visibility)

Inserting the mass flow relations from eq. 5.14 into the definition for the Reynolds number taking dynamic viscosity $\mu(T)$ from fig. 5.9:

$$Re = \frac{\rho U_{bulk} D_h}{\mu} = \frac{\dot{m} D_h}{\mu A_{cross}} = \frac{4\dot{m}}{\pi \cdot \mu(T) \cdot D_h} \tag{6.1}$$

It can be calculated, that for the inlet and outlet respectively:

$$Re_{inlet} = \cdot \frac{4 \cdot 1.007 \cdot 10^{-3} \frac{kg}{s}}{\pi \cdot 6.699 \cdot 10^{-6} Pa \cdot s \cdot 2.54 \cdot 10^{-3} m} = 75'352 \tag{6.2}$$

$$Re_{outlet} = \cdot \frac{4 \cdot 1.007 \cdot 10^{-3} \frac{kg}{s}}{\pi \cdot 3.042 \cdot 10^{-5} Pa \cdot s \cdot 2.54 \cdot 10^{-3} m} = 16'594 \tag{6.3}$$

The result in eqs. 6.2 and 6.3 confirm that turbulent behaviour is inevitable in the flow regime at hand. Fig. 6.10 indicates that the condition of $y^+ \in [30, 300]$ for the application of wall functions is satisfied on the entire fluid-solid interface for the fluid mesh used (Mesh Iteration 4.0). In fig. :

- All fields in fig. 6.11 appear to be fully developed at core station $z = 1$ m.

- $k$ and $\omega$ show a similar sharpening of the conic iso-line along the axial length of the channel of the conic profile .

- $\alpha_t$ and $\nu_t$ show a similar standing fluctuationg pattern between the inlet and core station $z = 0.5$ m due to their inherent relationship previously documented in eq. 3.3.1.

## 6.5 Axial Mesh Sensitivity

Despite the difference in total simulated time required to reach the termination criterion, all simulations required approximately the same computation time, most likely because the numerical stiffness of the linear algebraic system for the solution of the energy equation in solid regions increases with mesh resolution. For a variation of $\pm 20\%$ in axial cell length most simulation results in tab. 6.4 indicate relative deviations far below 1%, with the only exception being $\alpha_t$ for the mesh with axial cell length 1.0583mm. Simulations were run with $maxCo = 1$ but maximum time step intervals $\Delta t_{sim,max}$ were also set as an additional safeguard.

Table 6.4: Comparison between different axial mesh resolutions

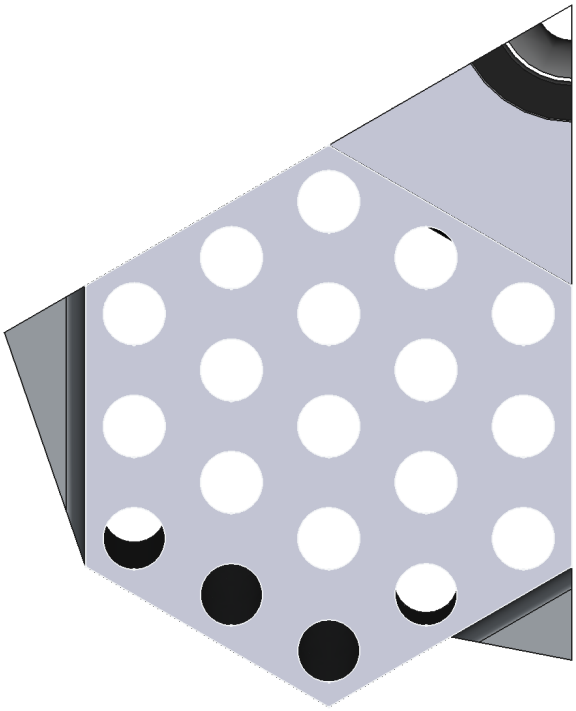
| **Axial Mesh Resolution** | 1.5875mm (1/16th inch) | 1.27mm (1/20th inch) | 1.0583mm (1/24th inch) |
|---|---|---|---|
| $\mathbf{t_{sim,max}}$ **[s]** | 2.5 | 2e-6 | 1.667e-6 |
| $\mathbf{t_{sim}}$ **[s]** | 16 | 16 | 13 |
| $\mathbf{max\{Co\}_{final}}$ | 0.967227 | 0.967255 | 0.967463 |
| $\mathbf{\dot{Q}_{final}}$ **[W]** | $39'323.35$ | $39'323.55$ | $39'323.34$ |
| **p [MPa]** | 0.46604 | 0.46574 | 0.46629 |
| $\mathbf{T_{Propellant,out}}$ **[K]** | $2544.83 \pm 3$ | $2544.445 \pm 2.555$ | $2544.835 \pm 2.925$ |
| $\mathbf{T_{Fuel,Station1}}$ **[K]** | 537.833 | 537.681 | 537.559 |
| $\mathbf{T_{Fuel,Station20}}$ **[K]** | 1788.09 | $1'788.13$ | 1788.19 |
| $\mathbf{T_{Fuel,Station26}}$ **[K]** | 2083.34 | $2'083.02$ | 2082.79 |
| $\mathbf{T_{Fuel,max}}$ **[K]** | $2'640.5$ | $2'640.42$ | $2'640.34$ |
| $\mathbf{z(T_{Fuel,max})}$ **[m]** | 1.24222 | 1.2414 | 1.24195 |
| $\mathbf{U_{z,outletBulk}}$ **[m/s]** | 609.659 | 609.802 | 609.799 |
| $\boldsymbol{\nu_{t,outletBulk}}$ $[\mathbf{m^2/s}]$ | $4.57673e-3$ | $4.57842e-3$ | $4.57829e-3$ |
| $\boldsymbol{\alpha_{t,outletBulk}}$ $[\mathbf{m^2/s}]$ | $1.7616e-3$ | $1.76155e-3$ | $1.7917e-3$ |

# Chapter 7
# Potential applications for the OpenFOAM Model at Cluster Scale

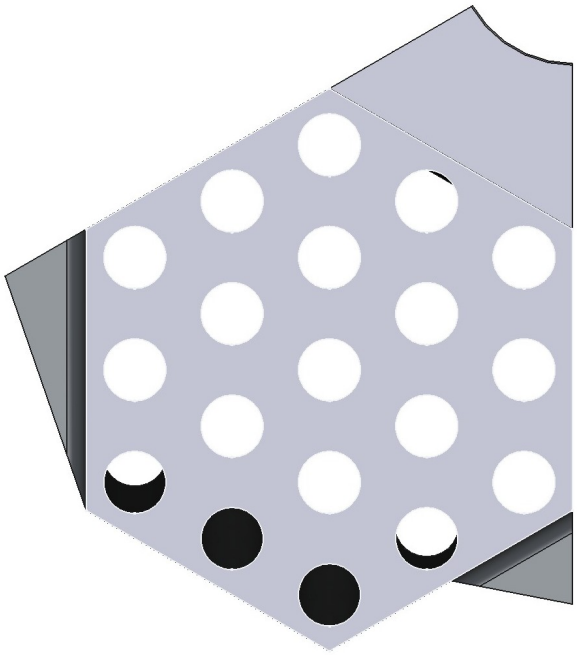
For further validation of the CFD model in this study and others against accurately placed thermocouple measurements and potentially thereafter validate alternative fuel variants in a similar overall reactor design economically, it is recommended to simulate two distinct implementations of the cluster to determine the minimal practical realization for a hot-cluster analysis. This can be done with a complete model of the entire cluster or with a reduced model as shown in the subsequent section, taking advantage of symmetries in temperature for a cluster near the center of the reactor core, where axial temperature gradients are small across a given fuel element.

## 7.1 Single-element Fuel Cluster Section

Such a model would contain a section with a single fuel element and a coangular cut from the central unfueled element and support block as well as other hardware modeled similarly to what is shown as shown in fig. 7.1a. For the meshing it is recommended that a small number of viscous layers around every channel range up to 1.4mm to 1.6mm from the channel centerline based on radial quasi-linearity of predicted fuel temperature near the fluid-solid interface shown in fig. 6.3. This model can be implemented by enforcing zero-gradient boundary conditions for temperature on the cut surfaces as well as the outer bounds of the system and restricting flow angularly in and out of the tie rod annulus section if the parts inside and including the pyrolytic graphite sleeve are to be simulated. A case can potentially be made for the elimination of these elements from a practical simulation of fuel heat transfer, namely that this section of the cluster is of much lower operating temperature compared to the rest of the reactor, reaching a maximum of only 416.7 K (see tab. B.1) (measured in the tie rod material in the core center region) because the pyrolytic sleeve acts as a radial insulator for heat transferred from the fuel. The center hole of the unfueled element can thus potentially also be modeled using a zero-gradient boundary condition for the temperature field for a geometry further reduced as shown in fig. 7.1b. If the tie rod section is to be modeled, the anisotropic thermal conductivity of the pyrolytic sleeve $\kappa$ can be modeled utilizing the corresponding material property dictionary functionality with polar coordinates in OpenFOAM v13 using for example fig. 7.2 by Sorell et al. as a source. A dedicated validation study using both cases is recommended.



(a) Partial implementation of a cluster a single fuel element and all parts coaxial centerpoint of the unfueled element cut into corresponding 1/6th sections



(b) Partial implementation of a cluster a single fuel element and central unfueled element as well as support block cut into corresponding 1/6th sections

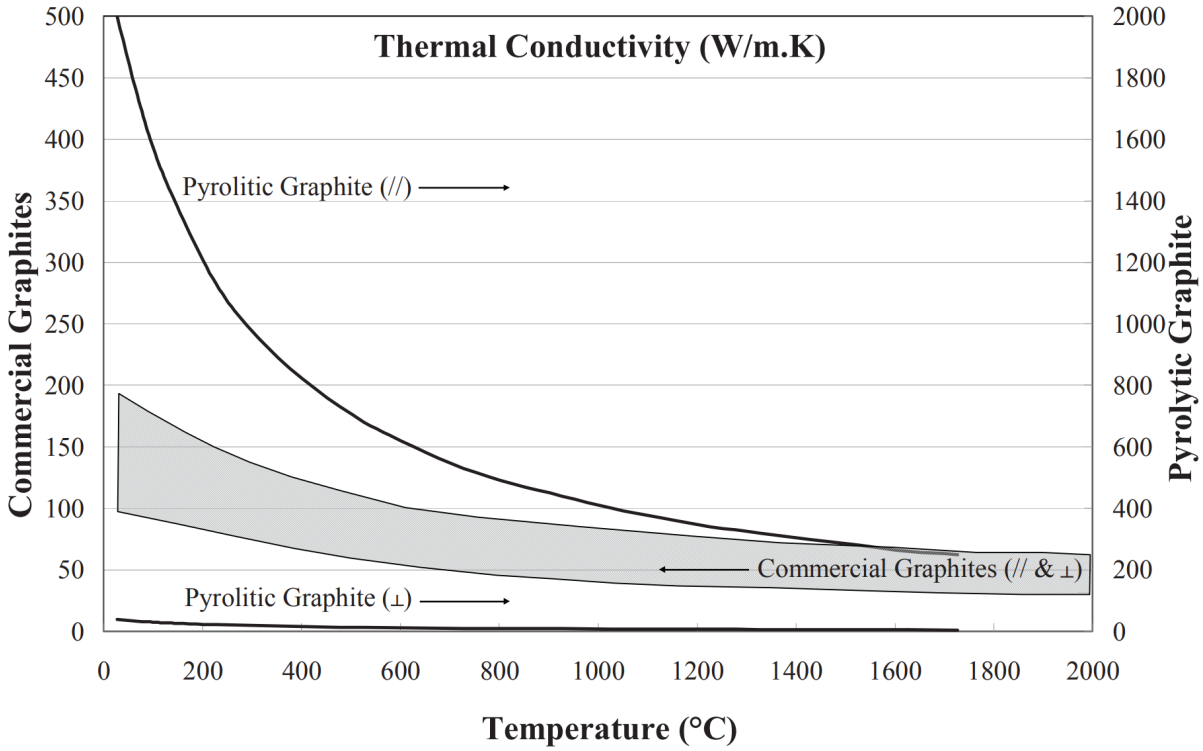



Figure 7.2: Thermal conductivity of pyrolytic and other forms of graphite - Source [34, fig. 6]

# Chapter 8
# Conclusion

The overall aim of implementing a functional simulation of CHT in an NTR has been achieved. OpenFOAM v13 has been found to be adaptable to this problem to produce results representative of this application, and the adopted turbulence model $k-\omega$ in a wall function implementation was found to be applicable to this problem.

The majority of fields were found to have been modeled within limited error margins. One major exception is the stark overestimation of the station 1 fuel thermocouple temperature, which is qualitatively consistent with findings made by Wang and Kotlyar though more pronounced. There is a considerable underestimation of reactor pressure drop based on the assumed pressure boundary conditions and further analysis and model iterations are required to identify whether the cause is intrinsic or extrinsic to the CFD model.

The simulation indicates the presence of significant turbulence inside a NERVA-type fuel channel within the center region of the reactor during operation at nominal steady state operating conditions.

The inner workings of foamMultiRun for the given problem have been mapped and theoretically outlined to allow for the accurate diagnosis through the course of further applications of this particular model.

The numerical formulation and solver configuration were developed and assessed with respect to stability and convergence for very-high-temperature CHT conditions, and the approach is applicable to the analysis of similar advanced reactor concepts.

Program ELM was evaluated as simulating an average channel within the NRX-A6 reactor and a model was developed based on pre-test predictions to scale power for different radial positions within the reactor.

Based on historical documentation gathered, the overall geometry of an NRX-A-type fuel cluster was reconstructed in a CAD model to aid contextualization of the fluid dynamic problem at hand and may be useful for future analysis.

## 8.1 Potential Research Applications

Fig. 8.1 outlines further research that could be conducted based on the findings in this study. First steps include the rectification of model errors based on potential causes identified in the discussion of the results and potential applications in fields relevant to NTR development and safety. Because of the limitations outlined with sourcing accurate thermal conductivity data for the graphite fuel, it is recommended that accurate material data is sourced or otherwise experimentally established based on the extensive contemporary material studies and documentation published by the Oak Ridge and Los Alamos National Laboratories. [31][37].

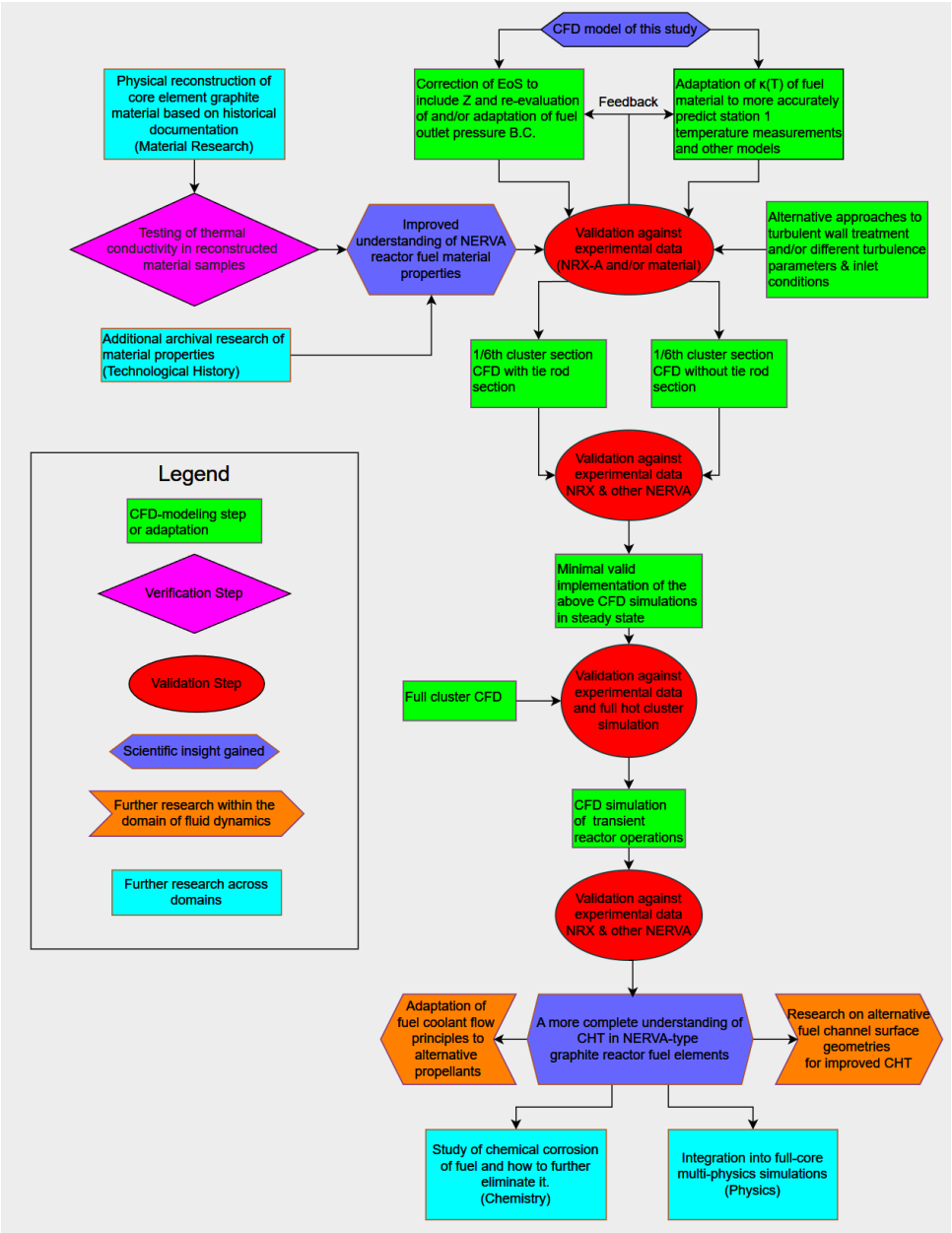



Figure 8.1: Potential research paths based on the findings of this study

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# Appendices

# Appendix A
# Documentation of Fluid Property Extraction

Source for figs. A.2, A.3, A.4, and fig. A.5: Hydrogen Technological Survey – Thermophysical Properties by Robert D. McCarty, 1975
Notes:

- In fig. A.2 the descending linear curves against a logarithmic T & $\rho$scale show ideal gas equation of state (EoS).
- The Nuclear sub-System (NSS) operates at pressures approximately between 4 to 5 MPa or 580 to 710 psi. This is crucial to demonstrating pressure independence for $\kappa$ in A.4 & $\mu$ in A.3 evidenced by the convergence of curves for the temperature regime in question (100-2500 K)
- With the aforementioned justification, the linear fit for specific heat of hydrogen for the fluid model seen in A.1 was taken directly from "Multiphysics Design and Simulation of a tungsten-Cermet Nuclear Thermal Rocket" by Brad Appel

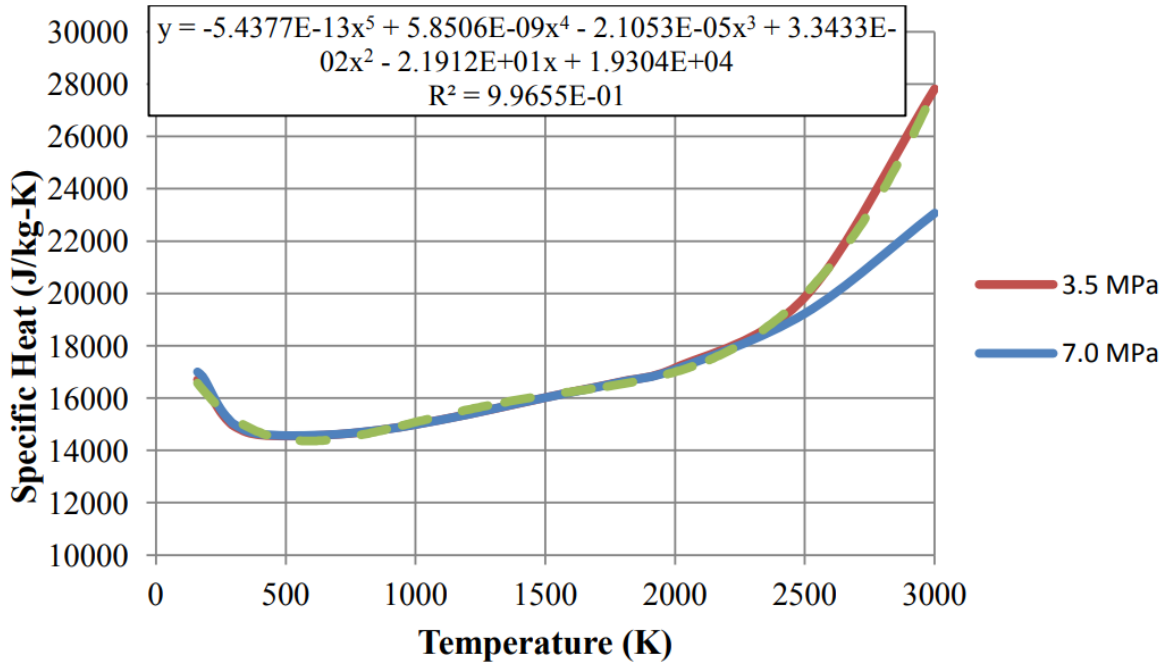



Figure A.1: Polynomial fit for specific heat $c_p$ of hydrogen - Source: Brad Appel

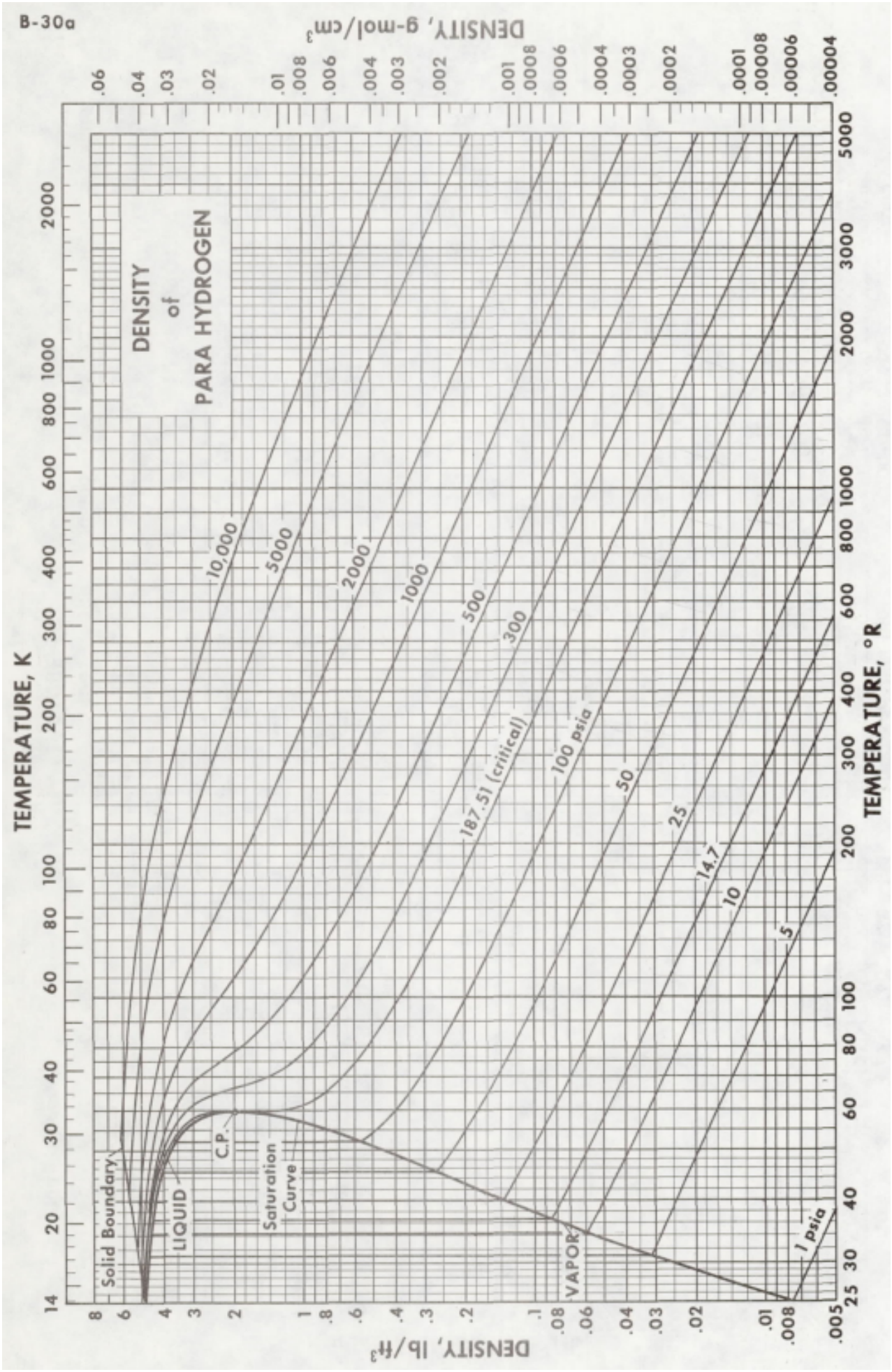



Figure A.2: Density of hydrogen as a function of temperature. Source: McCarty, p. 241.

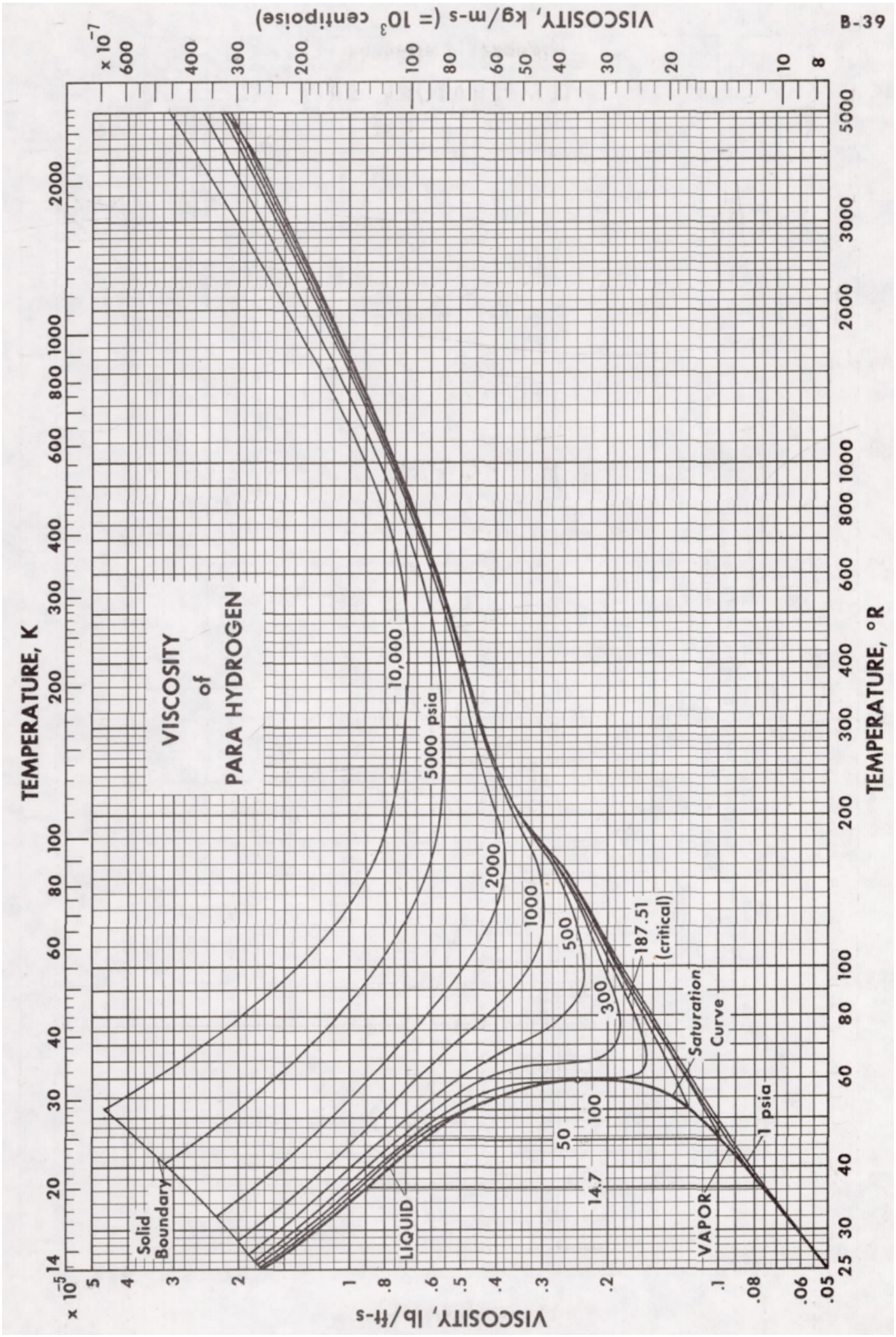



Figure A.3: Dynamic viscosity of hydrogen as a function of temperature.

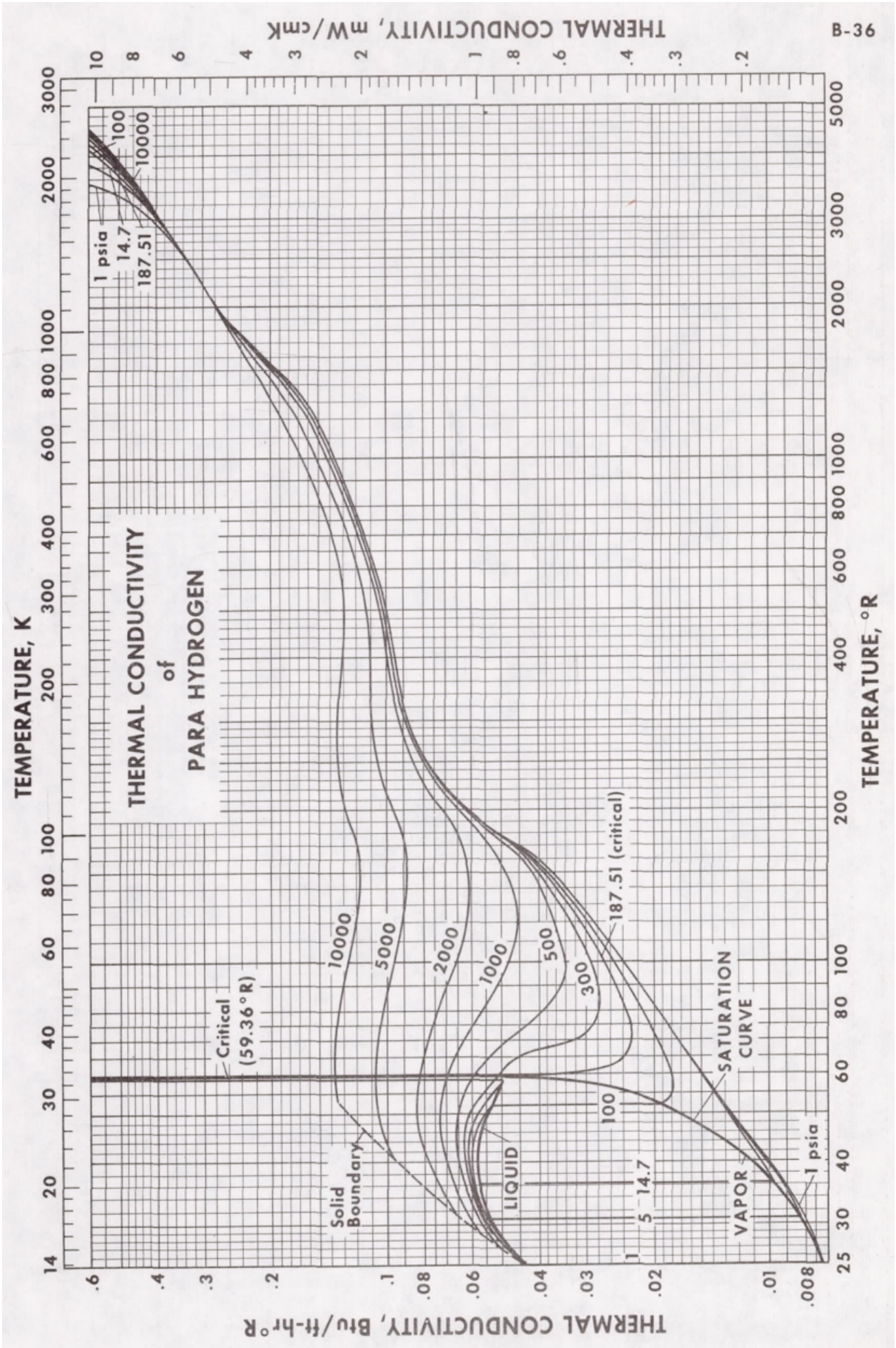



Figure A.4: Thermal conductivity of hydrogen as a function of temperature.

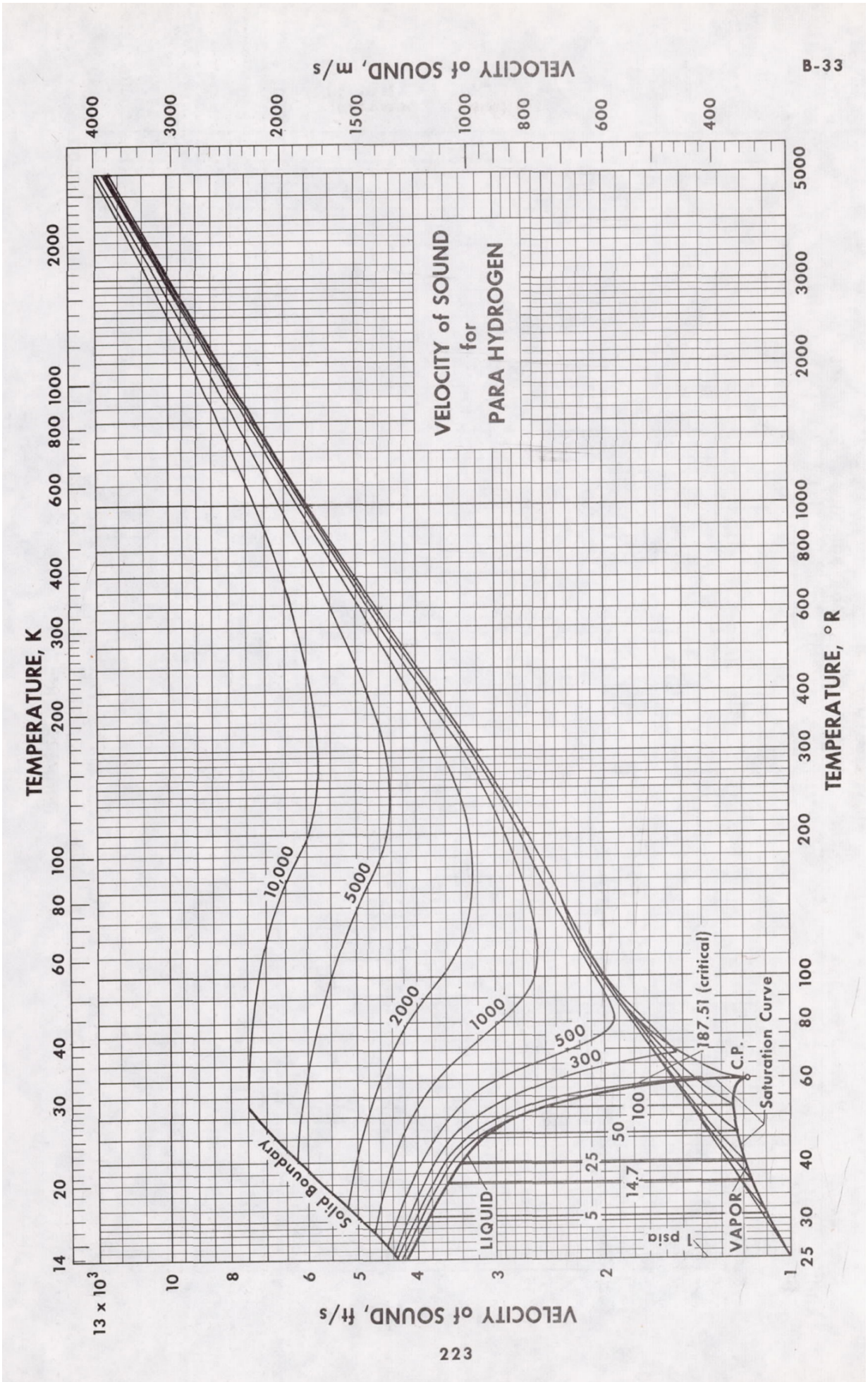



Figure A.5: Speed of sound in hydrogen

# Appendix B
# Documentation of System Reference Values

Note: Original documents are in greyscale, annotations by the thesis author were made in color.

## B.1 Overview of NERVA Test Reactors and technical Data

Source for table B.1: Overview of the Rover Engine Tests Final Report by J.L. Finseth, Feb 1991

Source for table B.2 and fig. B.3: Westinghouse Astronuclear Laboratory Contribution to the 27th high Temperature Fuels Committee Meeting by L.R. Fleischer & C.R. Simmons, Dec 1968 (WANL-TME-1878)

Note: Entry for NRX-A6 Avg. full Power (MW) in table B.1 is incorrect. This value instead represents peak power following startup, see table B.2 and fig. B.3.
It is read from the graph in fig. B.3, that the nominal reactor power range was 1120-1160 MW averaging ca. 1140MW, though the same report states in writing that power was held at 1150-1160MW.

| TEST DATE | REACTOR | NUMBER OF STARTS | AVG FULL POWER (MW) | TIME AT FULL POWER (s) | AVG PROPELLANT TEMPERATURE (K) CHAMBER | FUEL EXIT | CHAMBER PRESSURE (kPa) | FLOW RATE (kg/s) | IDEAL VACUUM SPECIFIC IMPULSE (s) |
|---|---|---|---|---|---|---|---|---|---|
| | *- RESEARCH REACTORS -* | | | | | | | | |
| 7/1959 | KIWI A | 1 | 70 | 300 | | 1778 | ? | 3.2 | 724 |
| 7/1960 | KIWI A' | 1 | 88 | 307 | | 2206 | 1125 | 3.0 | 807 |
| 10/1960 | KIWI A3 | 1 | 112.5 | 259 | | 2172 | 1415 | 3.8 | 800 |
| 12/1961 | KIWI B1A | 1 | 225 | 36 | | 1972 | 974 | 9.1 | 763 |
| 9/1962 | KIWI B1B | 1 | 880 | SEVERAL | | >2278 | 2413 | 34.5 | >820 |
| 11/1962 | KIWI B4A | 1 | 450 | SEVERAL | | 1556 | 1814 | 19.0 | 677 |
| 5/1964 | KIWI B4D | 1 | 915<br>990 PEAK | 64 | 2006<br>2156 PEAK | 2378<br>2539 PEAK | 3606 | 31.1 | 837<br>865 PEAK |
| 8/1964 | KIWI B4E | 2 | 937 | 480 | 1972 | 2356 | 3427 | 31.0 | 834 |
| 6/1965 | PHOEBUS 1A | 1 | 1090 | 630 | 2278 | 2444 | 3772 | 31.4 | 849 |
| 2/1967 | PHOEBUS 1B | 2 | 1290<br>1450 PEAK | 1800 | 2094<br>2244 PEAK | 2306<br>2456 PEAK | 5075 | 38.1 | 825<br>851 PEAK |
| 6-7/1968 | PHOEBUS 2A | 4 | 4082 | 744 | 2256 | 2283 | 3827 | 119.0 | 821 |
| 11-12/1968 | PEWEE 1 | 3 | 503 | 2400 | 1803 | 2539<br>2750 PEAK | 4344 | 18.8 | 865<br>901 PEAK |
| 6/1972 | NF-1 | 5 | 44 | 6528 | | 2444 | | 1.7 | (849) |
| | *- DEVELOPMENT REACTORS -* | | | | | | | | |
| 9-10/1964 | NRX A2 | 2 | 1096 | 40 | 2119 | 2229 | 4006 | 34.3 | 811 |
| 4-5/1965 | NRX A3 | 3 | 1093 | 990 | 2189 | >2400 | 3930 | 33.3 | >841 |
| 2-3/1966 | NRX/EST | 11 | 1144 | 830 | 2292 | >2400 | 4047 | 39.3 | >841 |
| 6/1966 | NRX A5 | 2 | 1120 | 580 | 2287 | >2400 | 4047 | 32.6 | >841 |
| 11-12/1967 | NRX A6 | 2 | 1199 | 3623 | 2406 | 2558 | 4151 | 32.7 | 869 |
| 3/1969 | XE-PRIME | 28 | 1137 | 1680 | 2267 | >2400 | 3806 | 32.8 | >841 |

C-2

**TABLE C-1. REACTOR TEST SUMMARY**

Figure B.1: Comprehensive overview of Rover Tests by Finseth

(CRD) NRX-A2 THROUGH NRX-A6 MAJOR REACTOR PERFORMANCE SUMMARY (U)

| Reactor | NRX-A2 | NRX-A3 | NRX/EST | NRX-A5 | NRX-A6 |
|---|---|---|---|---|---|
| Chamber Temperature (°R) | Time within chamber temperature interval (minutes) | | | | |
| 3700-3800 | 0.8 | 0.2 | 3.6 | 0.7 | 0.1 |
| 3800-3900 | --- | 13.4 | 6.1 | 3.0 | 0.1 |
| 3900-4000 | --- | 3.0 | 5.3 | 11.3 | 0.1 |
| 4000-4100 | --- | --- | 7.3 | 12.8 | 0.6 |
| 4100-4200 | --- | --- | 7.8 | 2.3 | 54.6 |
| 4200-4300 | --- | --- | --- | --- | 6.8 |
| 4300-4400 | --- | --- | --- | --- | 0.6 |
| Equivalent Full Power Radiation (Minutes) | | | | | |
| a. Entire Test Series | 5.7 | 26.8 | 54.4 | 32.2 | 66.9 |
| b. Single Run | 5.1 | 14.3 | 14.5 | 16.4 | 66.6 |
| Maximum Flow Rate (Lb/Sec) | 75.6 | 73.5 | 87.9 | 71.9 | 72.0 |
| Maximum Power (MW) | 1096 | 1100 | 1155 | 1120 | 1200 |
| Number of Startup and Shutdown Cycles Above 2500°R | 1 | 3 | 6 | 2 | 1 |
| Core Average Corrosion Weight Loss, Grams Per Element | 1.0 | 10.6 | 32.4 | 27.0 | 11.2 |
| Observed Total Reactivity Loss, ($) | 0.1 | 0.9 | 3.2 | 2.1 | 0.7 |

Figure B.2: Coeval tabulation of NRX-A series technical data; note the similarity in flow rate between NRX-A5 and NRX-A6

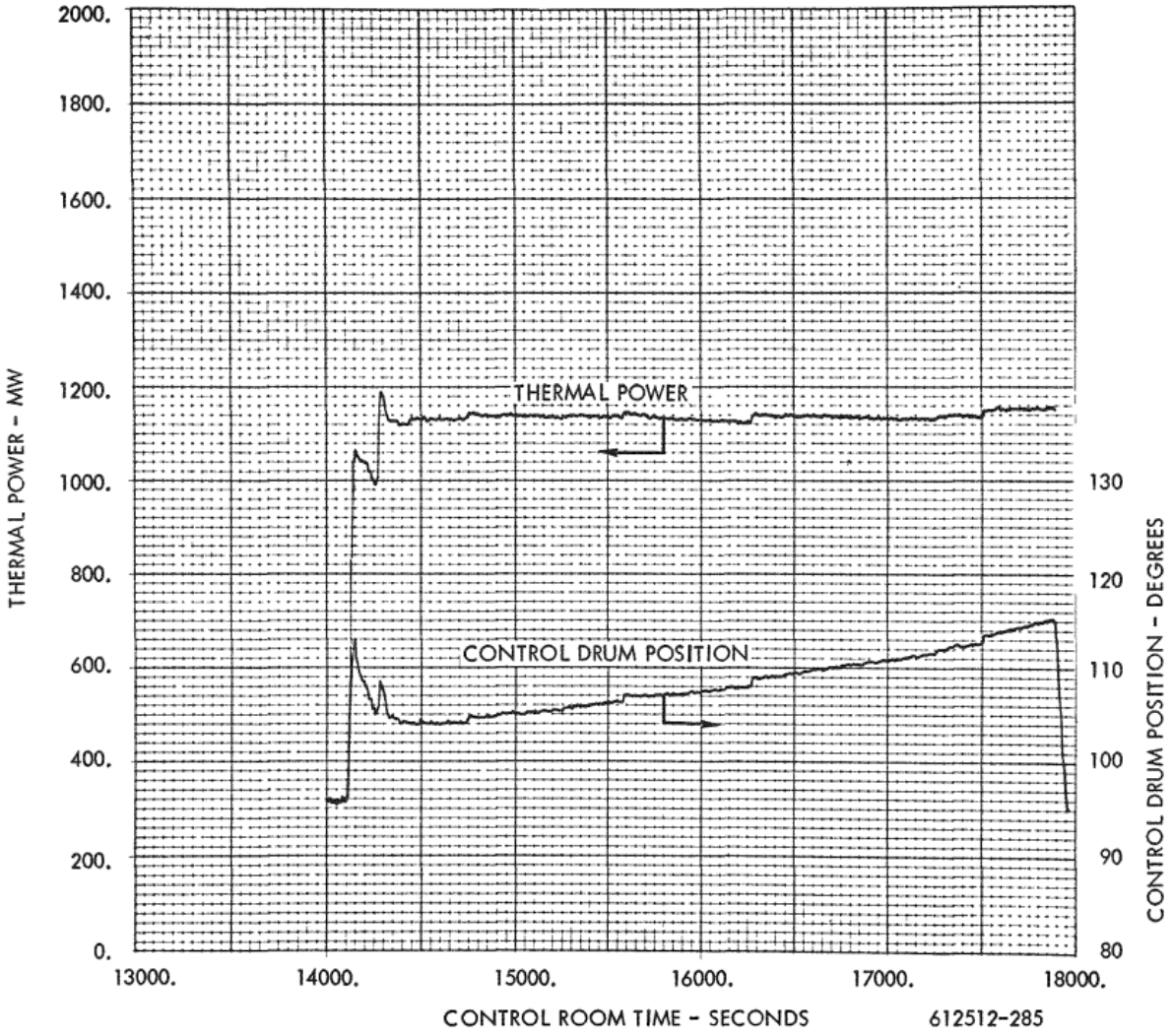



Figure B.3: NRX-A6 actual total reactor power over time

## B.2 Core Map showing Placement of thermocoupled Fuel Cluster and Cluster Designation Scheme

Source for fig. B.4 & B.5: Evaluation of the WANL Integrated Nuclear, Radiation, and Shielding Standard Design Method (WISDM) (U), Volume 2 by L.I. Ortenberg, Sep 1971 (WANL-TME-2689)

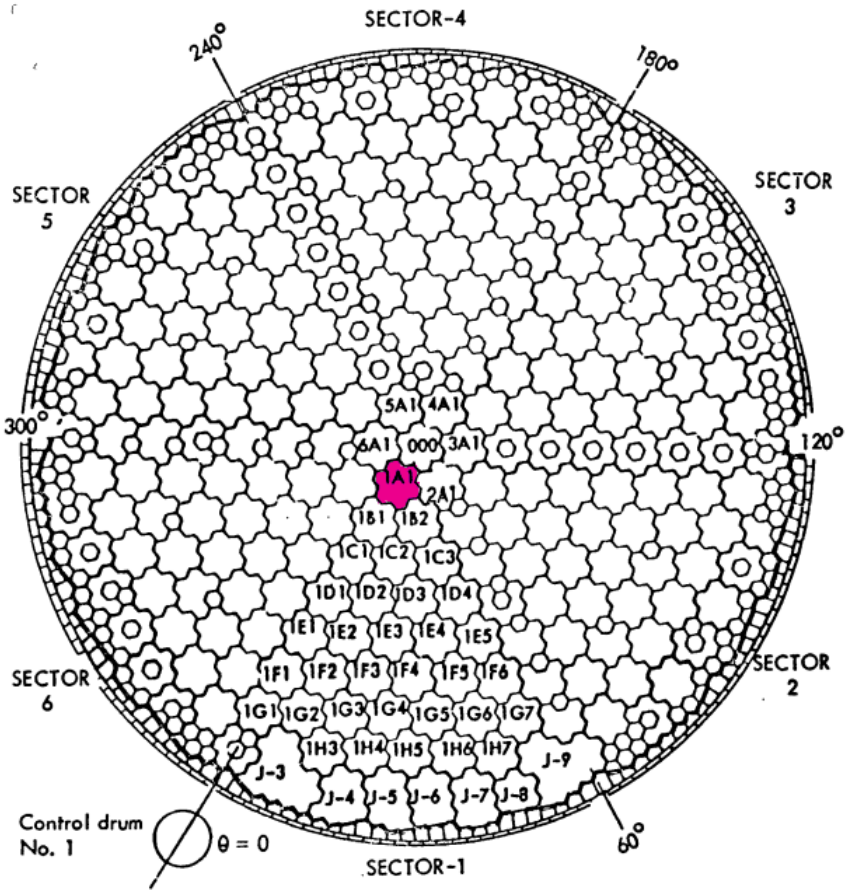
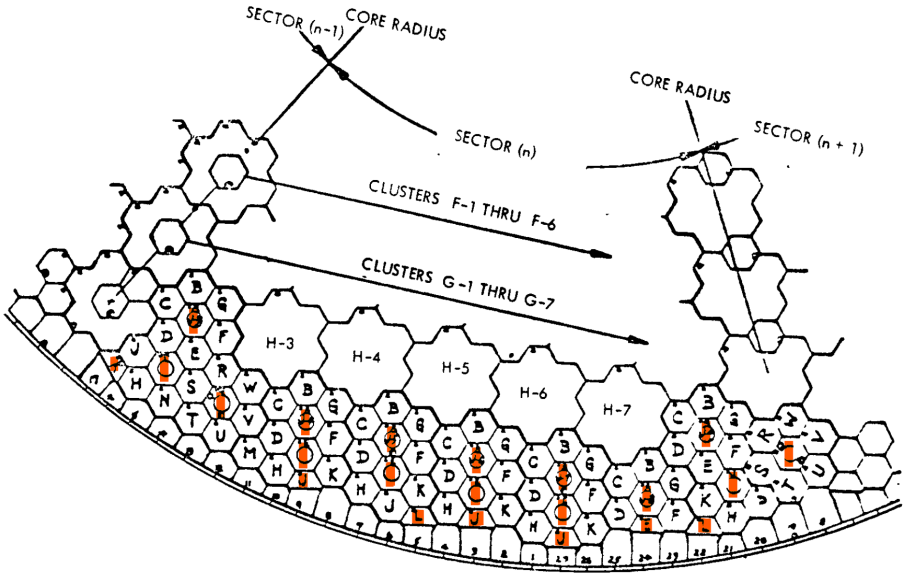



Figure B.4: NRX-A6 core map with thermocoupled hot cluster 1A1 in magenta




NOTES: 1) CORE VIEWED IN THE DIRECTION OF PROPELLANT FLOW.
2) ALL PARTIAL ELEMENTS IN NRX-A6 ARE UNFUELED.

Figure B.5: Detailed mapping of core periphery; Unfueled elements marked orange.

## B.3 Original Graphs predicting relative radial Power Distribution in NRX-A reactors to provide data for axial Power Distribution Scaling

Source for fig. B.6, B.7 & B.8: NRX-A6 Design Report Volume 2, thermal, Fluid Flow, and nuclear Analysis of the NRX-A6 prepared by Reactor Analysis Staff, Oct 1965 (WANL-TME-1290)

Note: Estimates were made using a method designated NSDM-II described in:
Improved Nuclear Standard Design Method for NERVA Calculations NSDM-II by J.M. Ravets, L.I. Kopp, A.L. Mowery and P.J. Sipush (WANL-TME-1091)
The method is validated in fig. B.6 based on NRX-A2 for a given control drum position. Incorporating further corrections made for the reactor periphery using data obtained from the PAX-A2 experimental reactor. [**25**, 4-21 to 4-47]

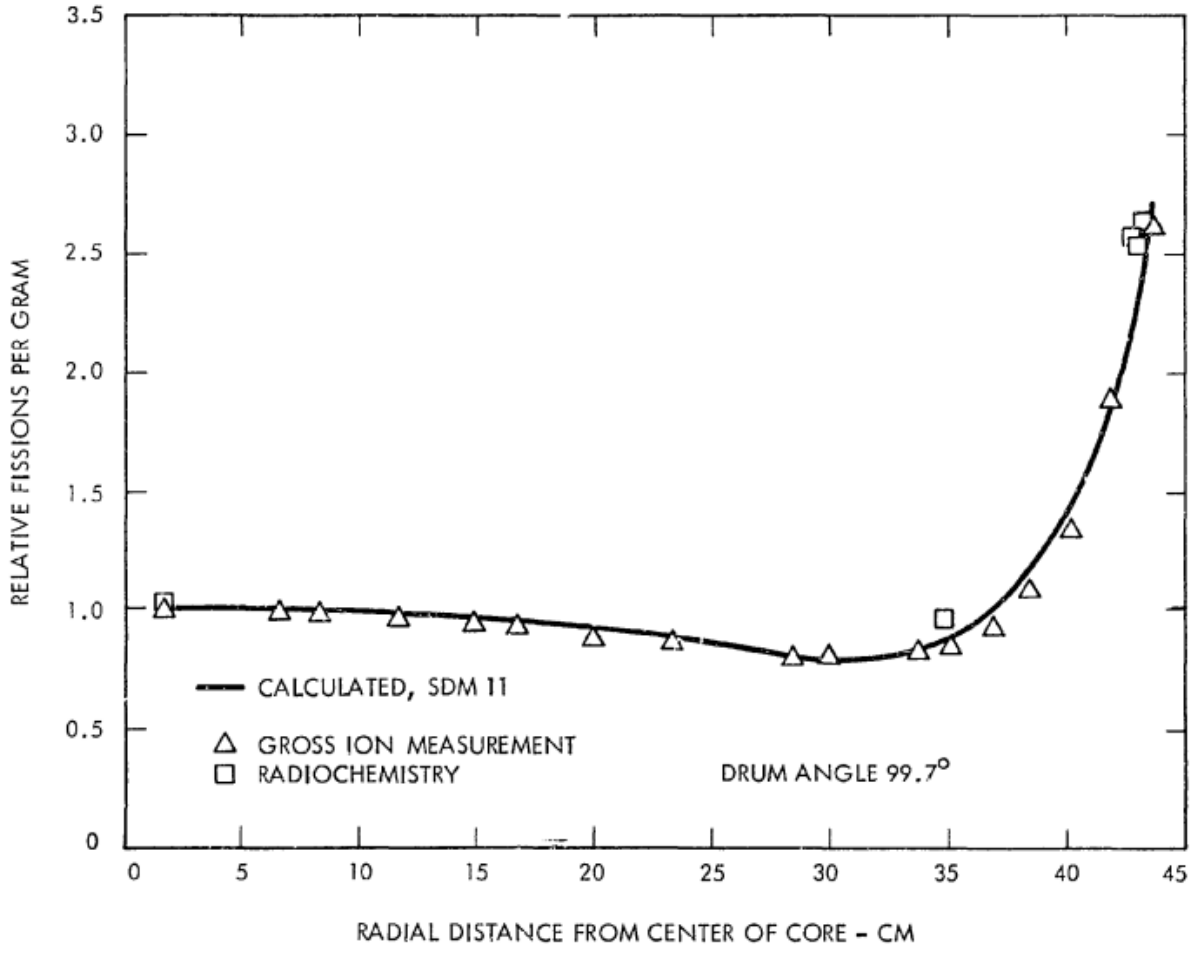



Figure 4-4. Comparison of the NRX-A2 Measured and Predicted Power Distributions through Traverse to J4M, Fissions Per Gram

Figure B.6: Validation of the NSDM-II method based on NRX-A2

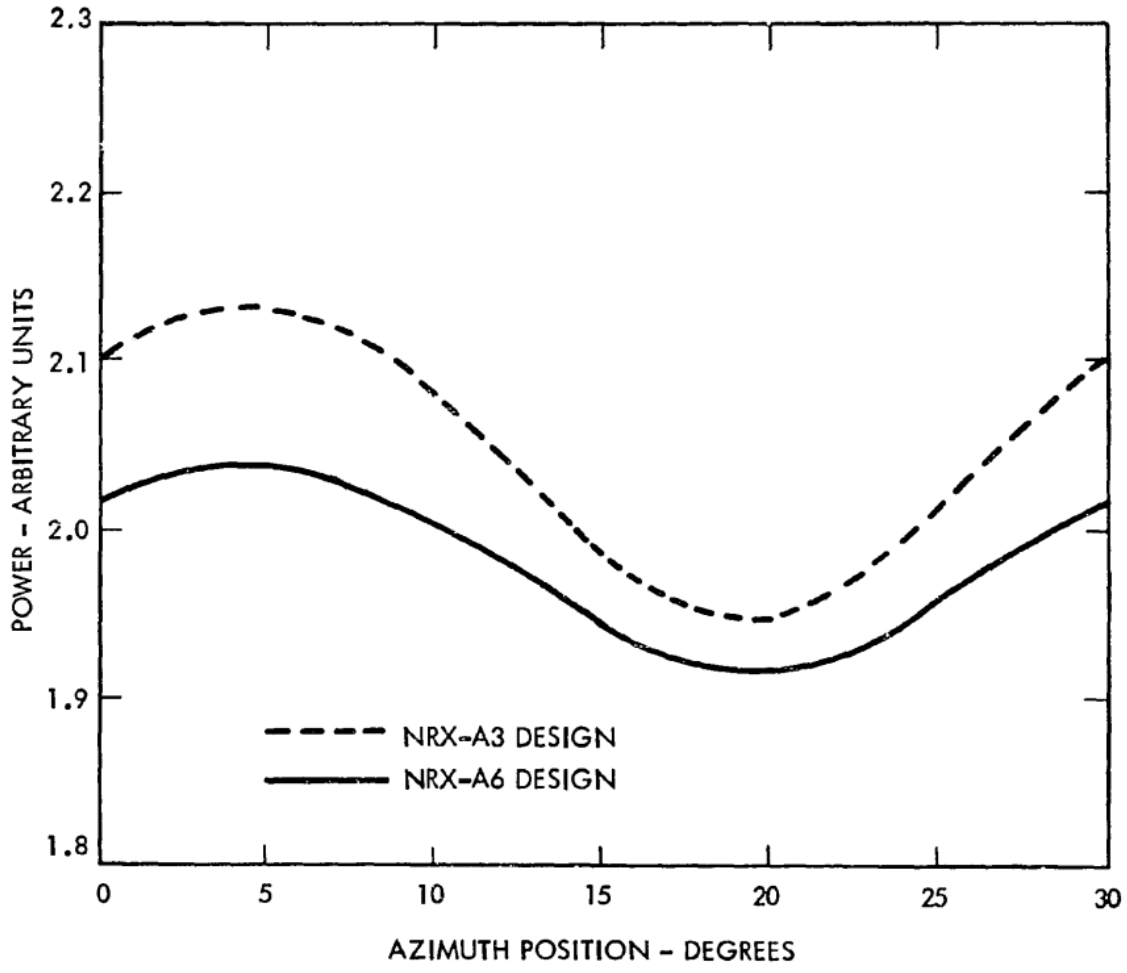



Figure 4-9. Comparison of the NRX-A3 and NRX-A6 Power Scalloping at the Periphery of the Cone, Exterminator Predictions

Figure B.7: Azimuthal sinusoidal scalloping of relative power along core periphery

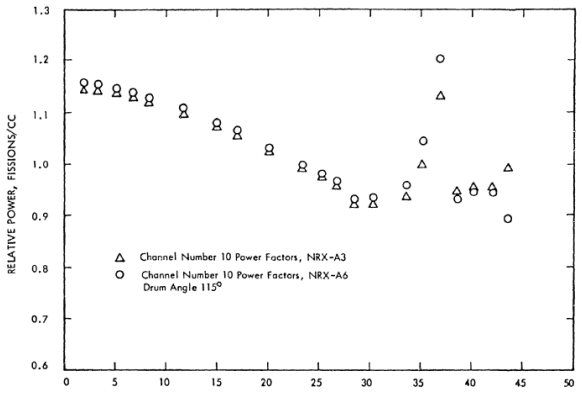



Figure 4-5. Comparison of the NRX-A3 and NRX-A6 Predicted Power Distributions through Traverse 00B to J4M, Fissions/cc

(a) Predicted relative fission rate per volume as a function of core radius

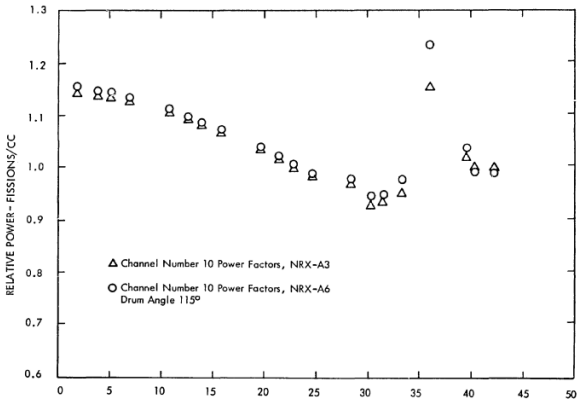



Figure 4-6. Comparison of the NRX-A3 and NRX-A6 Predicted Power Distributions through Traverse 00B to J5L, Fissions/cc

(b) Predicted relative fission rate per volume as a function of core radius

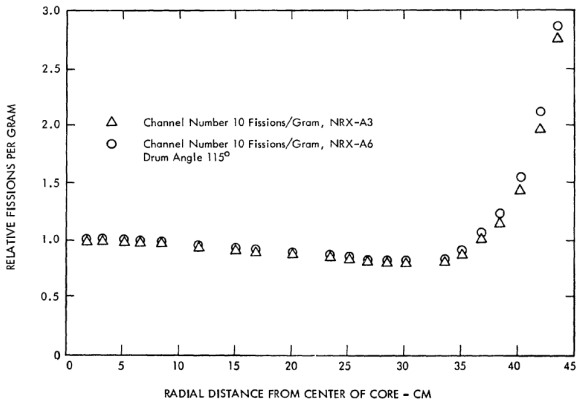



Figure 4-7. Comparison of the NRX-A3 and NRX-A6 Predicted Power Distributions through Traverse 00B to J4M, Fissions Per Gram

(c) Relative fission rate per mass as a function of core radius

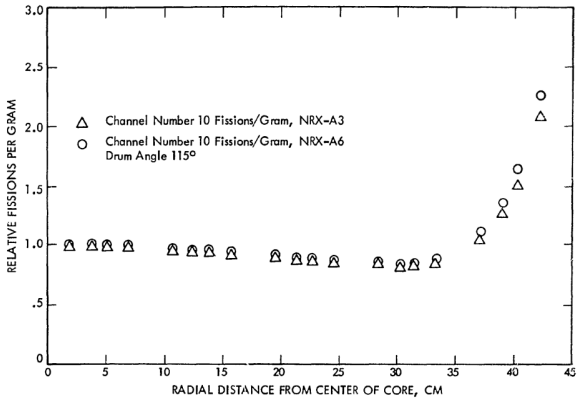



Figure 4-8. Comparison of the NRX-A3 and NRX-A6 Predicted Power Distributions through 00B to J5L, Fission Per Gram

(d) Relative fission rate per mass as a function of core radius

Figure B.8: Radial power distributions for the NRX-A3 & -A6 reactors.

## B.4 Scope, Justification, and Assembly for Use of NRX-A5 Data

Source: Single Cluster Corrosion Test Summary by L.A. Salvador, R.D. Foster, and L. Wagner, Apr. 1966 (WANL-TME-1375)

The NRX-A5 test reactor preceding NRX-A6 did not differ from A6 in overall core geometry[**14**, p.110] except that A6 featured axially shortened control blocks and adapted Support Block Hardware (See Appendix C.2). A5 and A6 had the exact same core propellant mass flow and approximately the same full nominal power, see appendix B.1. Therefore, these data are used to calculate nominal mass flow while support block/exit gas temperature is useful as an initial condition for the hot end temperature.

Table V

Comparison Between Typical HHT-4 and NRX-A5 Cluster Parameters at Full Power

Non-reactor Experiment

| Parameter | HHT-4 | NRX-A5(4) |
|---|---|---|
| Cluster Hydrogen Flow Rate, lb/sec | 0.0950 | 0.2677 0.121426677kg/s |
| Tie Rod Annulus Flow Rate, lb/sec | 0.0096 | 0.0146 6.622e-3kg/s |
| Cluster Pressure Drop, psi | 50 | 125 0.861845MPa |
| Tie Rod Exit Gas Temperature, °R | 1100 | 577 3.97827MPa |
| Cluster Exit Gas Temperature, °R | 4283 | 4283 2379.444K |
| Station 43 Temperature, °R | 4675* | 4280+ 2377.778K |
| Station 35 Temperature, °R | 4463* | 3925+ 2180.556K |
| Station 16 Temperature, °R | 4121* | 2193+ 1218.333K |
| Station 8 Temperature, °R | 2547* | 1441+ 800.5556K |
| Support Block Temperature, °R | 4283 | 4283 2379.444K |
| Power Input to Gas, mw | 1.50 | 4.12 |
| Total Power Input, mw | 2.00 | ---- |
| Tie Rod Diameter, Inches | 0.110 | 0.110 |
| Tie Rod Bushing Orifice, Inches | 0.052 | 0.0325 |

(Core) Station <z in inches>

*Surface

+Channel Wall

Figure B.9: Table V from the document

## B.5 NRX-A6 Validation Data

Source for fig. B.10, B.11, B.15, and figures in tab. B.1: NRX-A6 Transducer Evaluation Report by T.C. Burlas, E.F. Karako, R.C. Myers, Dec 1968 (WANL-TME-1863)
Note: Pages 6 and 24 of the above reference state that the peripheral thermocouples AT 703, 704, 705, & 706 as shown in fig. 1-36 of the same report indicate lower readings than the rest. It is therefore assumed that the clustered graphs of all other cluster exit gas thermocouples is indicative of measurements nearer to the center, as further details on radial placement cannot be found in the report.
Source for fig. 5.12: Westinghouse Astronuclear Laboratory to the 27th high Temperature Fuels Committee Meeting by L.R. Fleischer & C.R. Simmons, Dec 1968 (WANL-TME-1878)

| Measurement | Meas. No. 1 | Meas. No. 2 | Predicted (psi) | Measured (CRT = 14600) Xdcr 1 (psi) | Xdcr 2 (psi) |
|---|---|---|---|---|---|
| Reflector Inlet Plenum | AP 300 | AP 302 | 746 | 741.7 | 745.2 |
| Refl. Axial Pressure Drop | AP 301D | AP 303D | 24 | 15.6 | 15.5 |
| Refl. Outlet Plenum | AP 304 | - | 722 | 728.9 | - |
| Refl. Outlet to Core Inlet | P 501D | - | - | 13.2 | - |
| Core Inlet | AP 602 | AP 604 | 706 | 709.1 =4.889MPa | 709.5 =4.892MPa |
| Core Axial Press. Drop | AP 603D | AP 605D | 135 | 121.8 =0.84MPa | 120.2 =0.8287MPa |
| Core Exit | P702 | P704 | 571 | 585 =4.033MPa | 587.7 =4.052MPa |
| Reactor Pressure Drop | AP 701D | AP 703D | 175 | 153.4 | 151.6 |
| Seal Chamber 10 | AP 414 | - | - | 700.0 | - |
| Seal Chamber 8 to 10 | AP 415D | - | - | 7.1 | - |
| Seal Chamber 18 | AP 416 | - | - | 652 | - |
| Seal Chamber 9 to 10 | AP 417D | - | - | 3.25 | - |
| Seal Chamber 10 to 11 | AP 419D | - | - | 4.0 | - |
| Seal Chamber 10 to 12 | AP 421D | - | - | 8.2 | - |
| Seal Chamber 14 to 18 | AP 423D | - | - | 26.18 | - |
| Seal Chamber 16 to 18 | AP 425D | - | - | 8.25 | - |
| Seal Chamber 18 to 20 | AP 427D | - | - | 18.8 | - |
| Seal Chamber 18 to 21 | AP 429D | - | - | 26.2 | - |
| Seal Chamber 18 to 22 | AP 431D | - | - | 40.1 | - |

Figure B.10: NRX-A6 Pressure Data

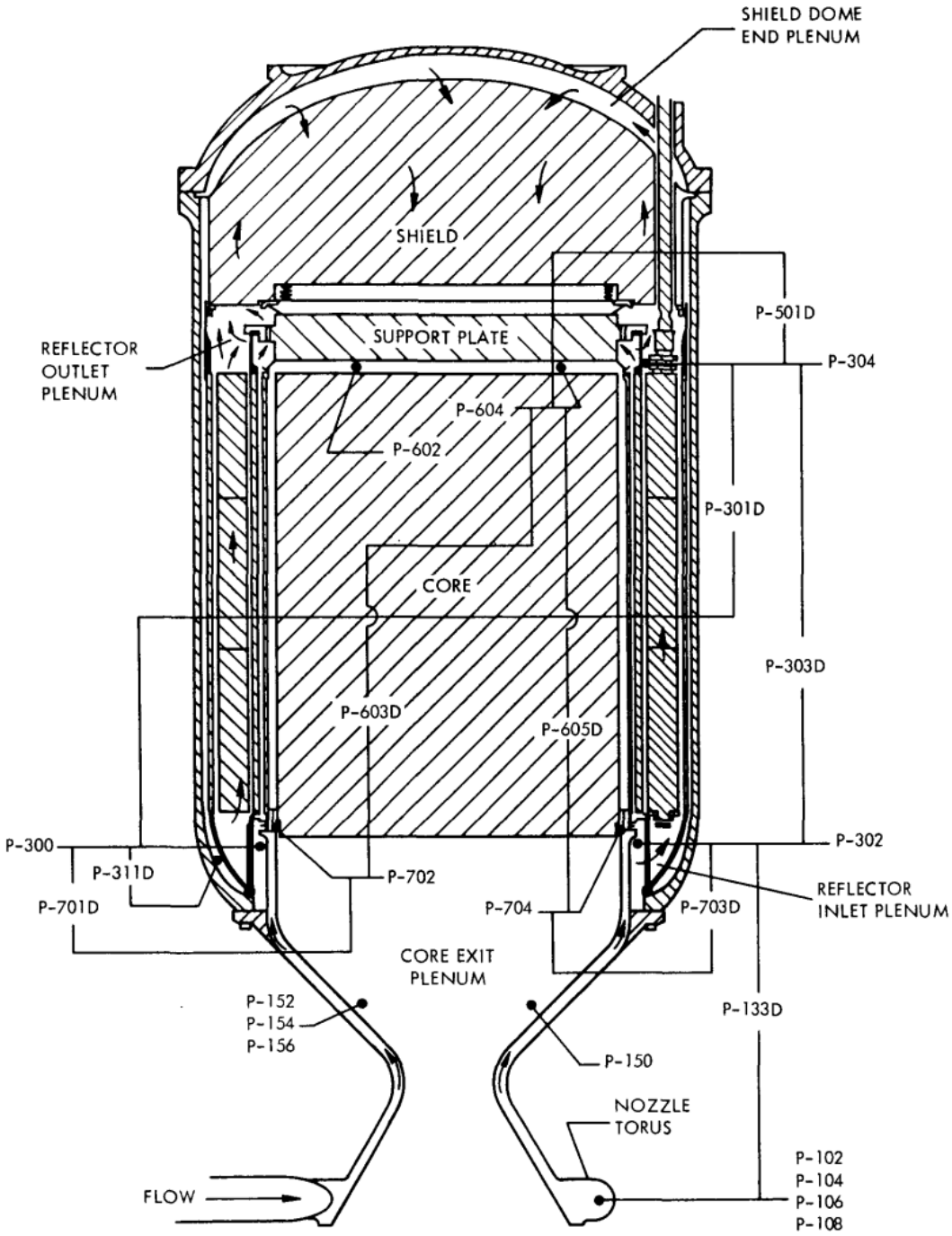



Figure 5-1. (U) NRX-A6 Pressure Measurement Locations

Figure B.11: Location of pressure transducer probes

Table B.1: Core center region measurements from Transducer Evaluation Report for hot cluster

| **Thermocouple** | **Designation** | **Core Radius** | **T [K]** | **T[°R]** | **CRT** | **Page** |
|---|---|---|---|---|---|---|
| Inlet Plenum | AT 776 | 25.4mm | 128.889 | 232 | 15'200s | 12 |
| Station 1 | AT 601 | 27.94mm | $324 \pm 46$ | $583 \pm 83$ | pwr. hold | 58 |
| Station 20 | AT 622 & AT 628 | 53.34mm | $1986.1 \pm 5.6$ | $3575 \pm 10$ | 17'800s | 22 |
| Cluster Exit Gas | AT 690-708 w/o AT 703-706 | various | $2422 \pm 97$ | $4360 \pm 175$ | pwr. hold | 66 |
| Tie Rod | AT 713 | 50.08mm | 398.4 | 710 | pwr. hold | 20 |
| Tie Rod | AT 720 | 100.16mm | 416.7 | 750 | pwr. hold | 20 |

Table B.2: Temperatures for for core center region from fig. B.12

| **Parameter [unit]** | **Conversion** | **Original value** |
|---|---|---|
| Station 20 | $1900 \pm 11\,K$ | $3420 \pm 20°R$ |
| Station 26 | $2172 \pm 11\,K$ | $3910 \pm 20°R$ |
| Mixed Cluster Exit Gas | $2486 \pm 30\,K$ | $4475 \pm 55°R$ |
| Total core flow | 33.1576kg/s | 71.3lb/s |

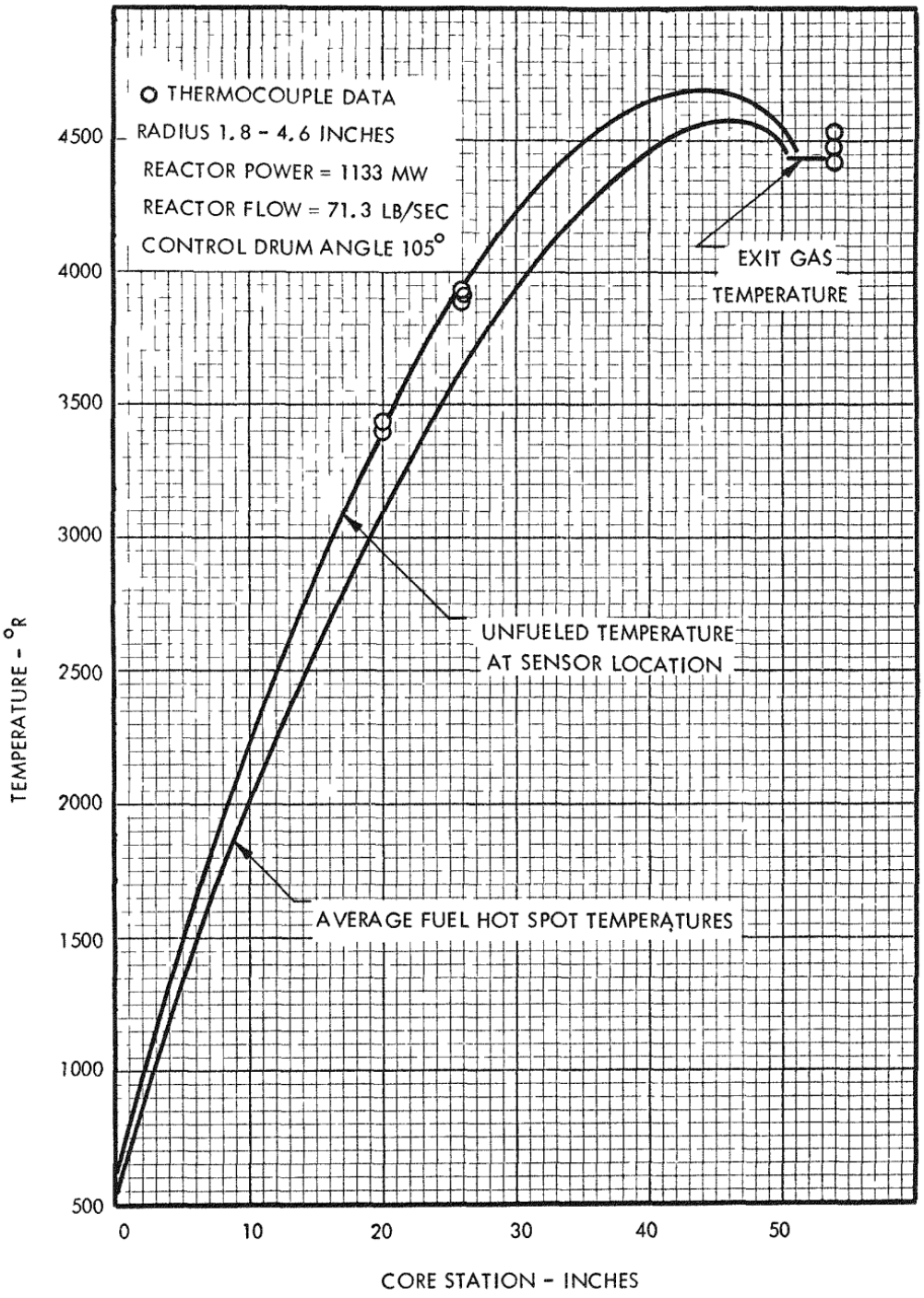



Figure 6. (U) Core Central Region, Axial Temperature Distributions During Power Hold

Figure B.12: Axial fuel temperature distribution for radius 45.72 to 116.84mm
Note that the curves are likely calculated, not measured as the reactor core elements were thermocoupled only at core stations 1, 20, and 26 inch

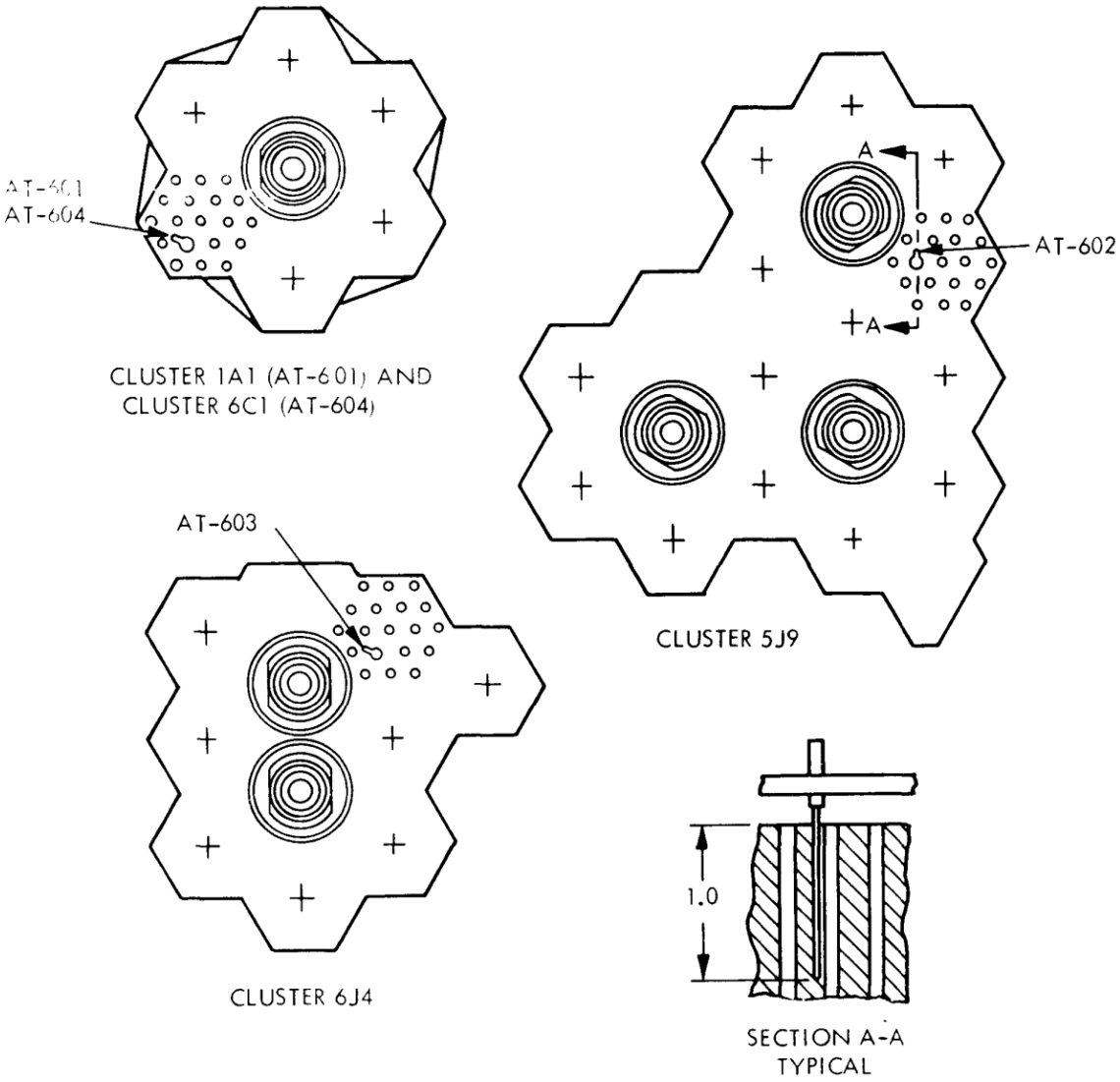
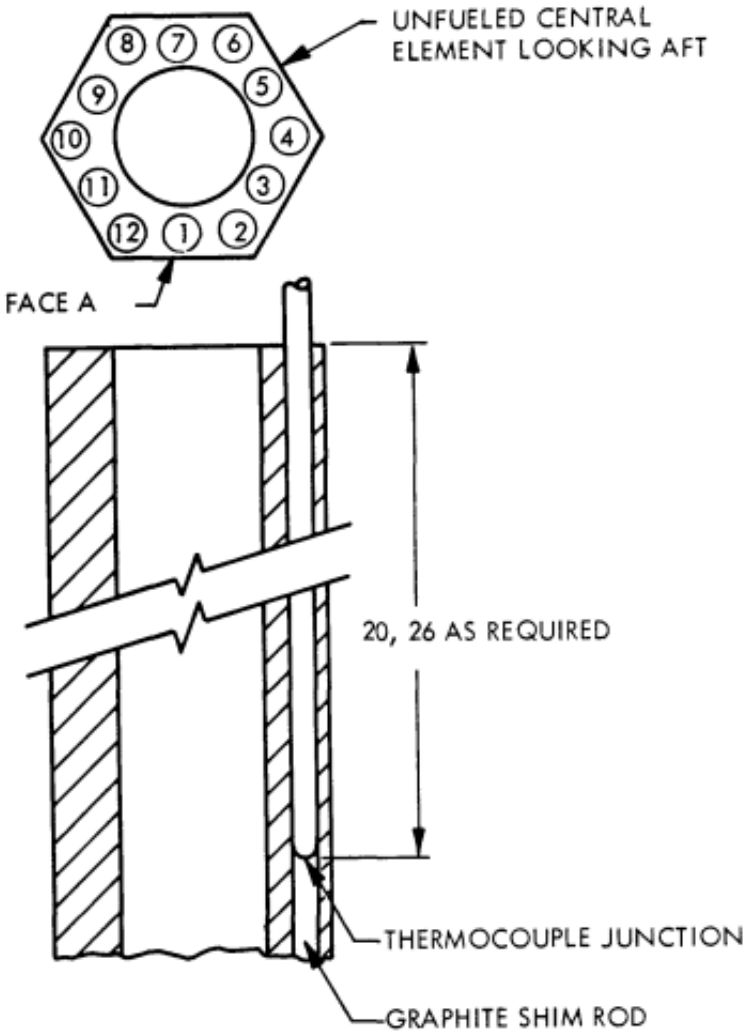



Figure 1-27. (CRD) NRX-A6 Station 1.0 Fueled Element Thermocouples (U)

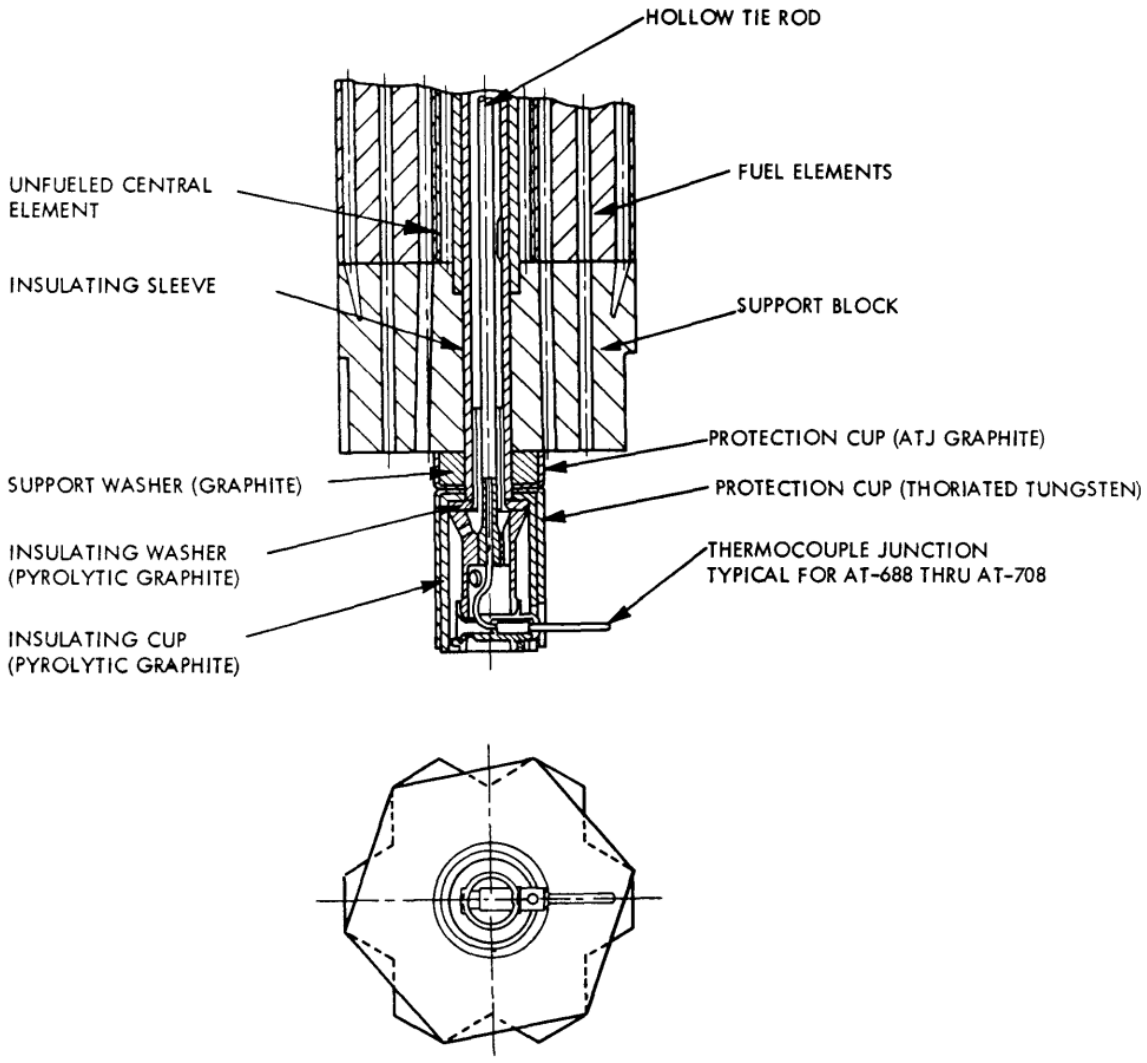
Figure B.13: Placement of Fuel Element thermocouples in the center point between between three adjacent channels of a fuel cluster (see fig. B.4 for placemenet within core)




NOTE: TWELVE HOLES ARE SHOWN ONLY FOR LOCATION IDENTIFICATION. ONLY THOSE HOLES REQUIRED FOR INSTRUMENTATION ARE EXTRUDED IN A GIVEN UNFUELED ELEMENT. THERMOCOUPLES ARE IN EVEN-NUMBERED (CORNER) HOLES ONLY.

Figure 1-31. (CRD) NRX-A6 Unfueled Element Thermocouple (U)

Figure B.14: This schematic figure from NRX-A6 Transducer Report explains that holes were only present in unfueled elements as needed for instrumentation




Fiaure 1-35. (CRD) NRX-A6 Cluster Exit Gas Thermocouple (U)

Figure B.15: Placement of the fuel exit gas thermocouple, note the insulating barrier between the probe and the colder tie rod annulus exhaust

# Appendix C
# Parts of the Fuel cluster

## C.1 Core Element Drawings

Source: Specifications and Drawings for Core Elements – The Plug Channel Core by C.C. Fields, Aug 1963 (WANL-TME-406)

Note: Measurements in the following technical drawings are simply designated ”BASIC TYP” or ”TYP BASIC”. They denote 1/1000inch=0.0254mm for lengths and degrees for angles. Crucial for this thesis is that:

●Coolant channel diameter: 0.1in=2.54mm and

●Core element length: 52in=1320.8mm.

●Channel pitch (distance between adjacent channel centers): 44mm [29, fig.3]

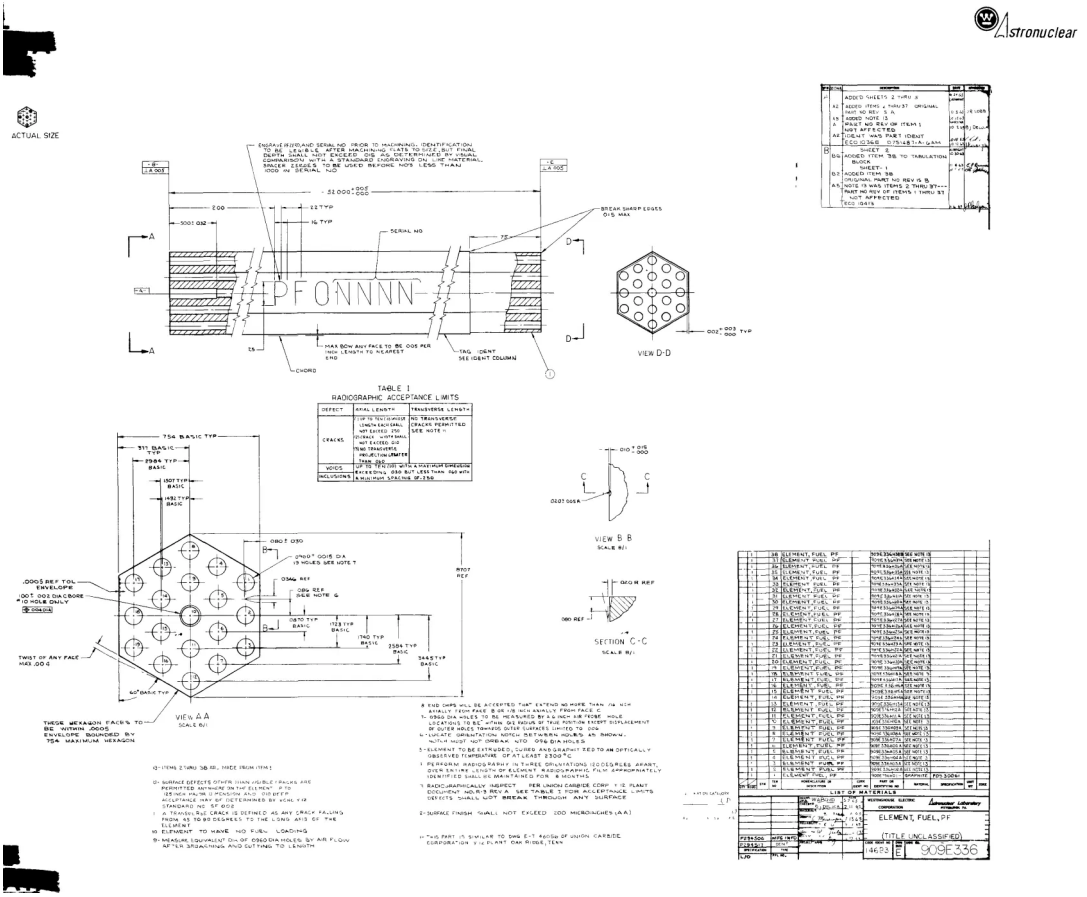
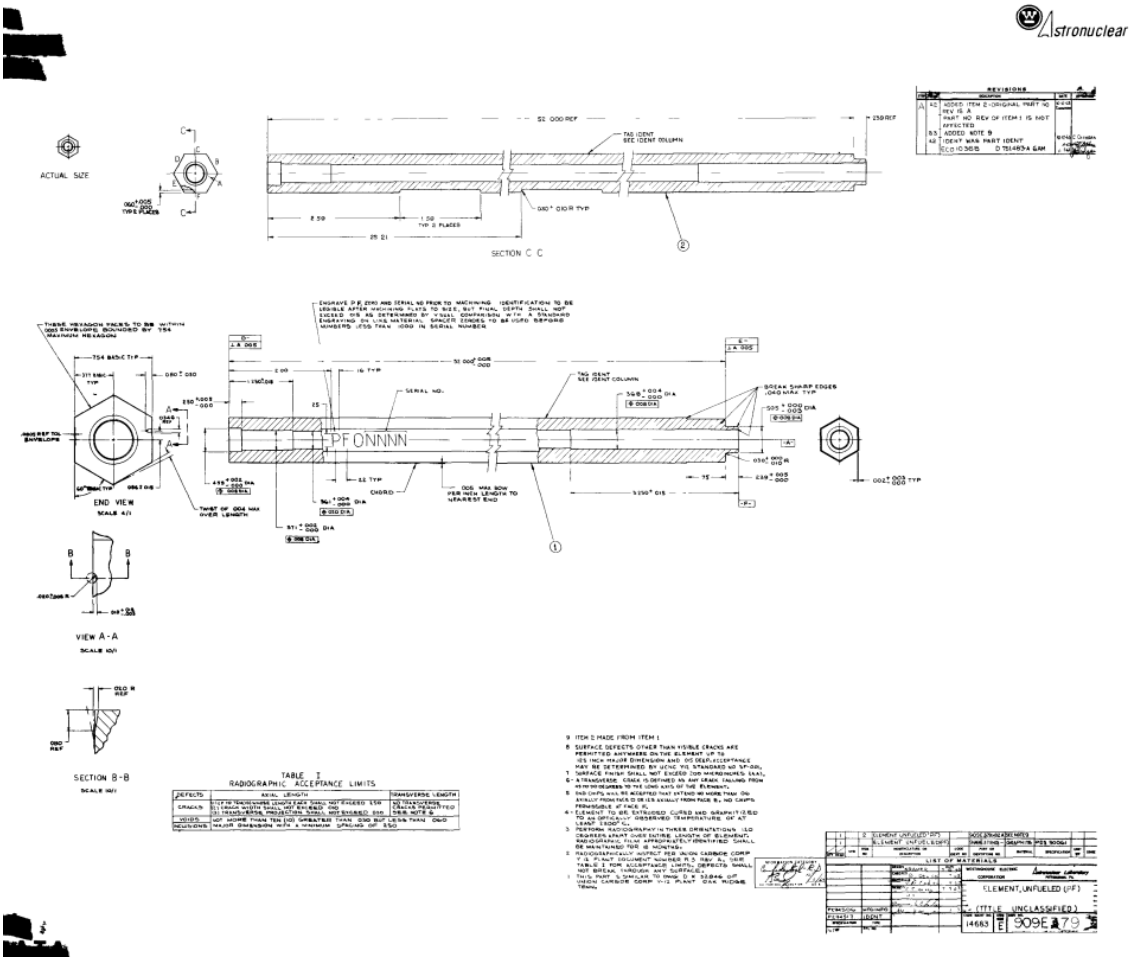
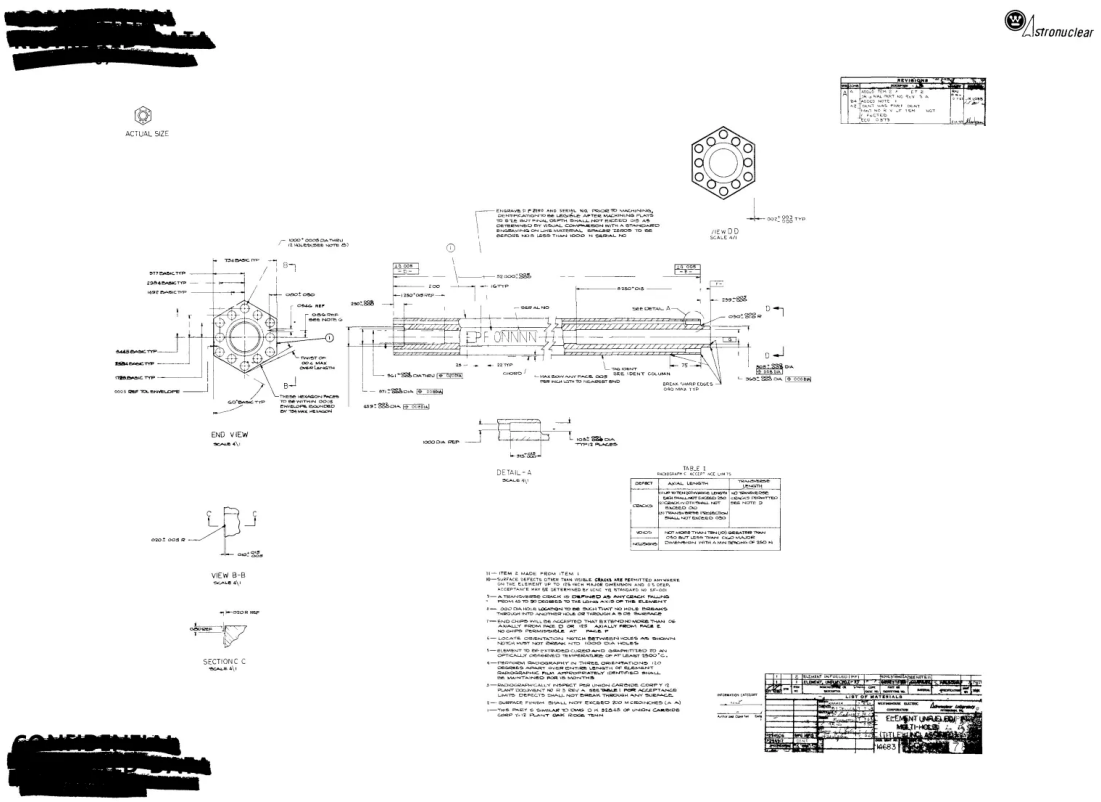



Figure C.1: Technical drawing of the standard fuel element




(a) Technical drawing of the unfueled element (without holes)




(b) Technical drawing with hole placements for thermocouples, see fig. B.14

Figure C.2: Technical drawings of the unfueled core element used for geometric reference.

## C.2 Support Block Reconstruction

From NRX-A5 to A6, the design of the support block and its hardware underwent an evolution which makes a precise reconstruction of every detail exceedingly difficult. However, measurements critical to the thermal problem could be reconstructed. Herafter follows a brief summary of the process reconstructing support block design.
The first indication is figure C.3 dated October 1966 after the NRX-A5 tests took place specifying a support block length of 2.25in=57.15mm.

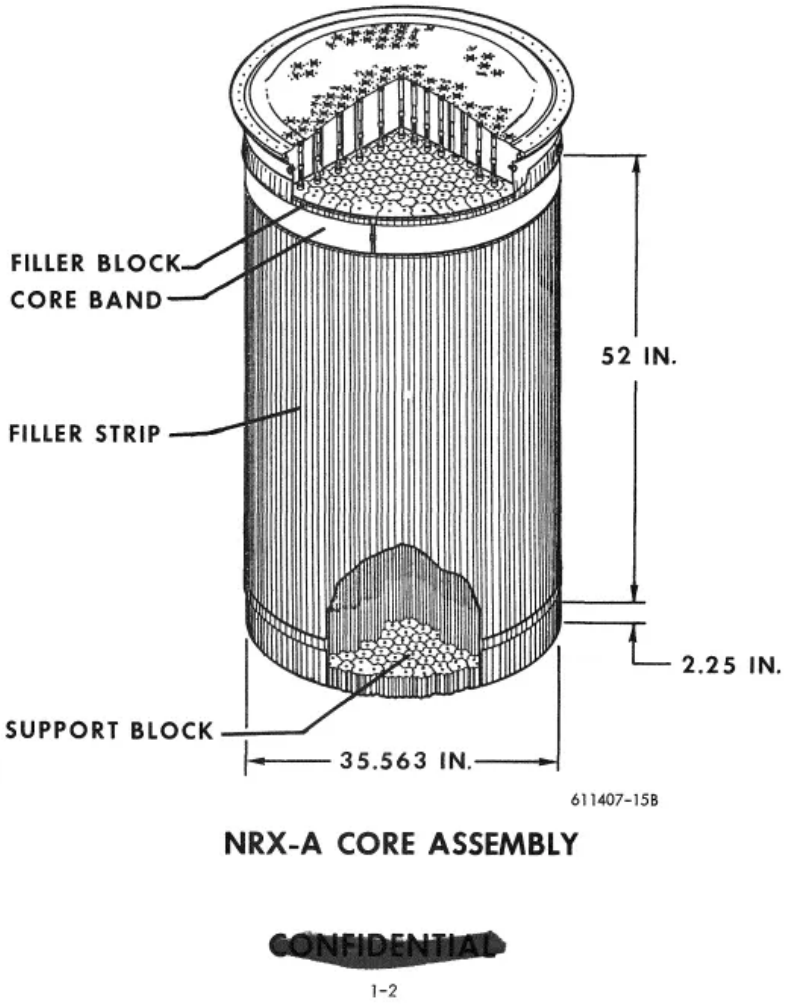



Figure C.3: Source: [30, p.1-2]

Knowing the full length of this support block and that the side adjacent to the core elements shares their geometry, the overall geometry could be reconstructed using single-view metrology on the photograph in fig.C.5, whereas fig. C.4, particluarly the hot-end view is used to establish the broad geometry.

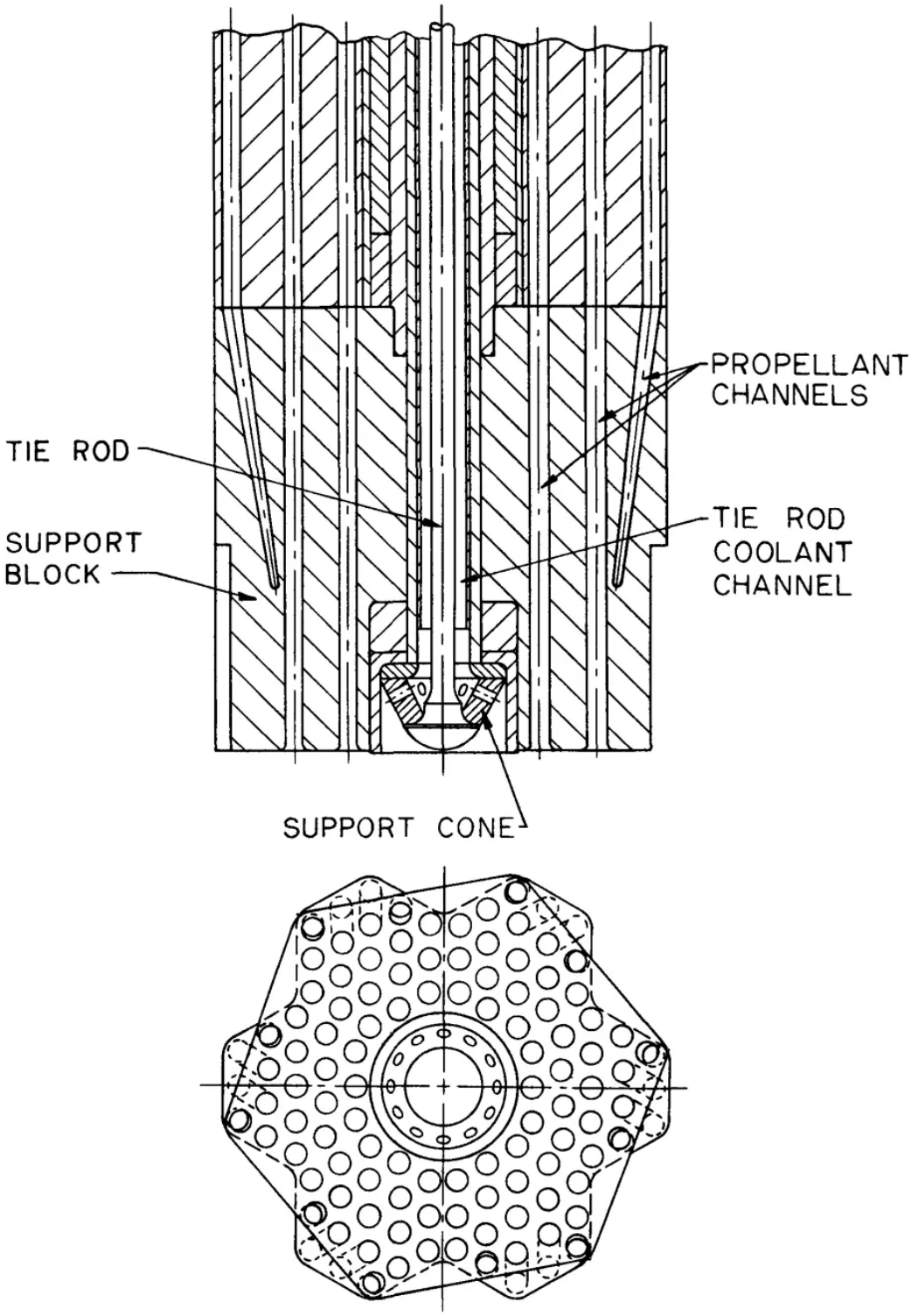



FIGURE 1 SUPPORT BLOCK END OF SEVEN-ELEMENT CLUSTER

Figure C.4: Schematic Drawing of the pre-NRX-A6 Core Support Block source WANL-TME-596 fig.1

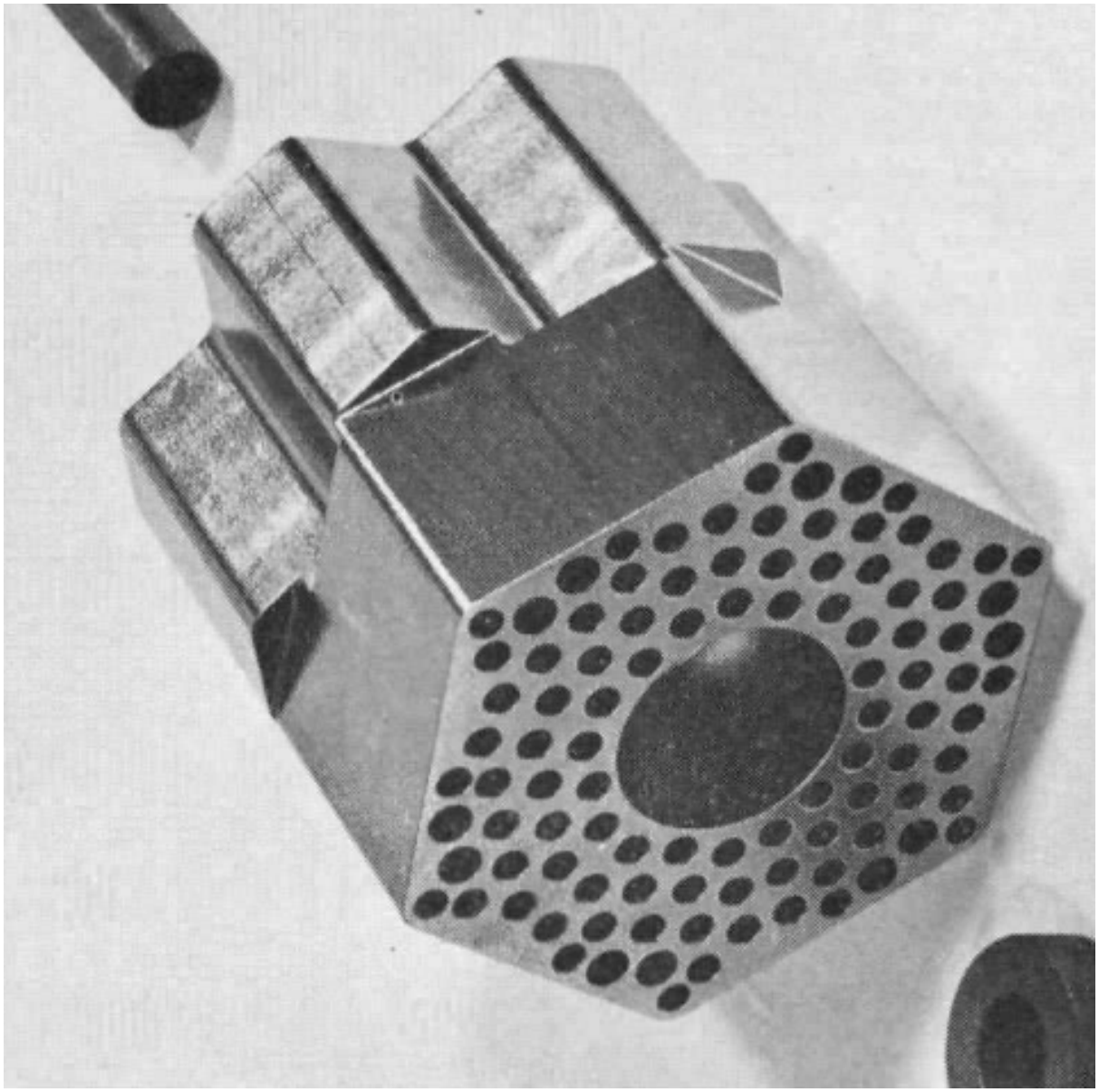
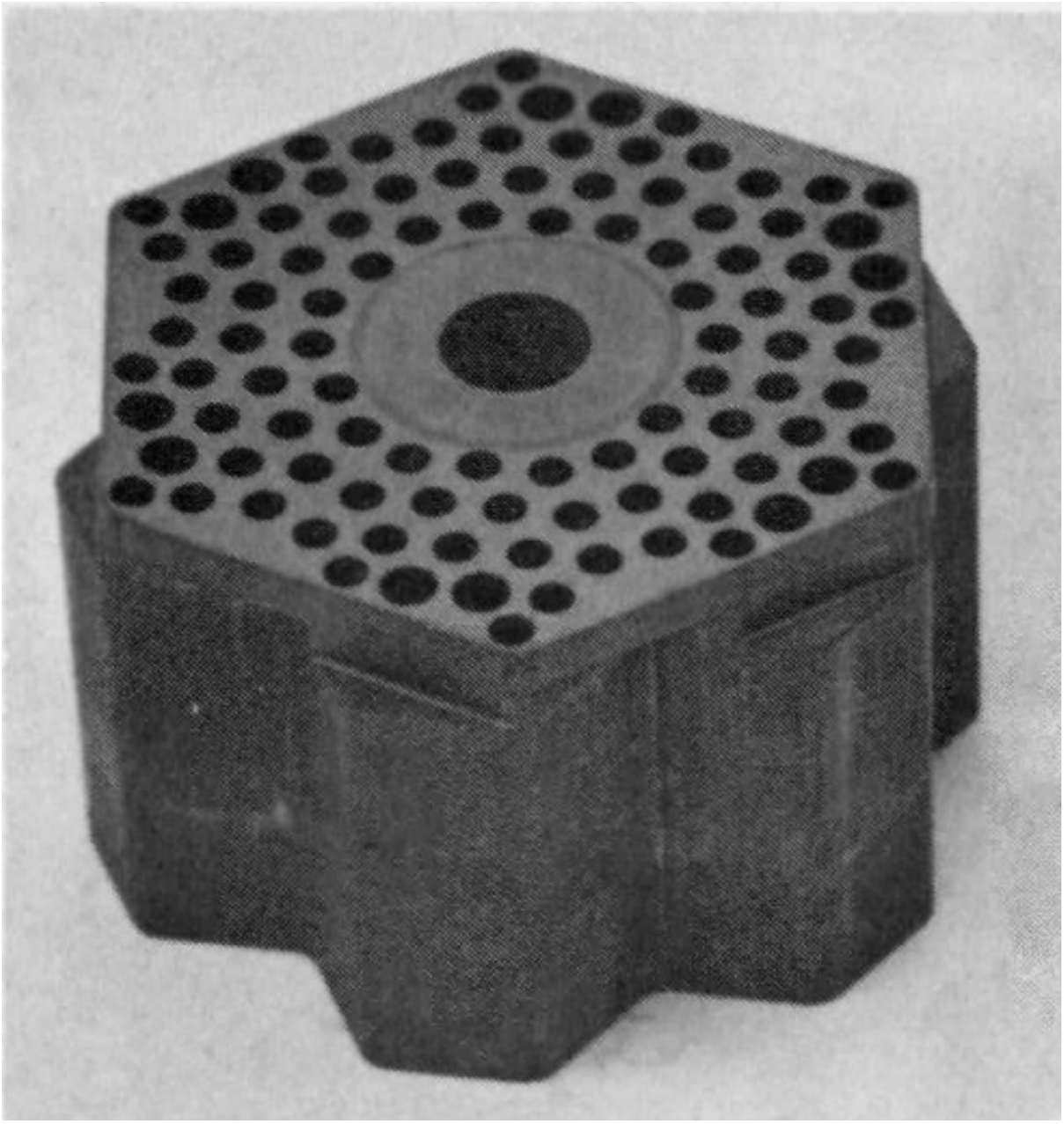
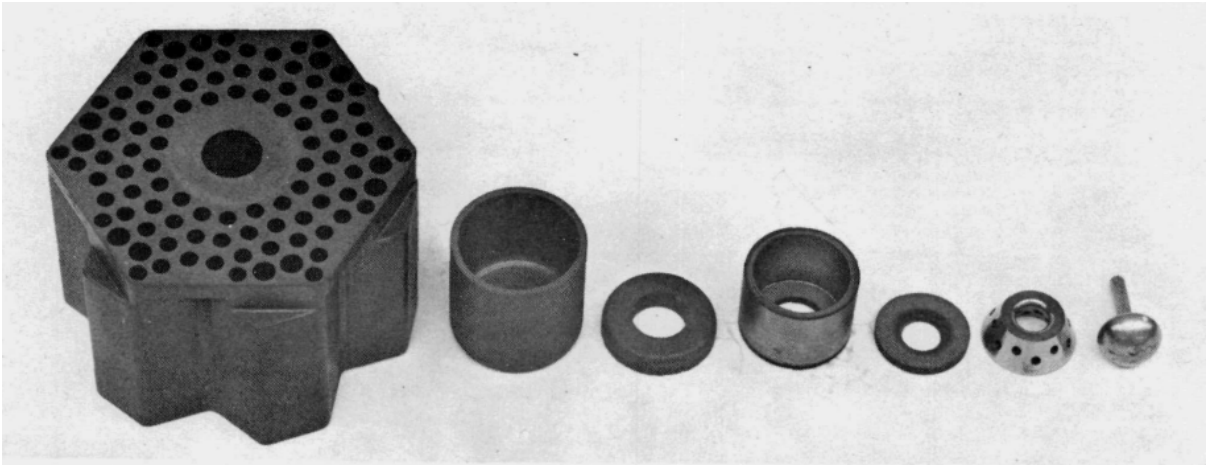


Figure C.5: NRX-A5 support block source WANL-TME-557 Fig. B.4

To then arrive at the ”skirtless” (shortened outlet section) NRX-A6 support block geometry, two more photographs are used. The support blocks in fig. C.5 are identical, the only difference is that the one in C.6a is shortened at the outlet by a known length and the one C.6a used in NRX-A6 by an unknown length. Here using single-view metrology and algebraic comparison of the ratios in the pictures, the exact length of the NRX-A6 Support Block is determined to be 1.5in=38.1mm. Destructive tests documented in reference [**44**] helped establish irregular channel arrangements where two channels at the edge unified into one enlarged outlet and single-view metrology established enlarged outlet diameter and the geometry of support block hardware seen in fig. C.6b, although the corrosion cup ultimately differed in reality, as seen in fig. C.7a & C.7b.

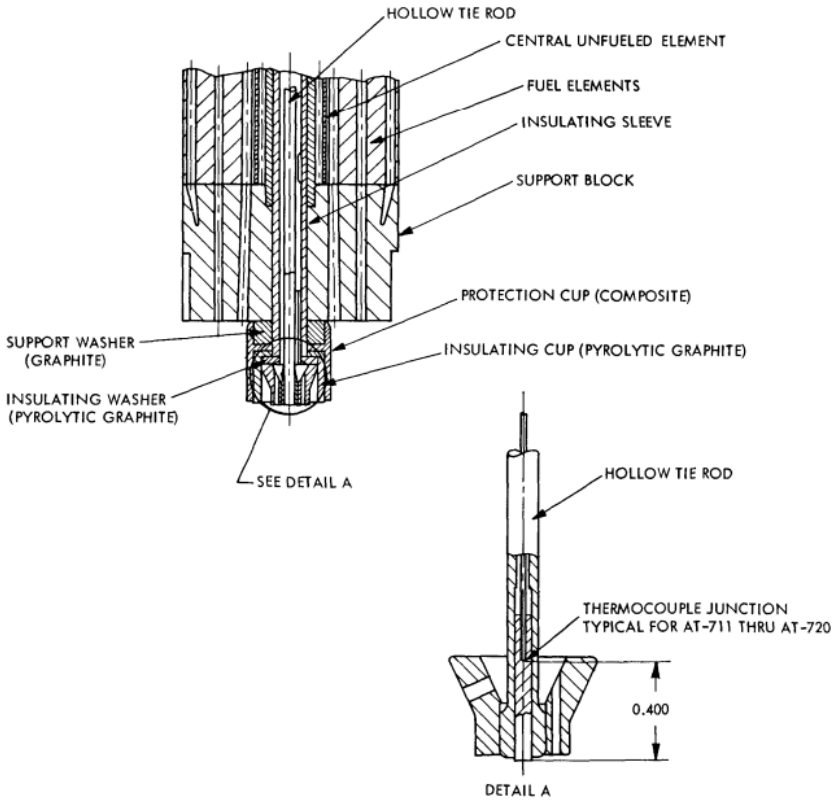


(a) Same support block as in fig. C.5 but with 19.05 mm (0.75in) machined away from TME-1373 fig. 3–2



(b) Support block designated as being for NRX-A6 in WANL-TME-1373 fig. 5–1

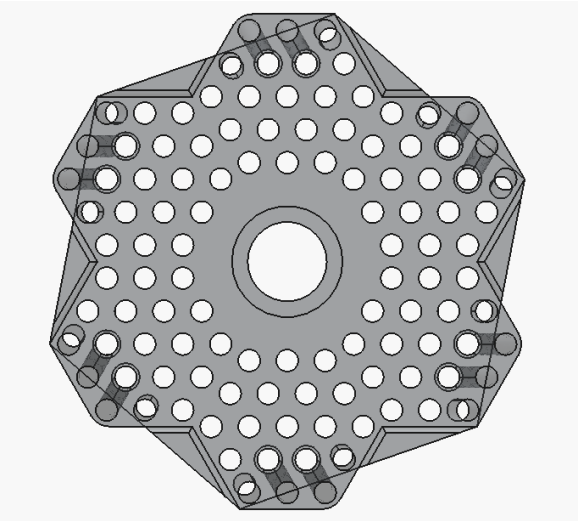
Figure C.6: Photographic documentation of support block geometry evolution used for reconstruction.



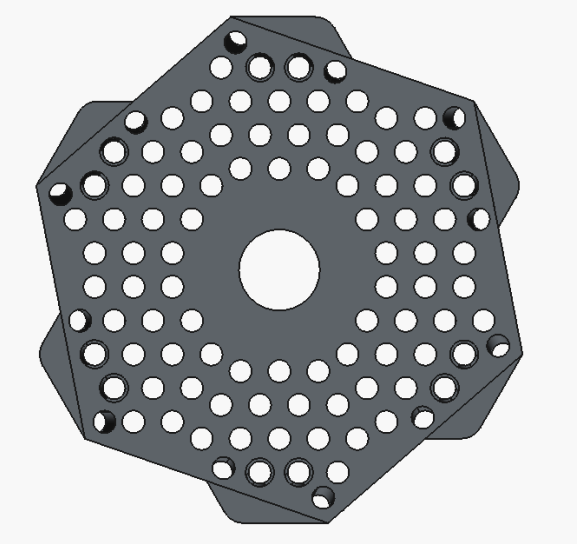

(a) Support block hardware detail used in NRX-A6 from the Transducer Evaluation Report; note that the tie rod is atypical due to the presence of a thermocouple.

(b) Probable design of the NRX-A6 support block with carbide composite anti-corrosion cup. Source: WANL-TME-1645.

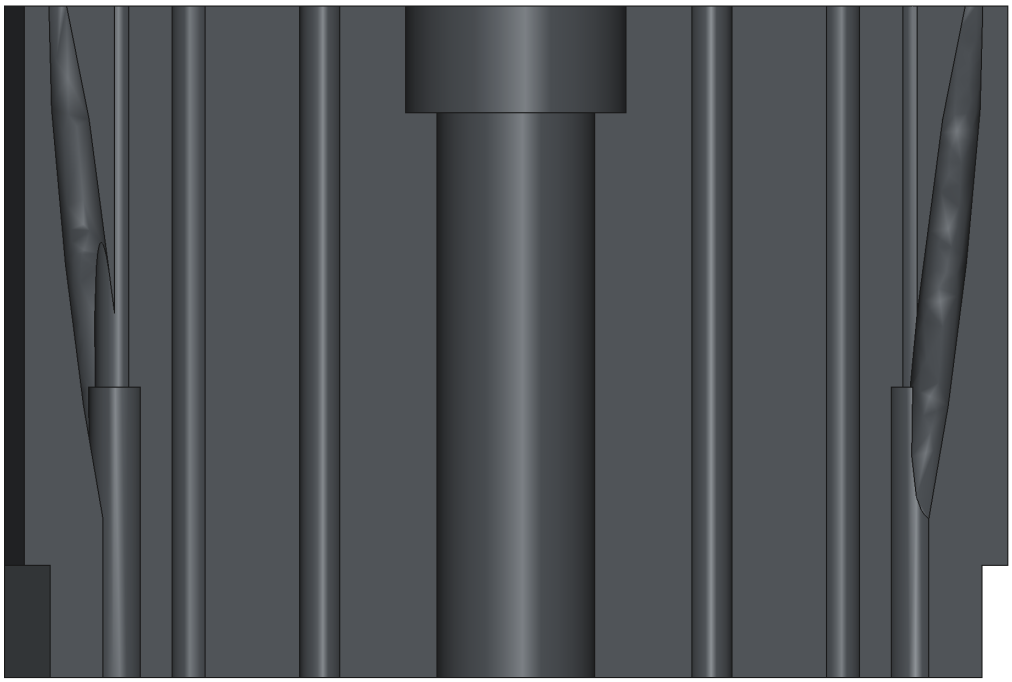
Figure C.7: Support block hardware configuration and reconstructed design details relevant to the NRX-A6 hot end.



(a) Transparent view from the side interfacing with core elements.



(b) Support block hot-end view in FreeCAD.



(c) Cut into Support Block showing the unification of channels

Figure C.8: Approximate reconstruction of NRX-A6 support block geometry rendered in FreeCAD. Not all edges could be filleted due to artifacts arising at intersections.

# Appendix D
# Codes Used

## D.1 Python Code used to generate average radial Power Distribution and average Value

```
#!/usr/bin/env python3
# -*- coding: utf-8 -*-
import traceback
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from pathlib import Path

# ----------------------------
# Parsing: block-format "CSV"
# ----------------------------
def read_block_csv(path: str | Path):
    path = Path(path)
    if not path.exists():
        raise FileNotFoundError(f"File not found: {path.resolve()}")

    lines = path.read_text(encoding="utf-8", errors="replace").
       splitlines()
    data = {}
    i, n = 0, len(lines)

    def is_blank(s: str) -> bool:
        return s is None or s.strip() == ""

    while i < n:
        while i < n and is_blank(lines[i]):
            i += 1
        if i >= n:
            break

        header = lines[i].strip()
        i += 1
        if "," not in header:
```

```
            raise ValueError(f"Expected '<category>,<subcategory>'
                at line {i}, got: {header}")
        category, subcat = [x.strip() for x in header.split(",", 1)]

        while i < n and is_blank(lines[i]):
            i += 1
        if i >= n:
            raise ValueError(f"Unexpected EOF after header '{header
                }'")
        _ = lines[i].strip()  # column header line
        i += 1

        rows = []
        while i < n and not is_blank(lines[i]):
            row = lines[i].strip()
            parts = [x.strip() for x in row.split(",")]
            if len(parts) >= 2:
                try:
                    r = float(parts[0])
                    p = float(parts[1])
                    rows.append((r, p))
                except ValueError:
                    pass
            i += 1

        if not rows:
            raise ValueError(f"No numeric data found for block '{
                category},{subcat}'")

        df = (
            pd.DataFrame(rows, columns=["r_cm", "relVolPower"])
            .sort_values("r_cm")
            .reset_index(drop=True)
        )
        data.setdefault(category, {})[subcat] = df

        while i < n and is_blank(lines[i]):
            i += 1

    return data


# ----------------------------
```

```
# Utilities
# ----------------------------
def piecewise_linear_interp(x_src, y_src, x_query):
    x_src = np.asarray(x_src, dtype=float)
    y_src = np.asarray(y_src, dtype=float)
    x_query = np.asarray(x_query, dtype=float)
    return np.interp(x_query, x_src, y_src)

def rel_err_percent(y, y_ref):
    y = np.asarray(y, dtype=float)
    y_ref = np.asarray(y_ref, dtype=float)
    out = np.full_like(y, np.nan, dtype=float)
    mask = np.isfinite(y) & np.isfinite(y_ref) & (np.abs(y_ref) > 0)
    out[mask] = (y[mask] - y_ref[mask]) / y_ref[mask] * 100.0
    return out

def combined_connector_df(df_a6: pd.DataFrame, df_peak: pd.DataFrame
   ) -> pd.DataFrame:
    """
    Build the connector polyline dataset by treating the peak point(
       s) as if they
    belong to the same dataset (but they can still be plotted in a
       different color).
    """
    df = pd.concat([df_a6, df_peak], ignore_index=True)
    # If duplicates exist at same r, keep last (peak should win if
       same r appears)
    df = df.sort_values("r_cm").drop_duplicates(subset=["r_cm"],
       keep="last").reset_index(drop=True)
    return df


# ----------------------------
# Main plotting
# ----------------------------
def main():
    SCRIPT_DIR = Path(__file__).resolve().parent
    csv_path = SCRIPT_DIR / "radial_power_template_4blocks.csv"
    data = read_block_csv(csv_path)

    cat_a, cat_b = "4-5", "4-6"
    if cat_a not in data or cat_b not in data:
```

```
        raise KeyError(f"Expected categories '{cat_a}' and '{cat_b}'
            in file; found: {list(data.keys())}")

    def find_subcat(dct, predicate):
        for k in dct.keys():
            if predicate(k):
                return k
        return None

    sub_a6_a = find_subcat(data[cat_a], lambda k: "nrx-a6" in k.
       lower().replace(" ", ""))
    sub_pk_a = find_subcat(data[cat_a], lambda k: ("nrx-a3" in k.
       lower().replace(" ", "")) and ("peak" in k.lower()))
    sub_a6_b = find_subcat(data[cat_b], lambda k: "nrx-a6" in k.
       lower().replace(" ", ""))
    sub_pk_b = find_subcat(data[cat_b], lambda k: ("nrx-a3" in k.
       lower().replace(" ", "")) and ("peak" in k.lower()))

    if not all([sub_a6_a, sub_pk_a, sub_a6_b, sub_pk_b]):
        raise KeyError(
            "Could not find required subcategories. "
            f"Found in 4-5: {list(data[cat_a].keys())}; found in
               4-6: {list(data[cat_b].keys())}"
        )

    df_45_a6 = data[cat_a][sub_a6_a]
    df_45_pk = data[cat_a][sub_pk_a]
    df_46_a6 = data[cat_b][sub_a6_b]
    df_46_pk = data[cat_b][sub_pk_b]

    # Connector polylines includes the peak points from alternative
       dataset
    df_45_conn = combined_connector_df(df_45_a6, df_45_pk)
    df_46_conn = combined_connector_df(df_46_a6, df_46_pk)

    # ----------------------------
    # Figure 1: points + connectors (including peak points)
    # ----------------------------
    fig1, ax1 = plt.subplots(figsize=(8.5, 5.5))

    ax1.plot(df_45_a6["r_cm"], df_45_a6["relVolPower"], linestyle="
       None", marker="x", color="red",
             label=f"{cat_a},{sub_a6_a}")
```

```
ax1.plot(df_45_pk["r_cm"], df_45_pk["relVolPower"], linestyle="
    None", marker="x", color="green",
          label=f"{cat_a},{sub_pk_a}")

ax1.plot(df_46_a6["r_cm"], df_46_a6["relVolPower"], linestyle="
    None", marker="x", color="blue",
          label=f"{cat_b},{sub_a6_b}")
ax1.plot(df_46_pk["r_cm"], df_46_pk["relVolPower"], linestyle="
    None", marker="x", color="purple",
          label=f"{cat_b},{sub_pk_b}")

# Connectors go through A6 + peak points (treated as one dataset
    )
ax1.plot(df_45_conn["r_cm"], df_45_conn["relVolPower"],
    linestyle="-", color="magenta",
          label=f"{cat_a} connectors (A6 & A3-peak)")
ax1.plot(df_46_conn["r_cm"], df_46_conn["relVolPower"],
    linestyle="-", color="cyan",
          label=f"{cat_b} connectors (A6 & A3-peak)")

ax1.set_title("Radial relative volumetric power: points + per-
    category connectors")
ax1.set_xlabel("r [cm]")
ax1.set_ylabel("relVolPower [-]")
ax1.grid(True, alpha=0.3)
ax1.legend(loc="center left", bbox_to_anchor=(1.02, 0.5),
    borderaxespad=0.0)
fig1.tight_layout()

# ----------------------------
# Figure 2: points + averaged curve + relative errors vs average
# (average is between the two connector polylines that include
    peak points)
# ----------------------------
fig2, ax2 = plt.subplots(figsize=(8.5, 5.5))

ax2.plot(df_45_a6["r_cm"], df_45_a6["relVolPower"], linestyle="
    None", marker="x", color="red",
          label=f"{cat_a},{sub_a6_a}")
ax2.plot(df_45_pk["r_cm"], df_45_pk["relVolPower"], linestyle="
    None", marker="x", color="green",
          label=f"{cat_a},{sub_pk_a}")
```

```
ax2.plot(df_46_a6["r_cm"], df_46_a6["relVolPower"], linestyle="
    None", marker="x", color="blue",
          label=f"{cat_b},{sub_a6_b}")
ax2.plot(df_46_pk["r_cm"], df_46_pk["relVolPower"], linestyle="
    None", marker="x", color="purple",
          label=f"{cat_b},{sub_pk_b}")

# common r-grid over overlap of the *connector* domains
rmin = max(df_45_conn["r_cm"].min(), df_46_conn["r_cm"].min())
rmax = min(df_45_conn["r_cm"].max(), df_46_conn["r_cm"].max())
if not (rmax > rmin):
    raise ValueError("No overlap in r between 4-5 and 4-6
        connector datasets; cannot form averaged curve.")

r_grid = np.linspace(rmin, rmax, 800)
y45_grid = piecewise_linear_interp(df_45_conn["r_cm"],
    df_45_conn["relVolPower"], r_grid)
y46_grid = piecewise_linear_interp(df_46_conn["r_cm"],
    df_46_conn["relVolPower"], r_grid)
y_avg = 0.5 * (y45_grid + y46_grid)

ax2.plot(r_grid, y_avg, linestyle="-", linewidth=2.0, color="
    black", label="Average(conn45, conn46)")

ax2.set_title("Radial power: points + averaged curve + relative
    error vs average")
ax2.set_xlabel("r [cm]")
ax2.set_ylabel("relVolPower [-]")
ax2.grid(True, alpha=0.3)

ax2r = ax2.twinx()

# errors of each connector vs average, sampled at that connector
    's knot points
yavg_at_45 = piecewise_linear_interp(r_grid, y_avg, df_45_conn["
    r_cm"].to_numpy())
yavg_at_46 = piecewise_linear_interp(r_grid, y_avg, df_46_conn["
    r_cm"].to_numpy())

err45 = rel_err_percent(df_45_conn["relVolPower"].to_numpy(),
    yavg_at_45)
err46 = rel_err_percent(df_46_conn["relVolPower"].to_numpy(),
    yavg_at_46)
```

```
    ax2r.plot(df_45_conn["r_cm"], err45, linestyle="--", color="
        magenta", label=f"{cat_a} error vs avg")
    ax2r.plot(df_46_conn["r_cm"], err46, linestyle="--", color="cyan
        ", label=f"{cat_b} error vs avg")
    ax2r.set_ylabel("Relative error vs average [%]")

    # combined legend outside
    h1, l1 = ax2.get_legend_handles_labels()
    h2, l2 = ax2r.get_legend_handles_labels()
    ax2.legend(h1 + h2, l1 + l2, loc="center left", bbox_to_anchor
        =(1.1, 0.5), borderaxespad=0.0)

    fig2.tight_layout()
    # ---- global average of the averaged curve (length-weighted
        mean over r) ----
    global_avg = np.trapezoid(y_avg, r_grid) / (r_grid[-1] - r_grid
        [0])

    print("\nTHE ANALYSIS IS FOR AN AVERAGE")
    print("COOLANT CHANNEL, SUCH AS AT A POINT IN THE CORE WHERE THE
         RADIAL")
    print("POWER OVER AVERAGE POWER RATIO IS UNITY.\n")
    print(
        f"Global average of Average(conn45, conn46) over r   "
        f"[{r_grid[0]:.3f}, {r_grid[-1]:.3f}] cm: {global_avg:.8f}"
    )

    # ---- confirmation prompt ----
    input("\nPress ENTER to open the plots...")

    plt.show()

if __name__ == "__main__":
    try:
        raise SystemExit(main())
    except SystemExit as e:
        if e.code not in (None, 0):
            input("\nPress ENTER to close...")
        raise
    except Exception:
        print("\n Python error (full traceback):")
        traceback.print_exc()
```

```
        input("\nPress ENTER to close...")
        raise
```

## D.2 Data used in the above code:

Note: The following data originate from figs. B.8a & B.8b .

```
4-5,NRX-A6
r_cm,relVolPower
1.887,1.157
3.396,1.154
5.189,1.145
6.792,1.139
8.453,1.129
11.81,1.109
15.00,1.081
17.17,1.067
20.06,1.033
23.4,1.0
25.38,0.9815
26.89,0.9689
28.52,0.9333
30.38,0.9363
33.7,0.96
35.28,1.046
38.77,0.9304
40.3,0.9452
42.19,0.9444
43.77,0.8933

4-5,NRX-A3 peak
r_cm,relVolPower
37.07,1.132

4-6,NRX-A6
r_cm,relVolPower
1.852,1.156
3.759,1.148
5.185,1.145
6.925,1.136
10.81,1.113
```

```
12.63,1.099
13.93,1.087
15.89,1.073
19.76,1.039
21.47,1.022
22.87,1.006
24.6,0.9871
28.36,0.9786
31.52,0.9486
33.42,0.9471
39.58,1.038
40.37,0.9907
42.22,0.9871

4-6,NRX-A3 peak
r_cm,relVolPower
36.12,1.155
```